Burrows-Wheeler Transform and Suffix Arrays

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Algorithms on Strings
Data Structures and Algorithms

This slide desk is incomplete.
For the complete set of frames,
please see our videos in the
Algorithms on Strings course on Coursera
(Algorithms and Data Structures Specialization)

Outline

- **Burrows-Wheeler Transform**
- Inverting Burrows-Wheeler Transform
- Using BWT for Pattern Matching
- Suffix Arrays
- Approximate Pattern Matching

Text Compression by Run-Length Encoding

- **Run-length encoding** compresses a run of *n* identical symbols:

Text

GGGGGGGGGGCCCCCCCCCCCCAAAAAAATTTTTTTTTTTTTTTTTTCCCCCCG

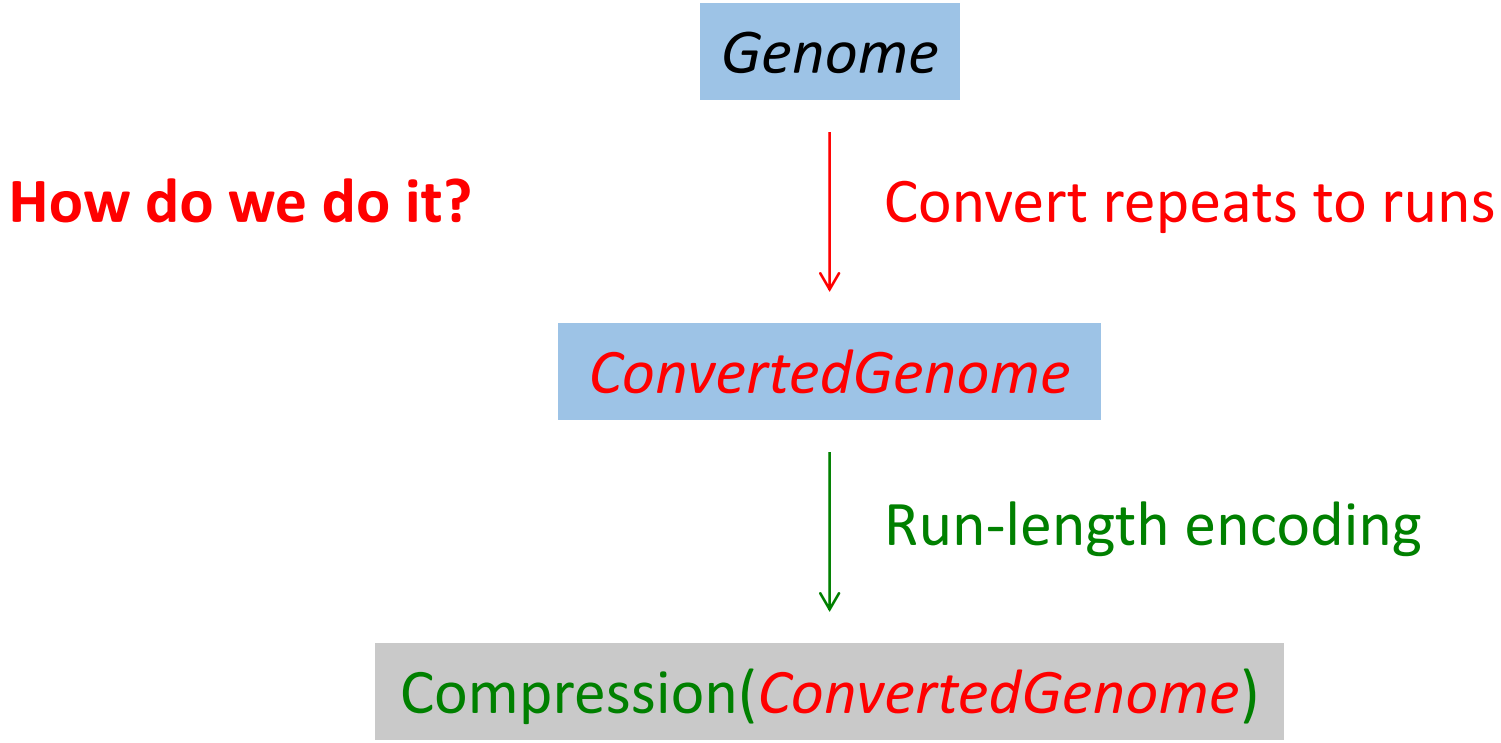


10G11C7A15T5C1G

- genomes don't have lots of runs… but they do have lots of repeats:

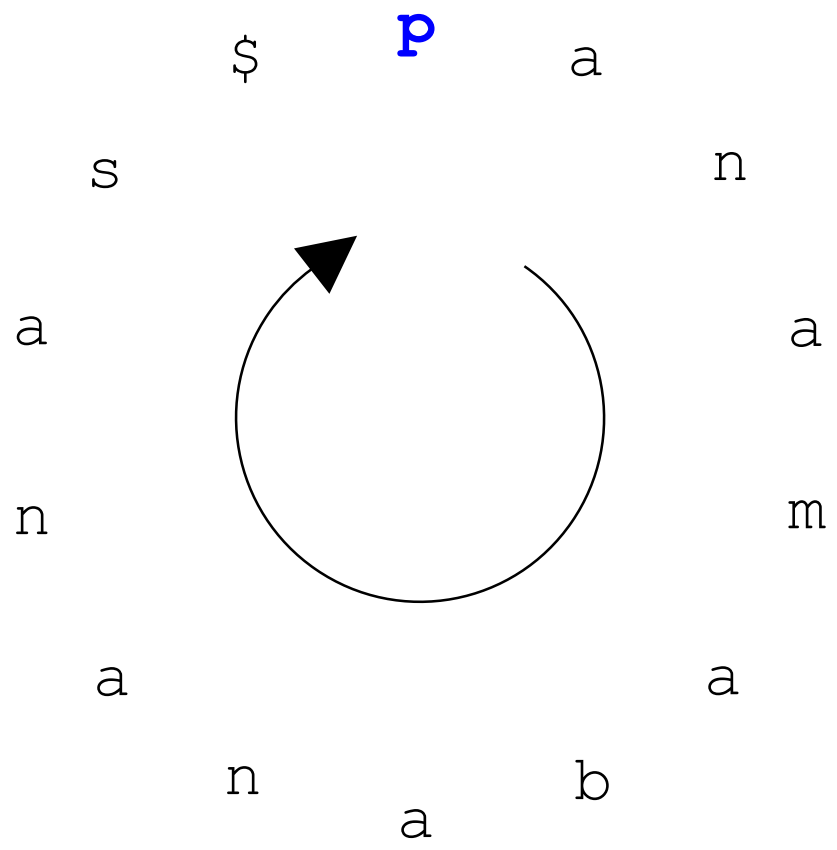
ACTGACCGAAACTGAGTATCCGACTGAAACTGATCAGTACTGACATTGC

Idea: Converting Repeats to Runs



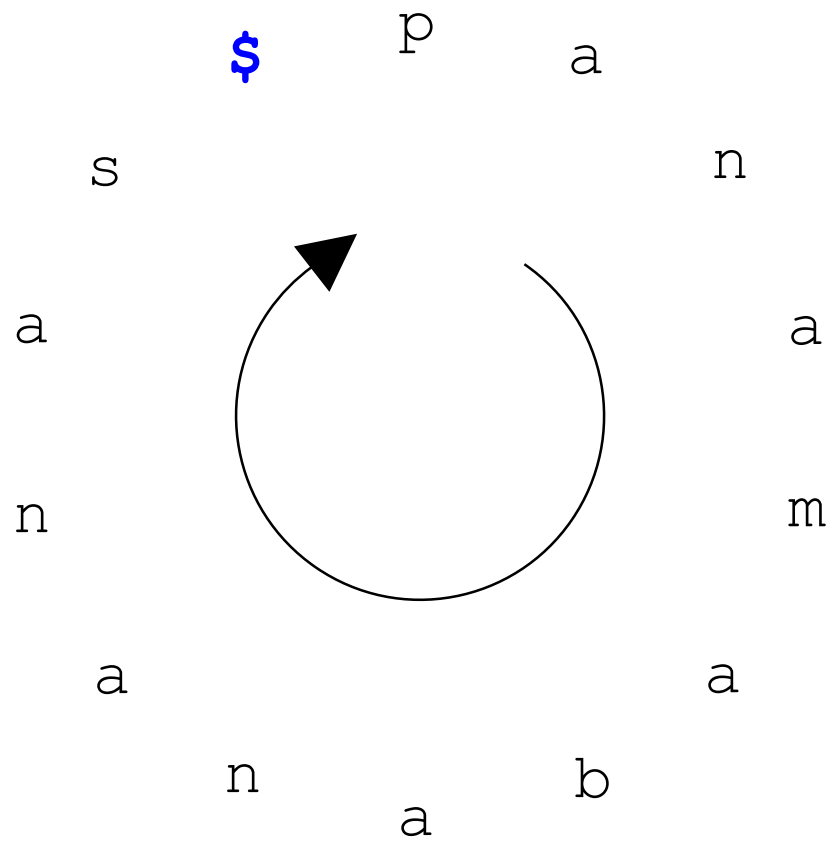
Forming All Cyclic Rotations of *Text*

panamabananas$



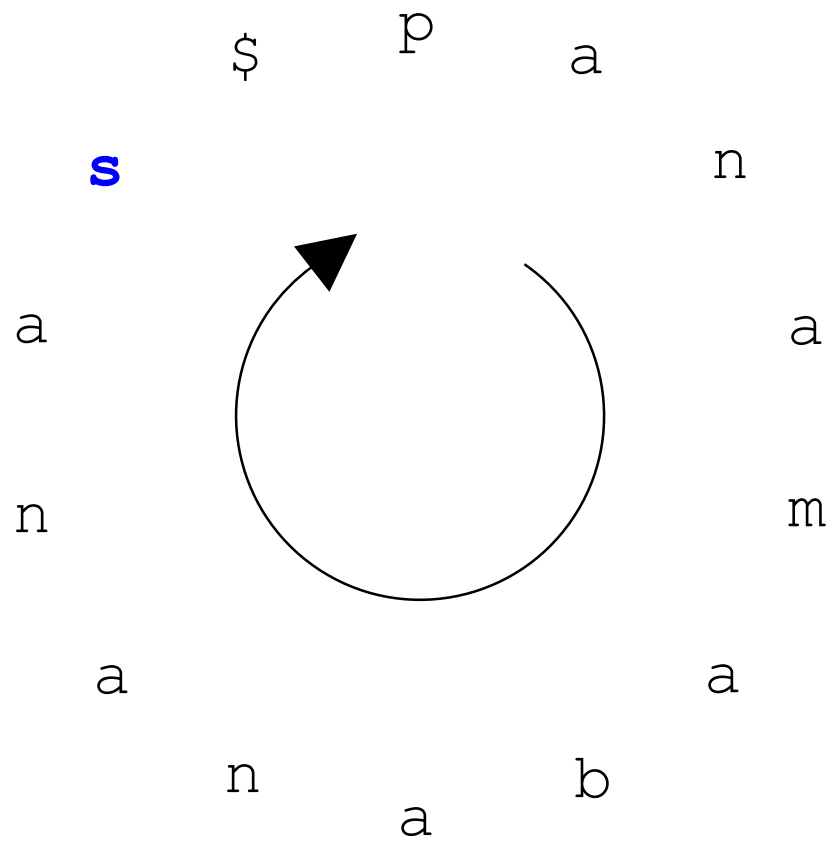
Cyclic Rotations

panamabananas$
$panamabananas



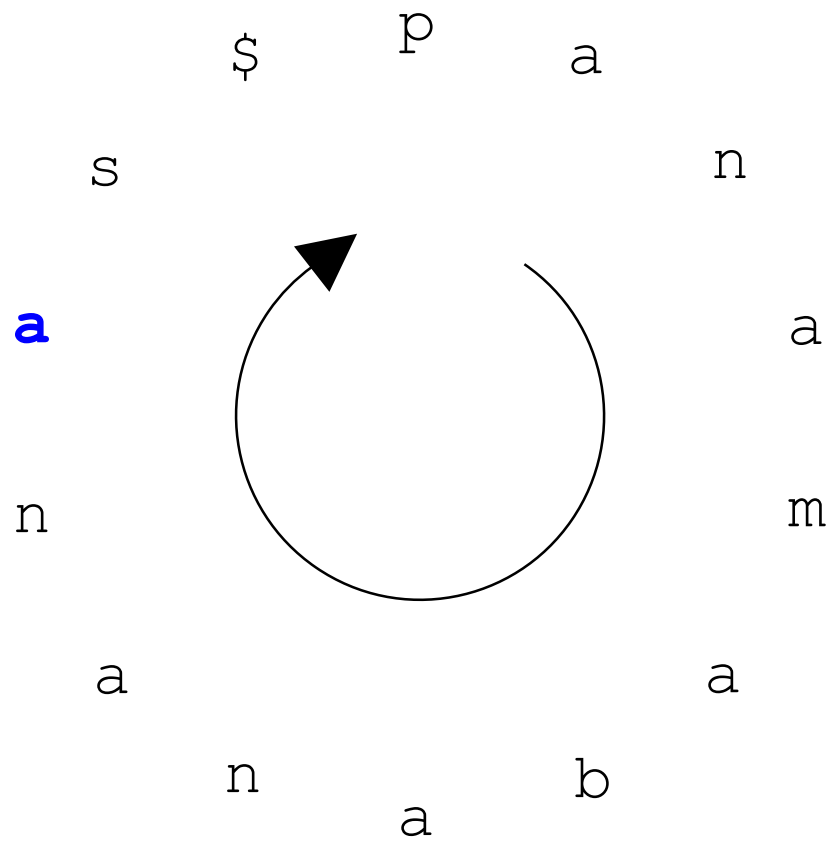
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana



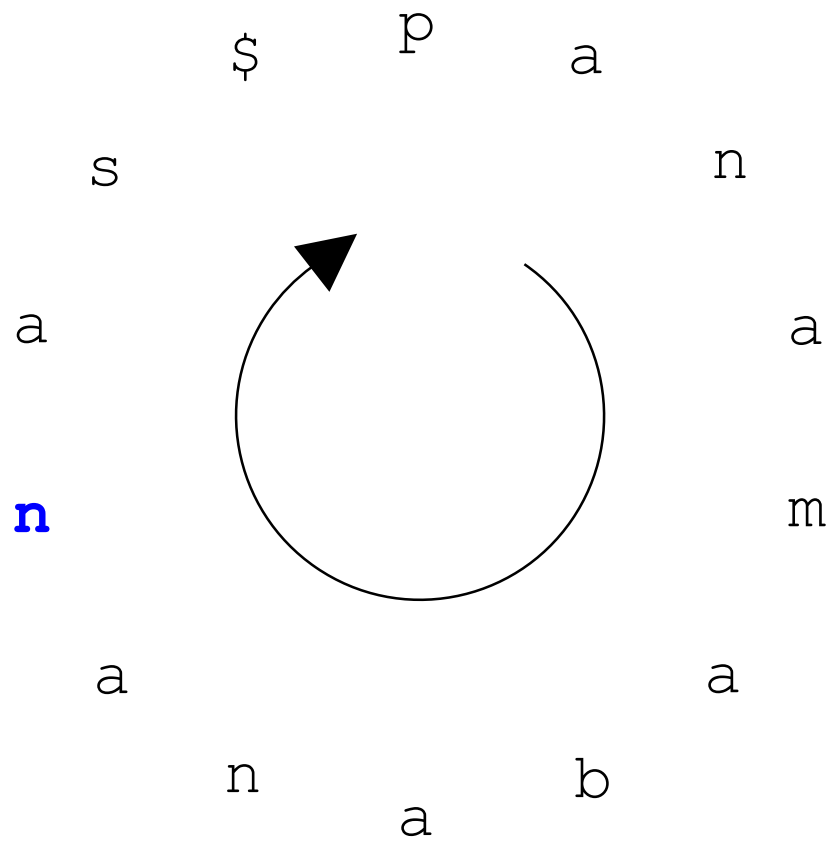
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan



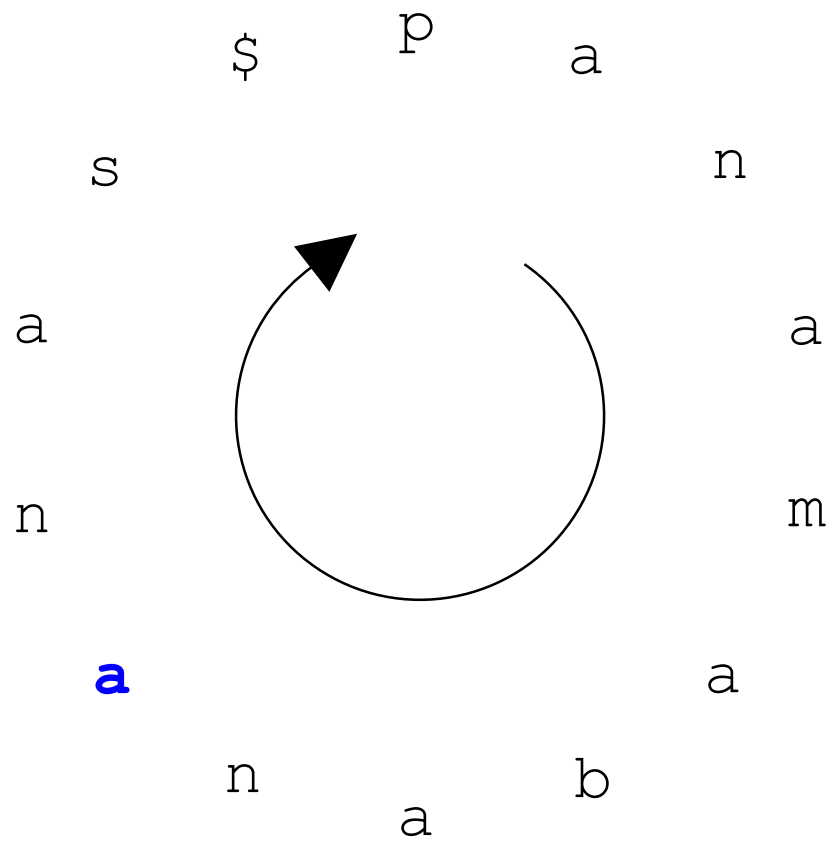
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana



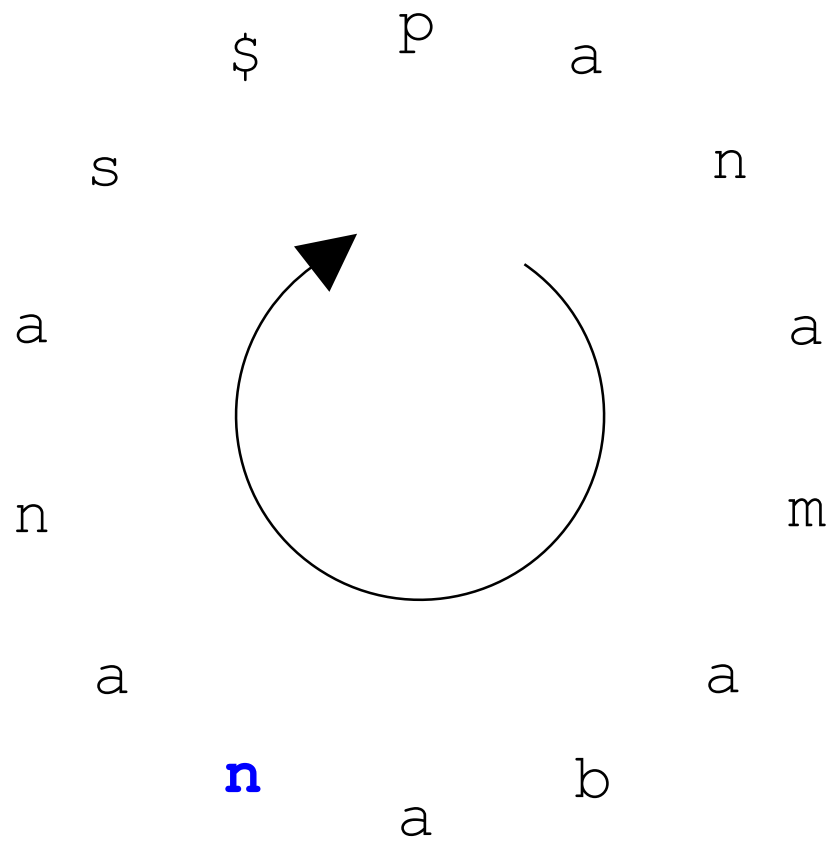
Cyclic Rotations

```
panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
```



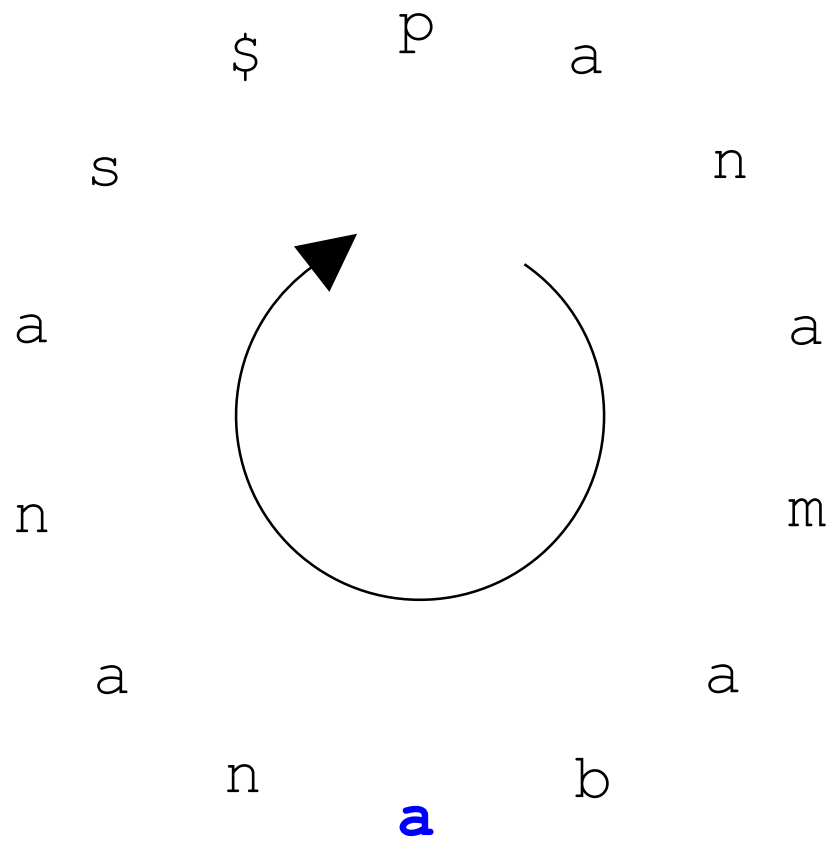
Cyclic Rotations

```
panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
```



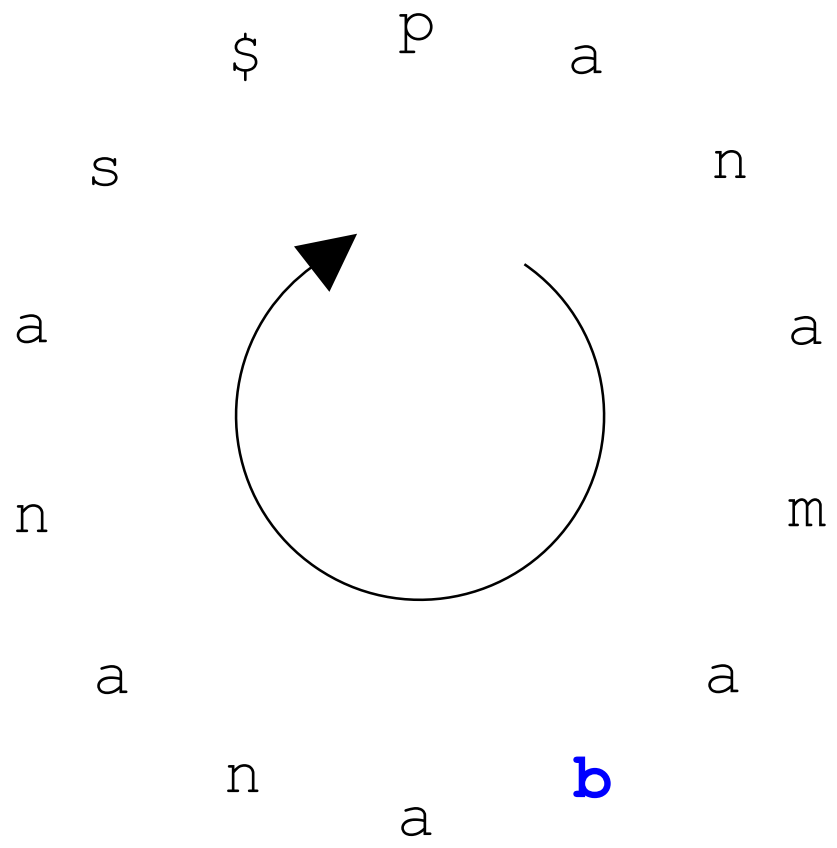
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab



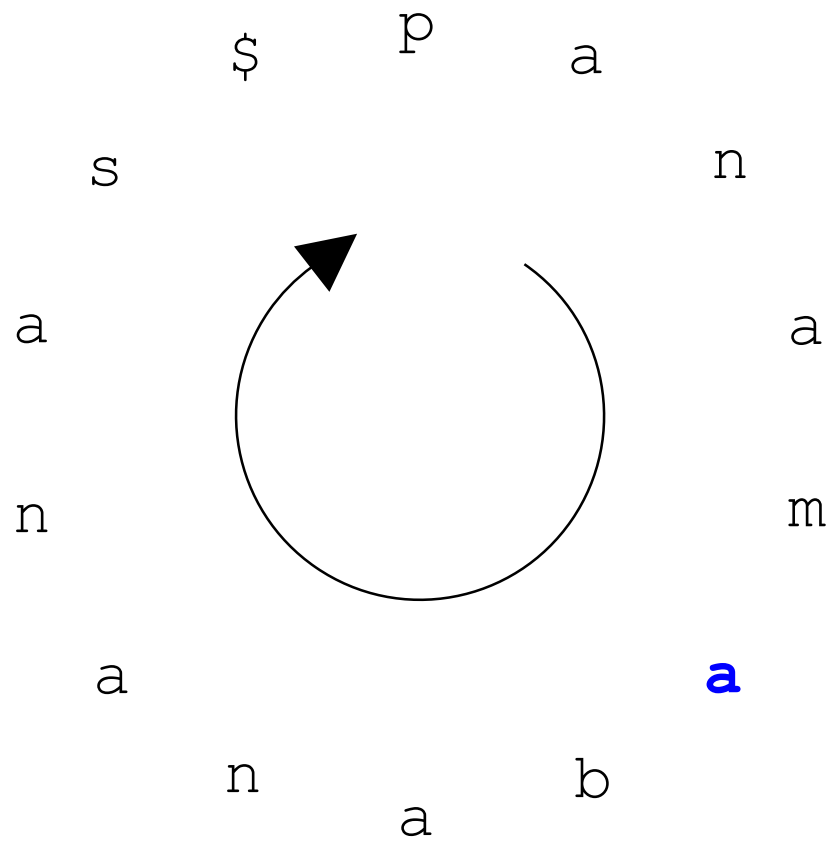
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama



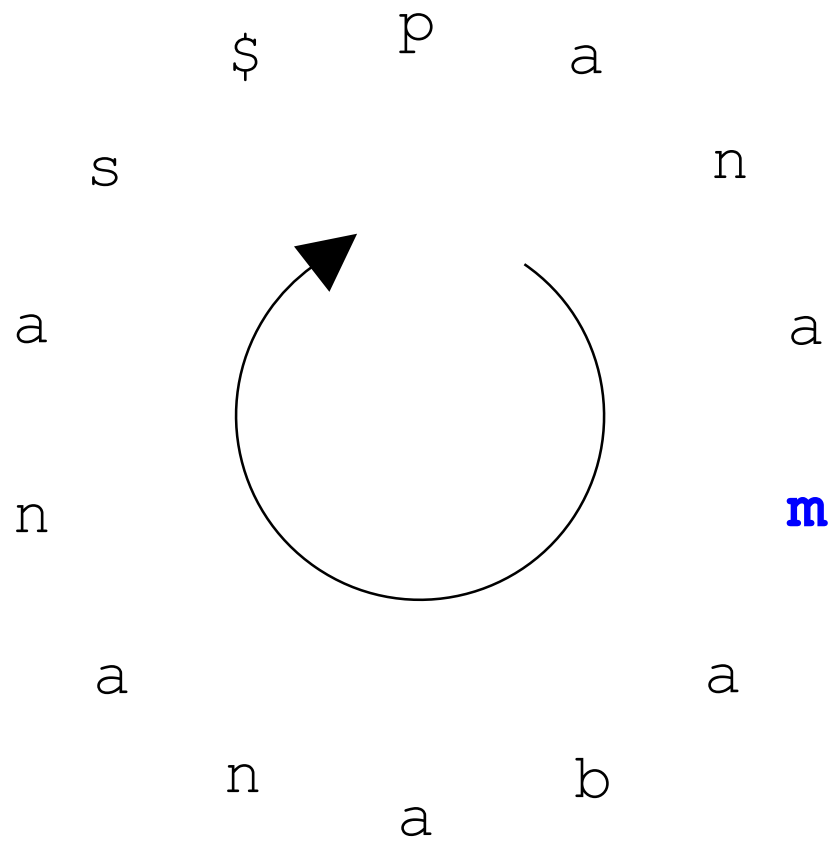
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama
abananas$panam



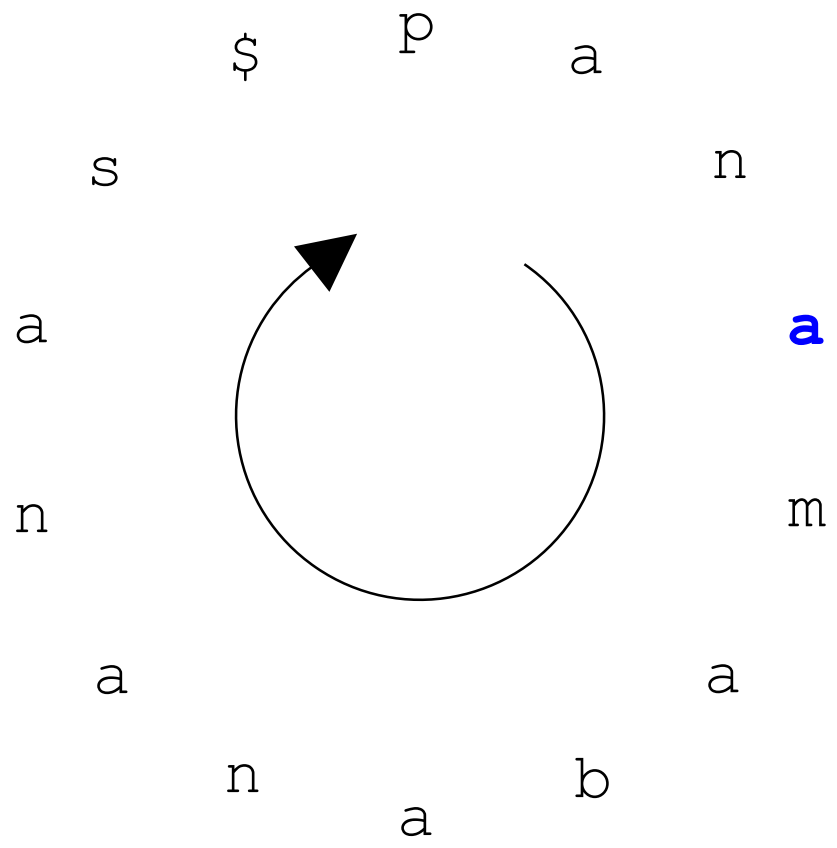
Cyclic Rotations

```
panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama
abananas$panam
mabananas$pana
```



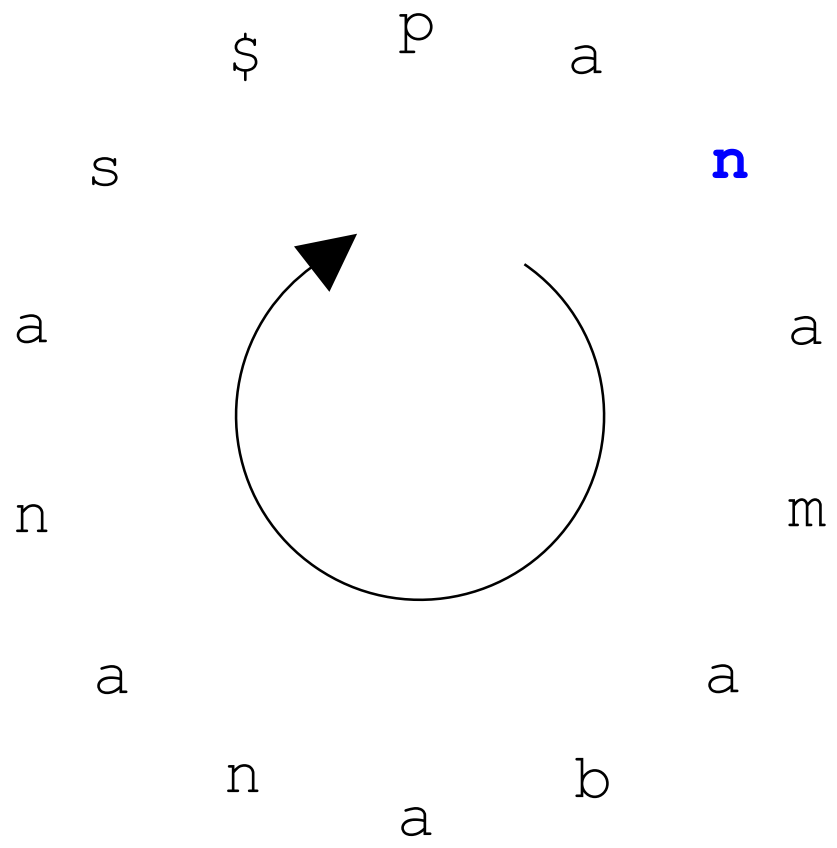
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama
abananas$panam
mabananas$pana
amabananas$pan



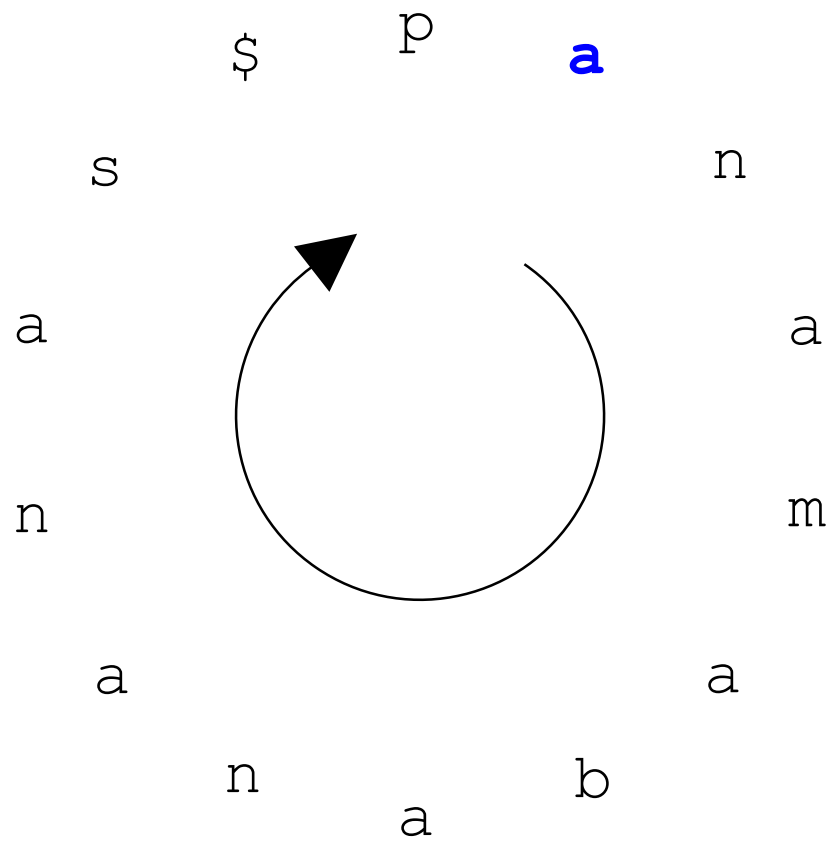
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama
abananas$panam
mabananas$pana
amabananas$pan
namabananas$pa



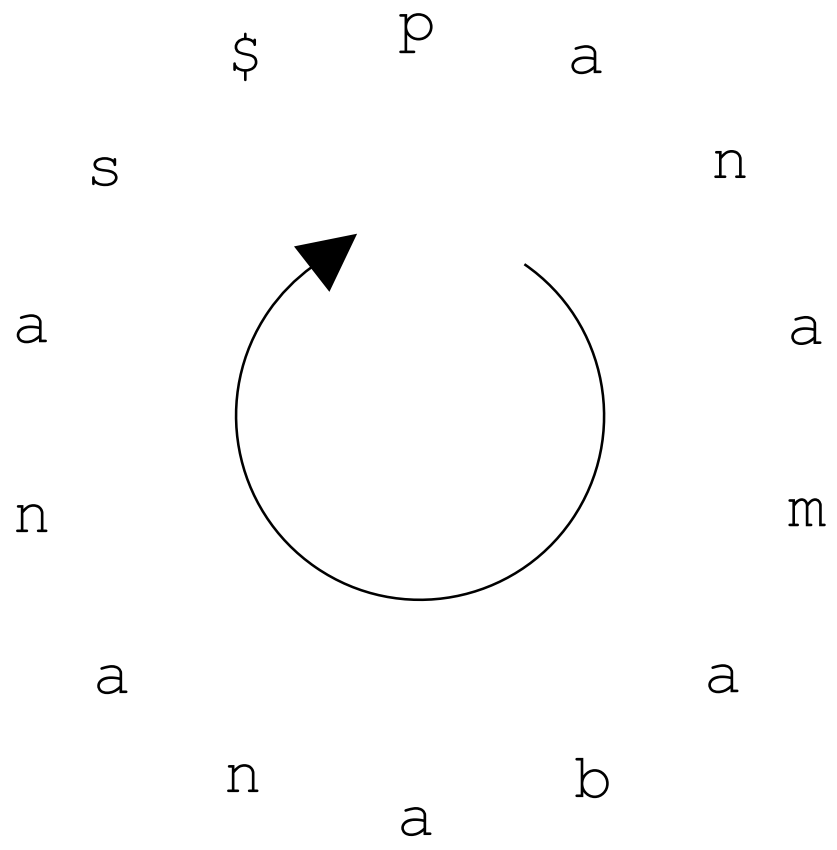
Cyclic Rotations

panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama
abananas$panam
mabananas$pana
amabananas$pan
namabananas$pa
anamabananas$p

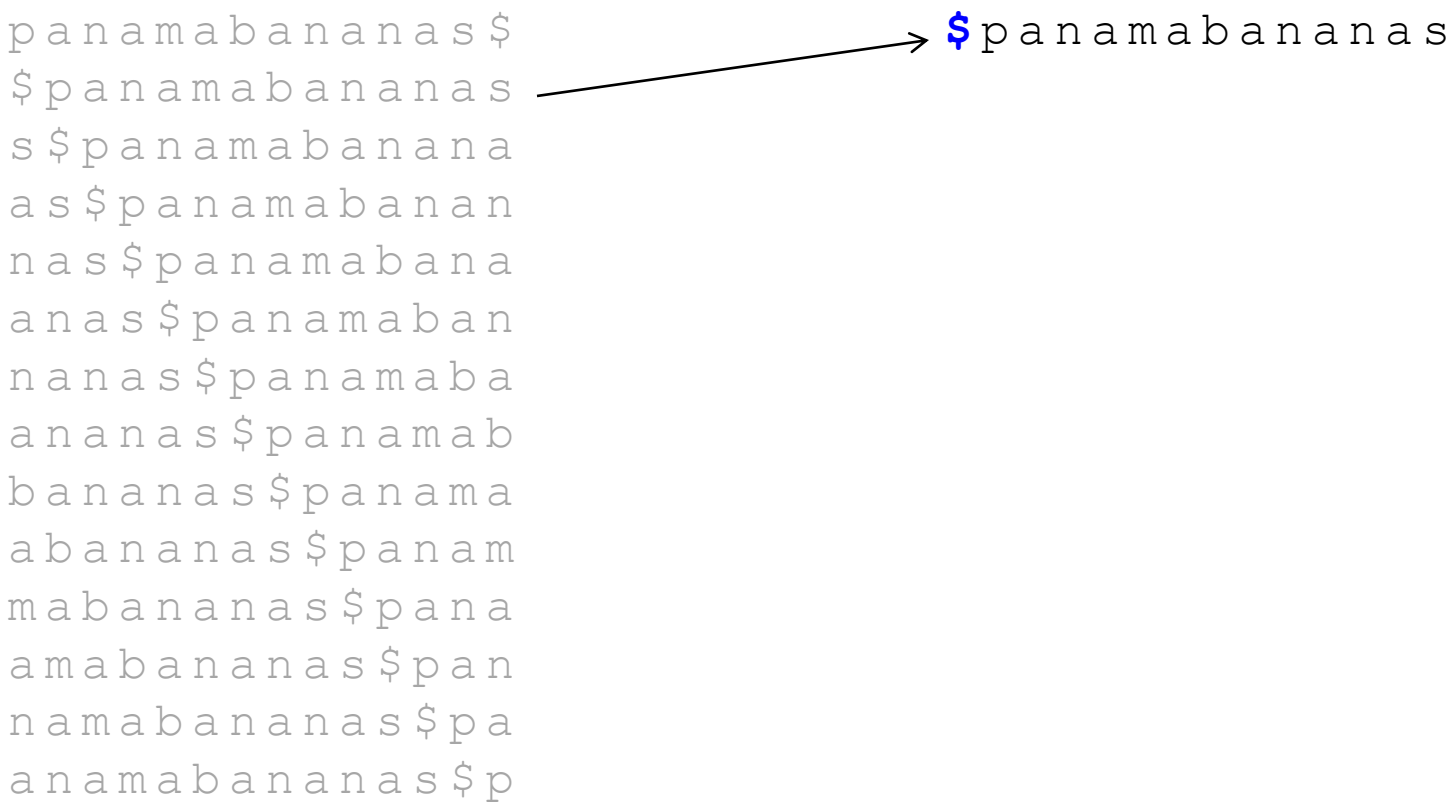


Cyclic Rotations

```
panamabananas$
$panamabananas
s$panamabanana
as$panamabanan
nas$panamabana
anas$panamaban
nanas$panamaba
ananas$panamab
bananas$panama
abananas$panam
mabananas$pana
amabananas$pan
namabananas$pa
anamabananas$p
```

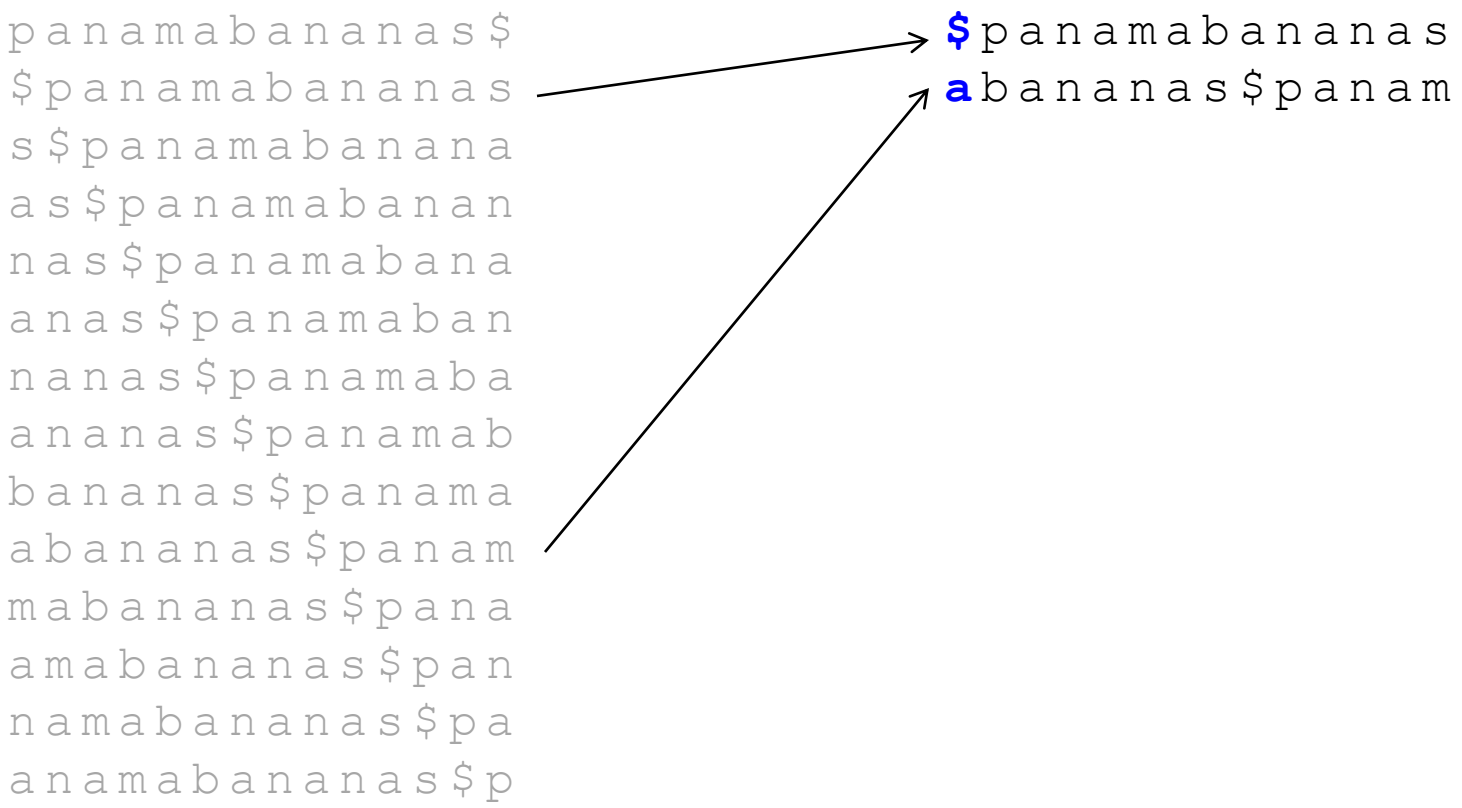


Sorting Cyclic Rotations



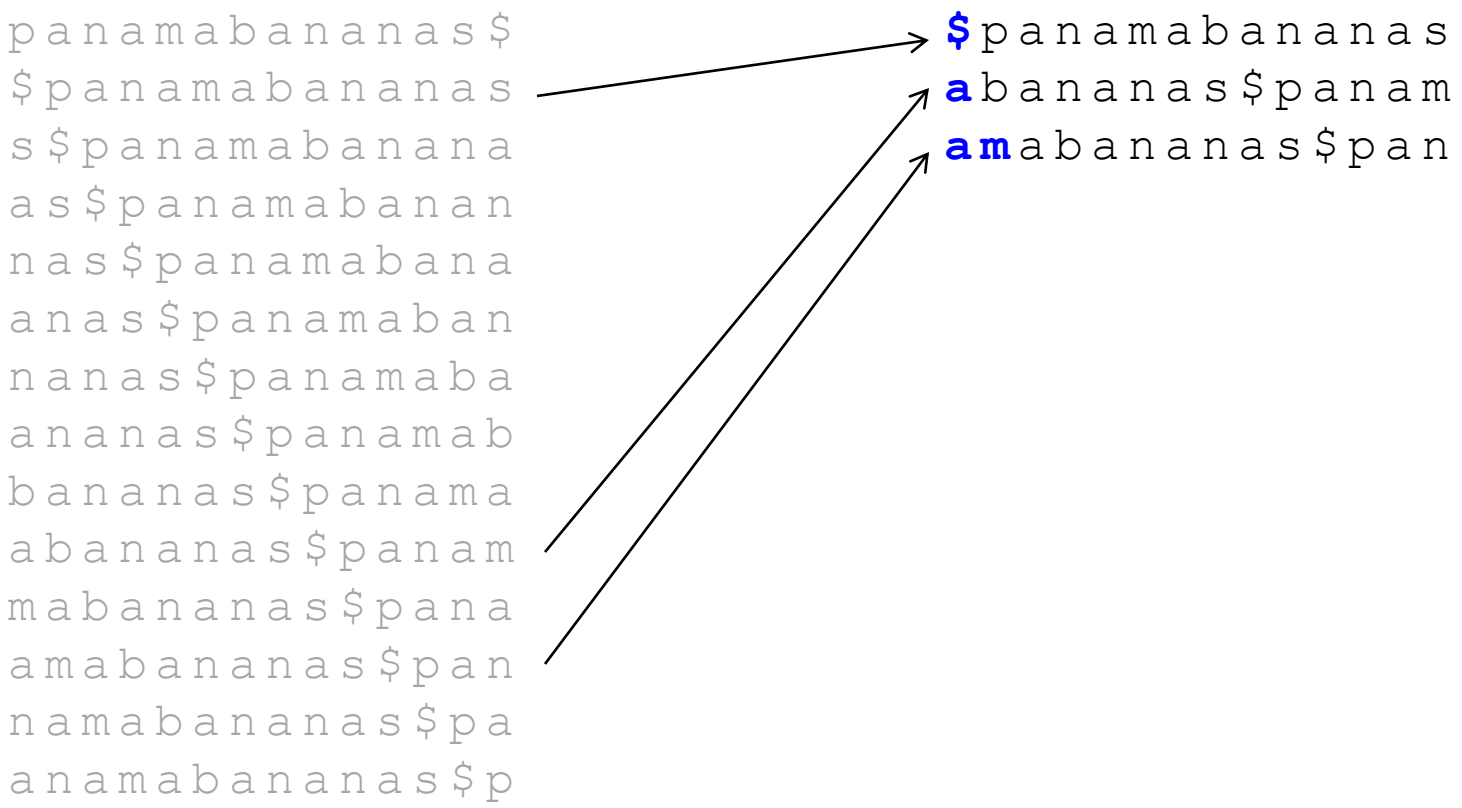
Sort the strings lexicographically ($ comes first)

Sorting Cyclic Rotations



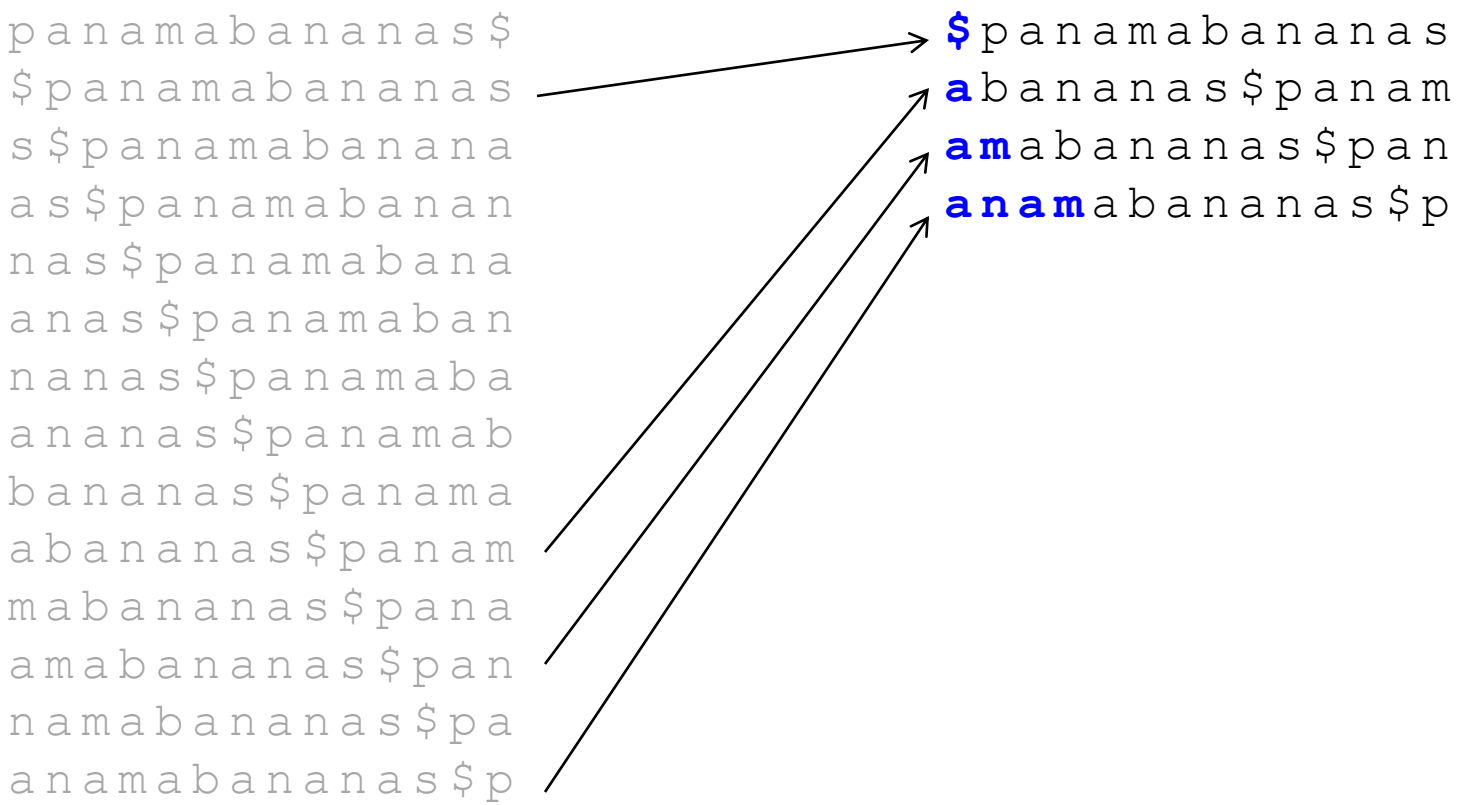
Sort the strings lexicographically ($ comes first)

Sorting Cyclic Rotations



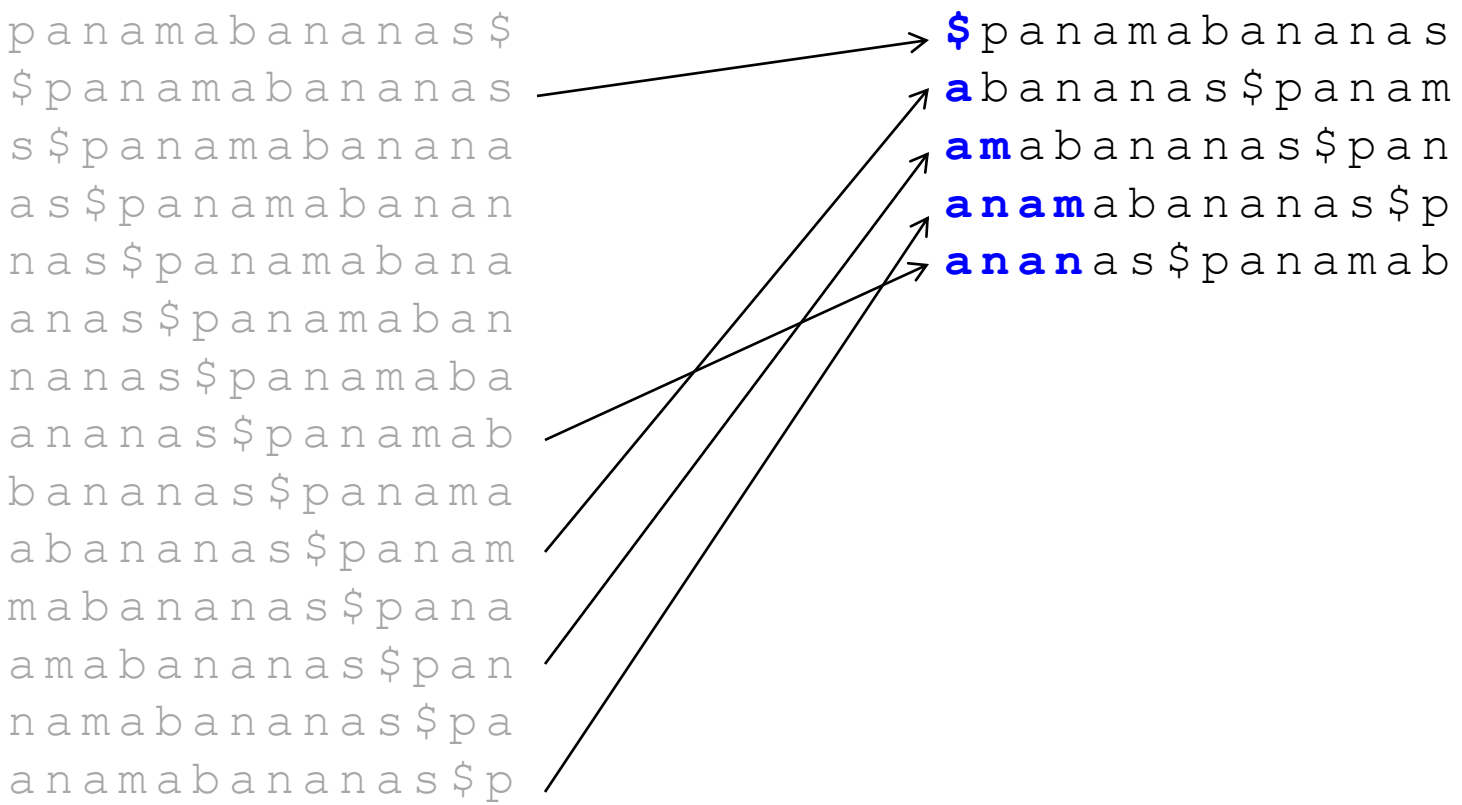
Sort the strings lexicographically ($ comes first)

Sorting Cyclic Rotations



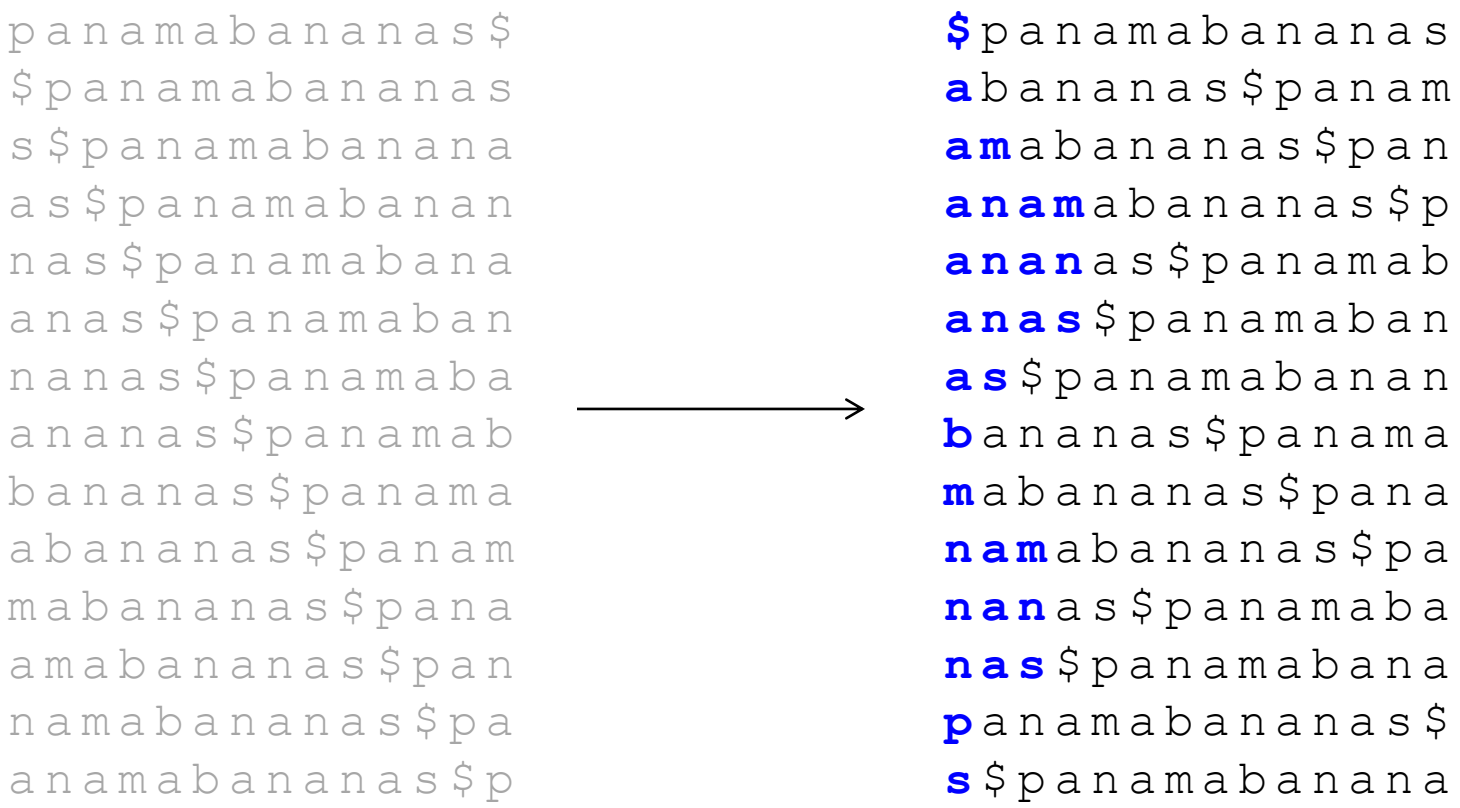
Sort the strings lexicographically ($ comes first)

Sorting Cyclic Rotations



Sort the strings lexicographically ($ comes first)

Sorting Cyclic Rotations



Sort the strings lexicographically ($ comes first)

BWT(panamabananas$)=smnpbnnaaaaa$a

panamabananas$	$panamabanana**s**
$panamabananas	abananas$pana**m**
s$panamabanana	amabananas$pa**n**
as$panamabanan	anamabananas$**p**
nas$panamabana	ananas$panama**b**
anas$panamaban	anas$panamaba**n**
nanas$panamaba	as$panamabana**n**
ananas$panamab	bananas$panam**a**
bananas$panama	mabananas$pan**a**
abananas$panam	namabananas$p**a**
mabananas$pana	nanas$panamab**a**
amabananas$pan	nas$panamaban**a**
namabananas$pa	panamabananas**$**
anamabananas$p	s$panamabanan**a**

All cyclic rotations of
“panamabananas$”

Burrows-Wheeler Transform (BWT):
Last column = **smnpbnnaaaaa$a**

BWT(panamabananas$)=smnpbnnaaaaa$a

panamabananas$	$panamabanana**s**
$panamabananas	abananas$pana**m**
s$panamabanana	amabananas$pa**n**
as$panamabanan	anamabananas$**p**
nas$panamabana	ananas$panama**b**
anas$panamaban	anas$panamaba**n**
nanas$panamaba	as$panamabana**n**
ananas$panamab	bananas$panam**a**
bananas$panama	mabananas$pan**a**
abananas$panam	namabananas$p**a**
mabananas$pana	nanas$panamab**a**
amabananas$pan	nas$panamaban**a**
namabananas$pa	panamabananas**$**
anamabananas$p	s$panamabanan**a**

All cyclic rotations of “panamabananas$”

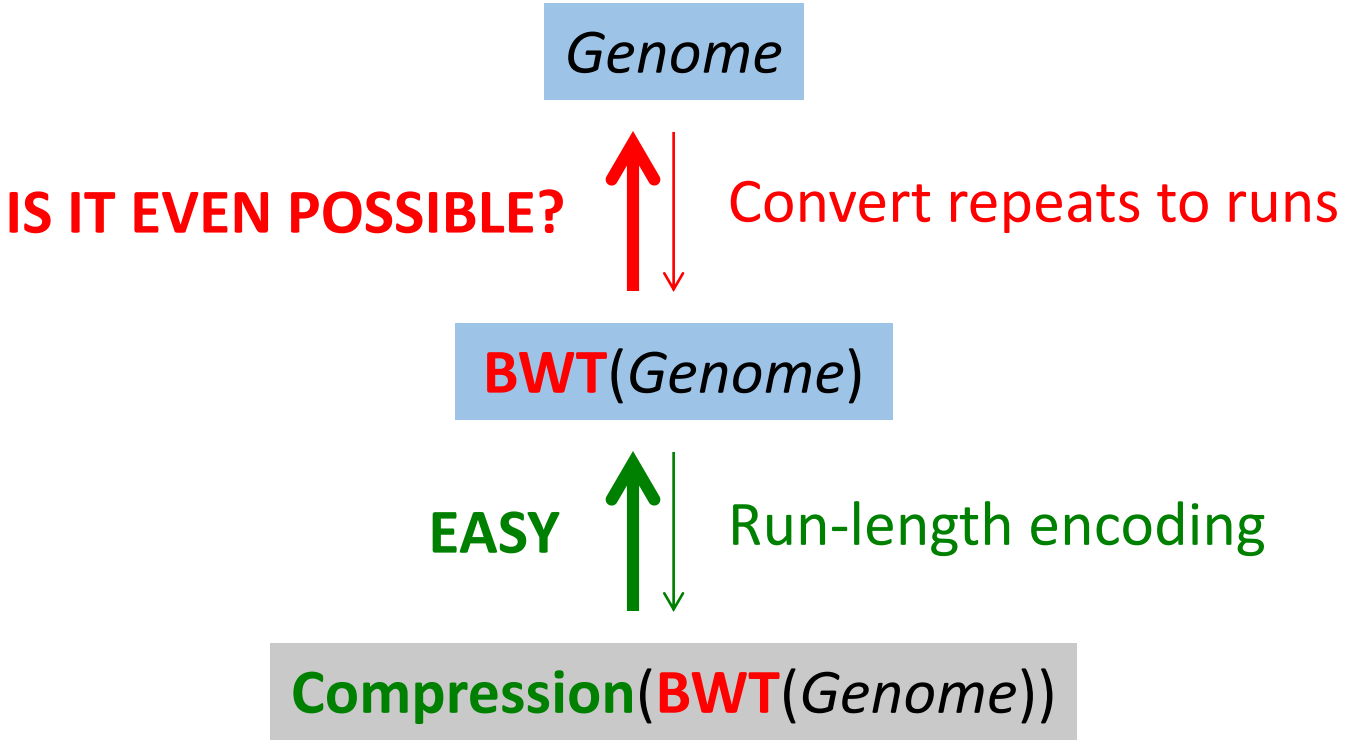
Burrows-Wheeler Transform (BWT):
Last column = **smnpbnnaaaaa$a**

Applying BWT to the Double Helix Paper by Watson&Crick

```
nd Corey (1).  They kindly made their manuscript availa ..... a
nd criticism, especially on interatomic distances.  We  ..... a
nd cytosine.  The sequence of bases on a single chain d ..... a
nd experimentally (3,4) that the ratio of the amounts o ..... u
nd for this reason we shall not comment on it.  We wish ..... a
nd guanine (purine) with cytosine (pyrimidine).  In oth ..... a
nd ideas of Dr.  M. H. F. Wilkins, Dr.  R. E. Franklin  ..... a
nd its water content is rather high.  At lower water co ..... a
nd pyrimidine bases.  The planes of the bases are perpe ..... a
nd stereochemical arguments.  It has not escaped our no ..... a
nd that only specific pairs of bases can bond together  ..... u
nd the atoms near it is close to Furberg's 'standard co ..... a
nd the bases on the inside, linked together by hydrogen ..... a
nd the bases on the outside.  In our opinion, this stru ..... a
nd the other a pyrimidine for bonding to occur.  The hy ..... a
nd the phosphates on the outside.  The configuration of ..... a
nd the ration of guanine to cytosine, are always very c ..... a
nd the same axis (see diagram).  We have made the usual ..... u
nd their co-workers at King's College, London.  One of  ..... a
```

"and" is a frequent repeat in English texts

Going Back From BWT(*Genome*) to *Genome*



Outline

- Burrows-Wheeler Transform
- **Inverting Burrows-Wheeler Transform**
- Using BWT for Pattern Matching
- Suffix Arrays
- Approximate Pattern Matching

Reconstructing banana from annb$aa

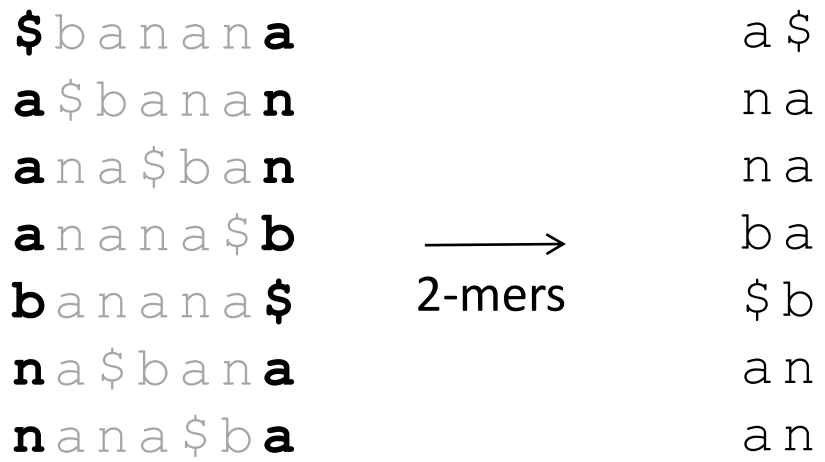
$banan**a**
a$bana**n**
ana$ba**n**
anana$**b**
banana**$**
na$ban**a**
nana$b**a**

Reconstructing banana

$banan**a**
a$banan**n**
ana$ba**n**
anana$**b**
banana**$**
na$ban**a**
nana$b**a**

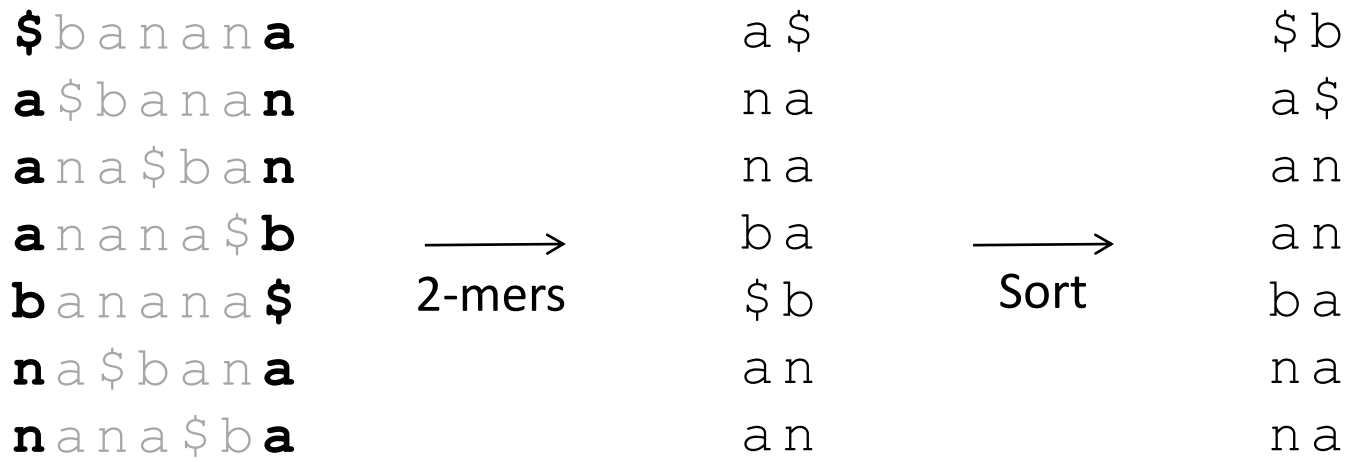
- Sorting all elements of “annb$aa” gives first column of BWT matrix.

Reconstructing banana



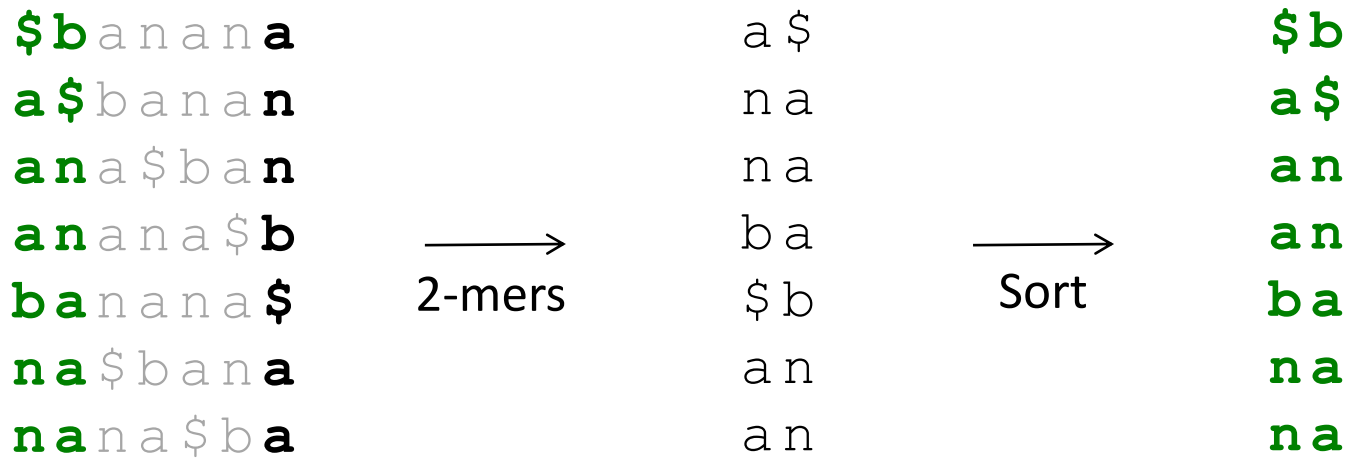
- We now know 2-mer composition of the circular string banana$

Reconstructing banana



- We now know 2-mer composition of the circular string banana$
- Sorting gives us the first 2 columns of the matrix.

Reconstructing banana

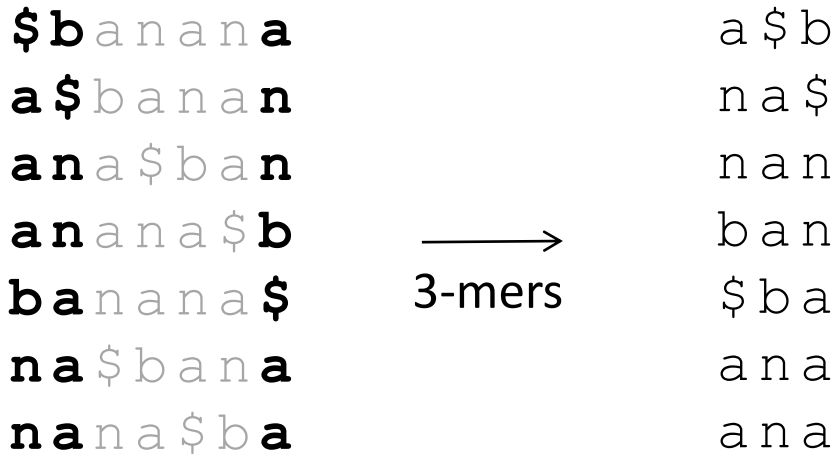


- We now know 2-mer composition of the circular string banana$
- Sorting gives us the first 2 columns of the matrix.

Reconstructing banana

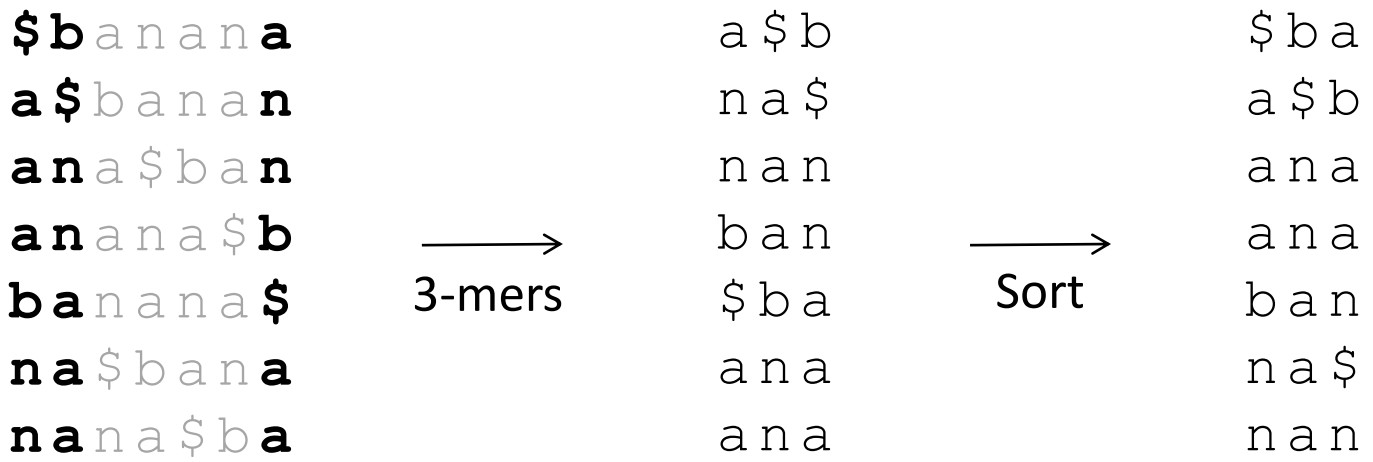
$banan**a**
a$banan**n**
ana$ba**n**
anana$**b**
banana**$**
na$ban**a**
nana$b**a**

Reconstructing banana



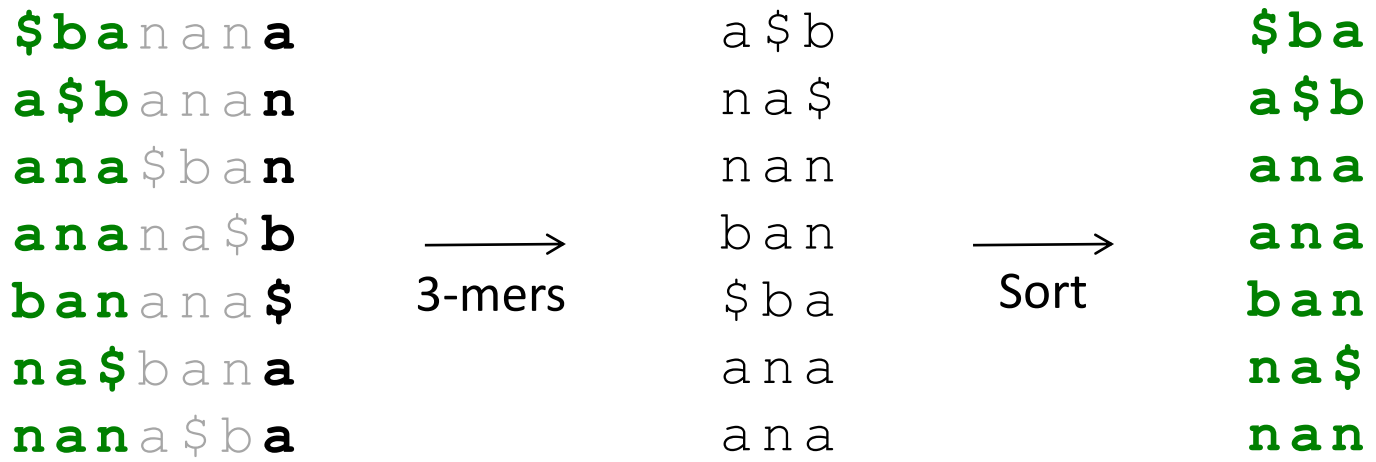
- We now know 3-mer composition of the circular string banana$

Reconstructing banana



- We now know 3-mer composition of the circular string banana$
- Sorting gives us the first 3 columns of the matrix.

Reconstructing banana

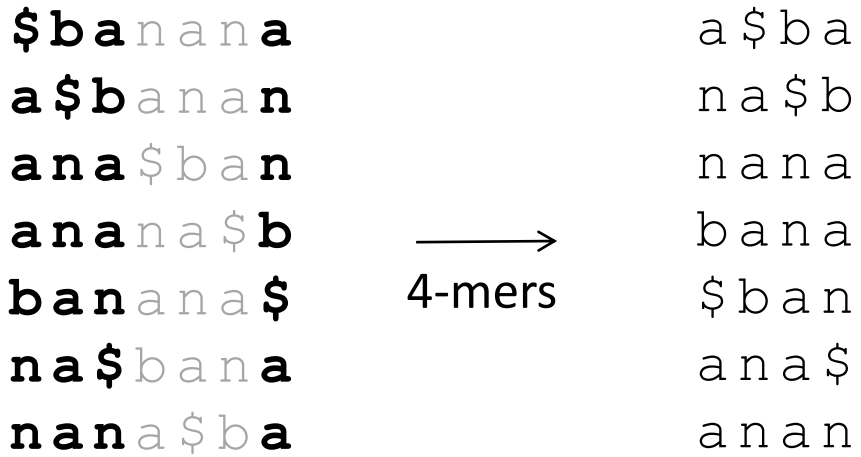


- We now know 3-mer composition of the circular string banana$
- Sorting gives us the first 3 columns of the matrix.

Reconstructing banana

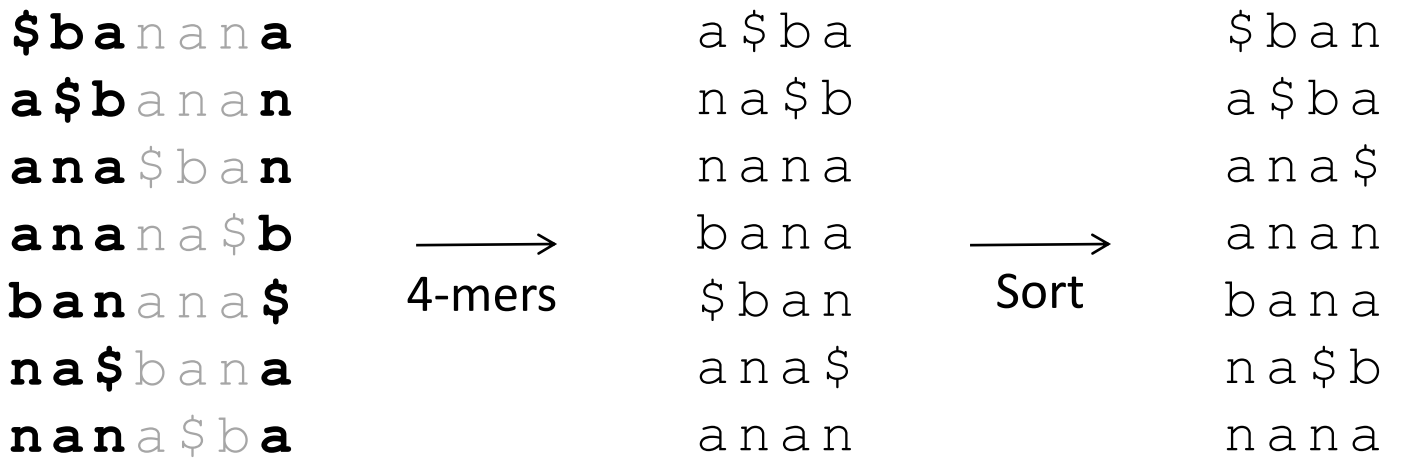
$banana
a$banan**a**
ana$ba**n**
anana$**b**
banana**$**
na$ban**a**
nana$b**a**

Reconstructing banana



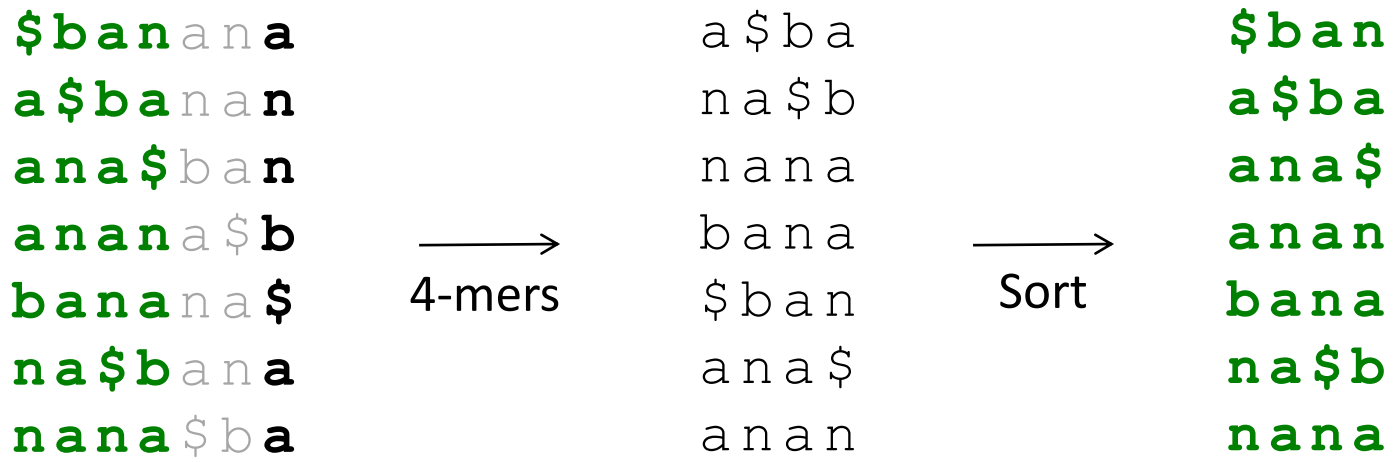
- We now know 4-mer composition of the circular string banana$

Reconstructing banana



- We now know 4-mer composition of the circular string banana$
- Sorting gives us the first 4 columns of the matrix.

Reconstructing banana

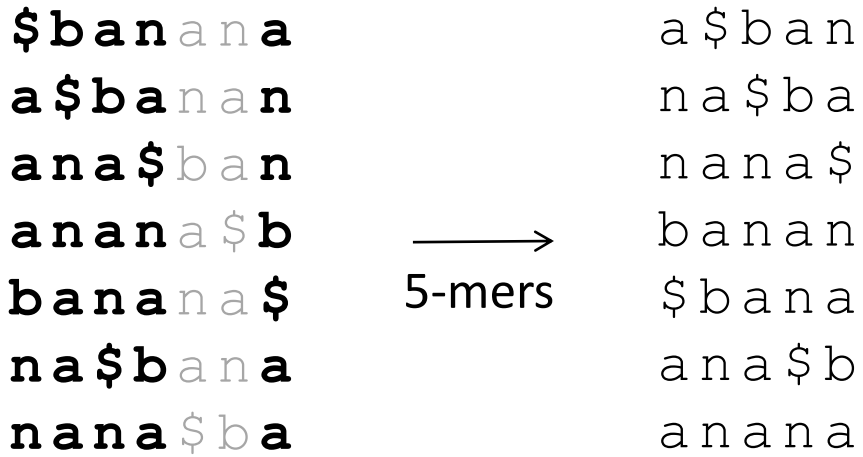


- We now know 4-mer composition of the circular string banana$
- Sorting gives us the first 4 columns of the matrix.

Reconstructing banana

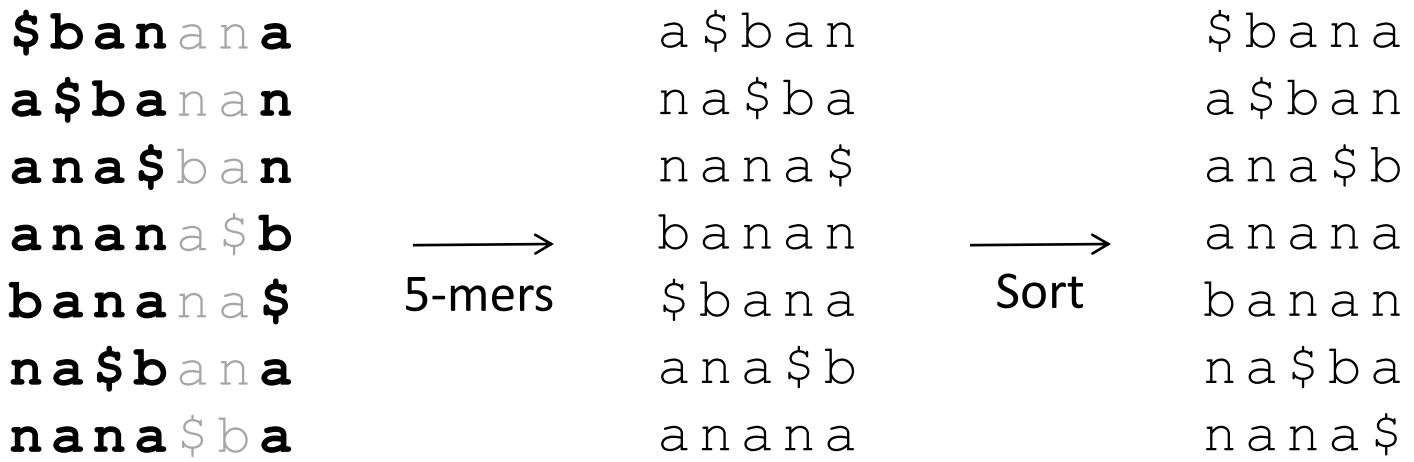
$banana
a$banan
ana$ban
anana$**b**
banana**$**
na$bana
nana$b**a**

Reconstructing banana



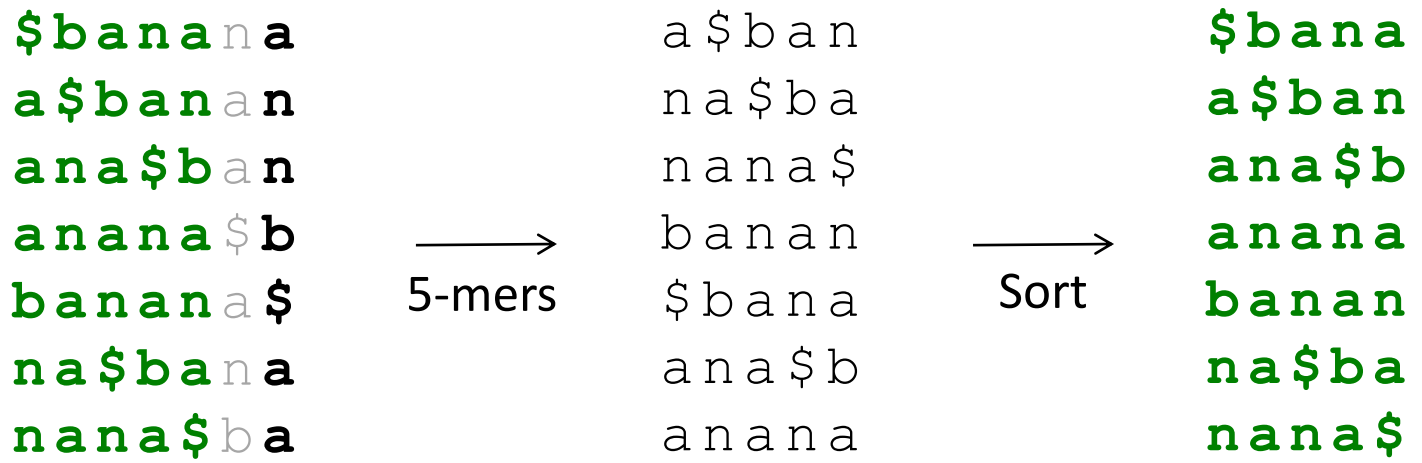
- We now know 5-mer composition of the circular string banana$

Reconstructing banana



- We now know 5-mer composition of the circular string banana$
- Sorting gives us the first 5 columns of the matrix.

Reconstructing banana

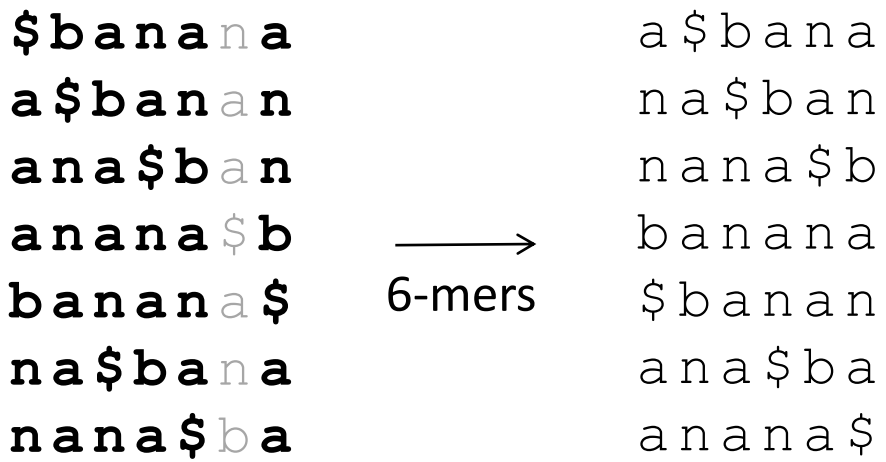


- We now know 5-mer composition of the circular string banana$
- Sorting gives us the first 5 columns of the matrix.

Reconstructing banana

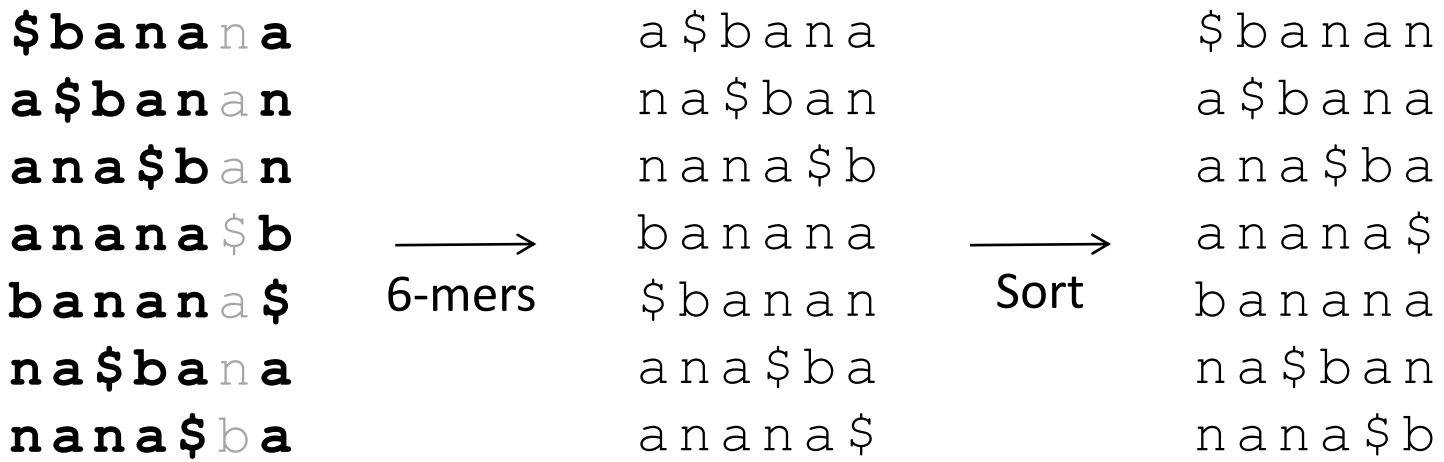
$banana
a$banan
ana$ban
anana$b
banana$
na$bana
nana$ba

Reconstructing banana



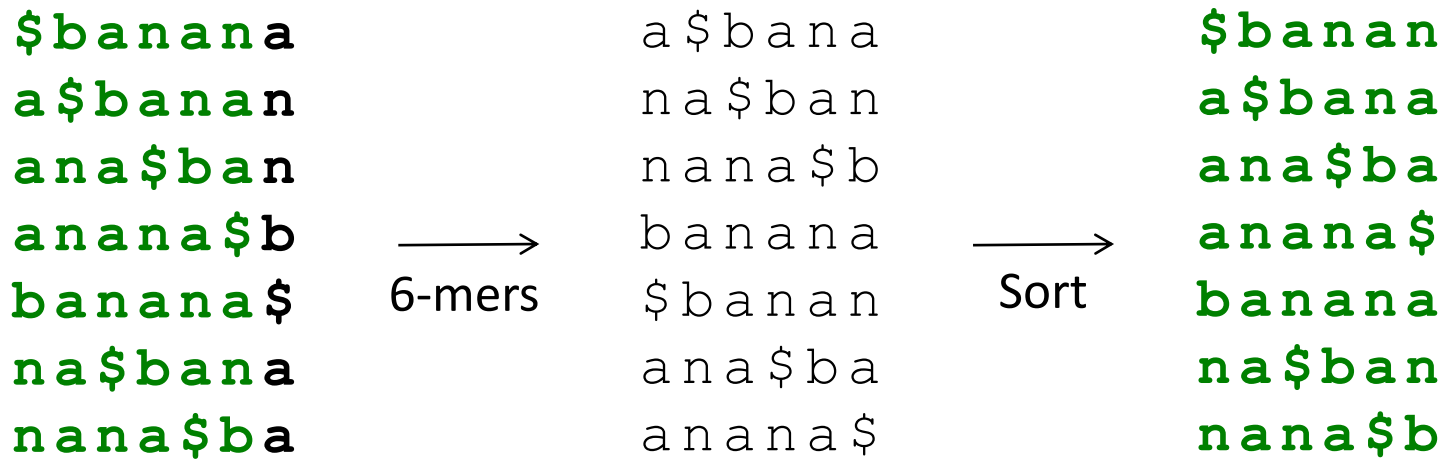
- We now know 6-mer composition of the circular string banana$

Reconstructing banana



- We now know 6-mer composition of the circular string banana$
- Sorting gives us the first 6 columns of the matrix.

Reconstructing banana



- We now know 6-mer composition of the circular string banana$
- Sorting gives us the first 6 columns of the matrix.

Reconstructing banana

```
$banana
a$banan
ana$ban
anana$b
banana$
na$bana
nana$ba
```

- We now know the entire matrix!

Reconstructing banana

$banana
a$banan
ana$ban
anana$b
banana$
na$bana
nana$ba

- We now know the entire matrix!

- Symbols in the first row (after $) spell **banana**.

More Memory Issues

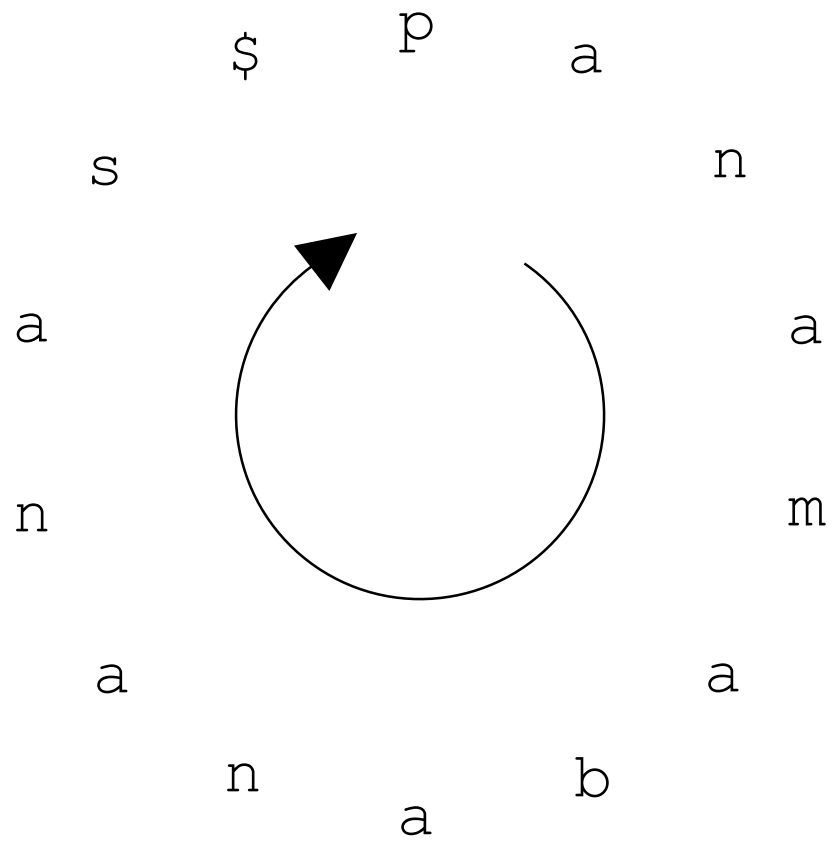
- Reconstructing *Text* from *BWT*(*Text*) required us to store |*Text*| cyclic rotations of |*Text*|.

```
$banana
a$banan
ana$ban
anana$b
banana$
na$bana
nana$ba
```

- Can we invert BWT(*Text*) with less space and without |*Text*| rounds of sorting?

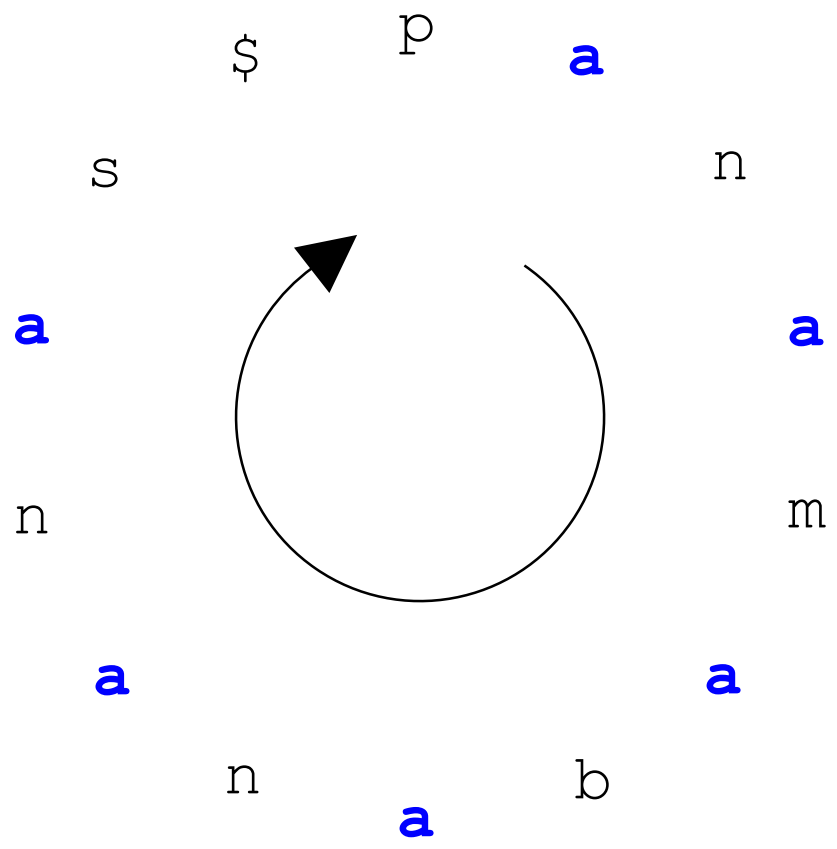
A Strange Observation

```
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana
```



A Strange Observation

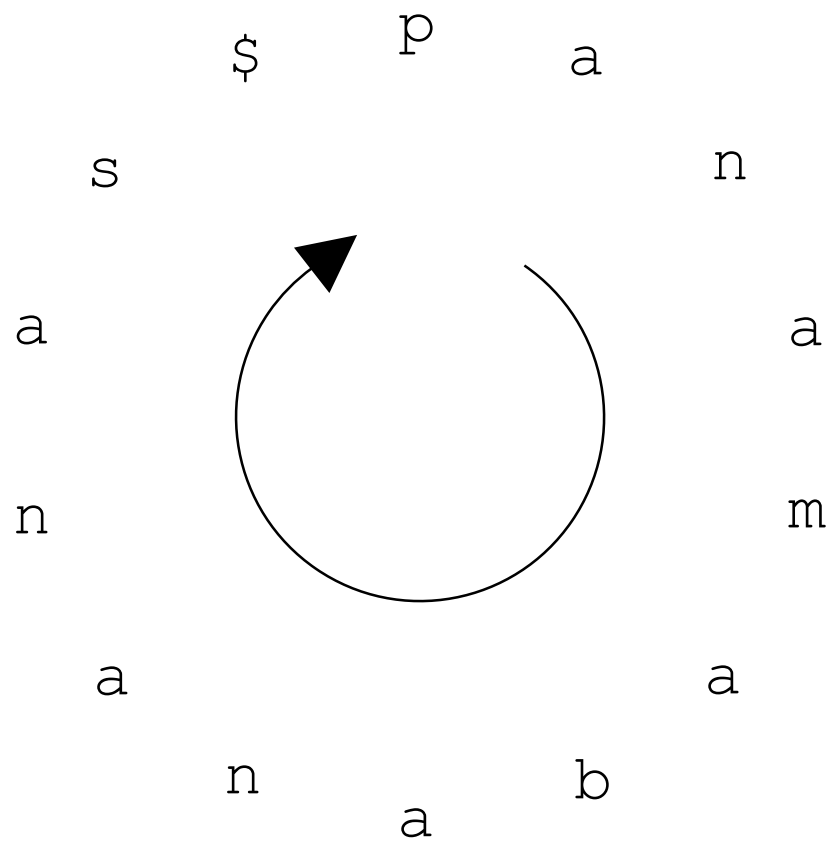
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamab
nas$panamaban
panamabananas$
s$panamabanana



A Strange Observation

Where is first "a" hiding inside the circle?

$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana

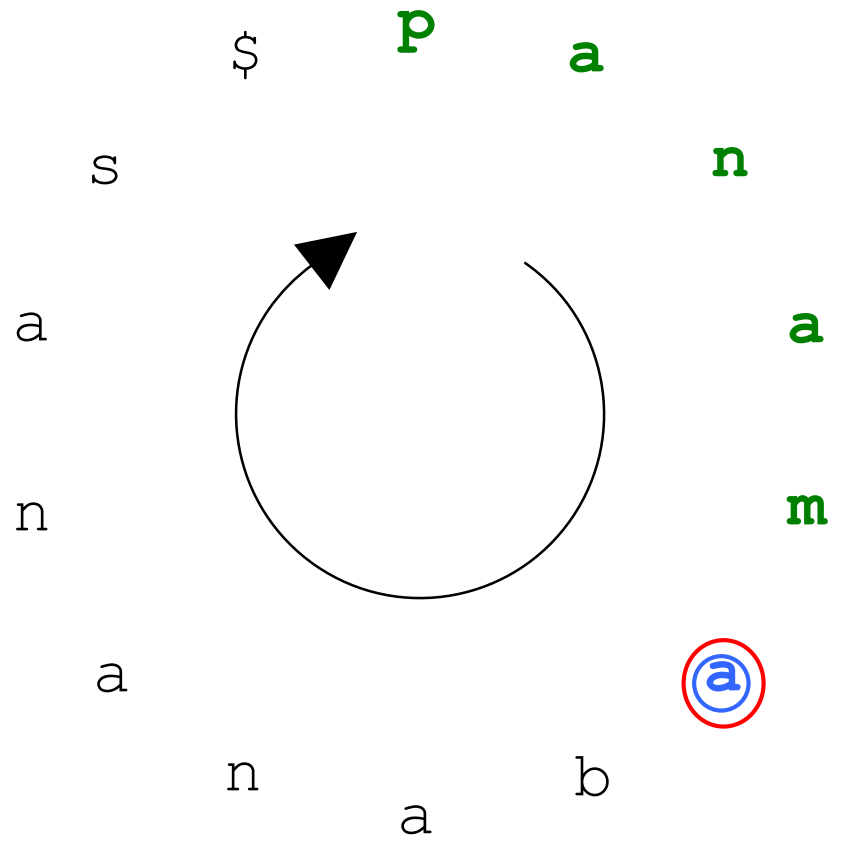


A Strange Observation

Where is first "a" hiding inside the circle?

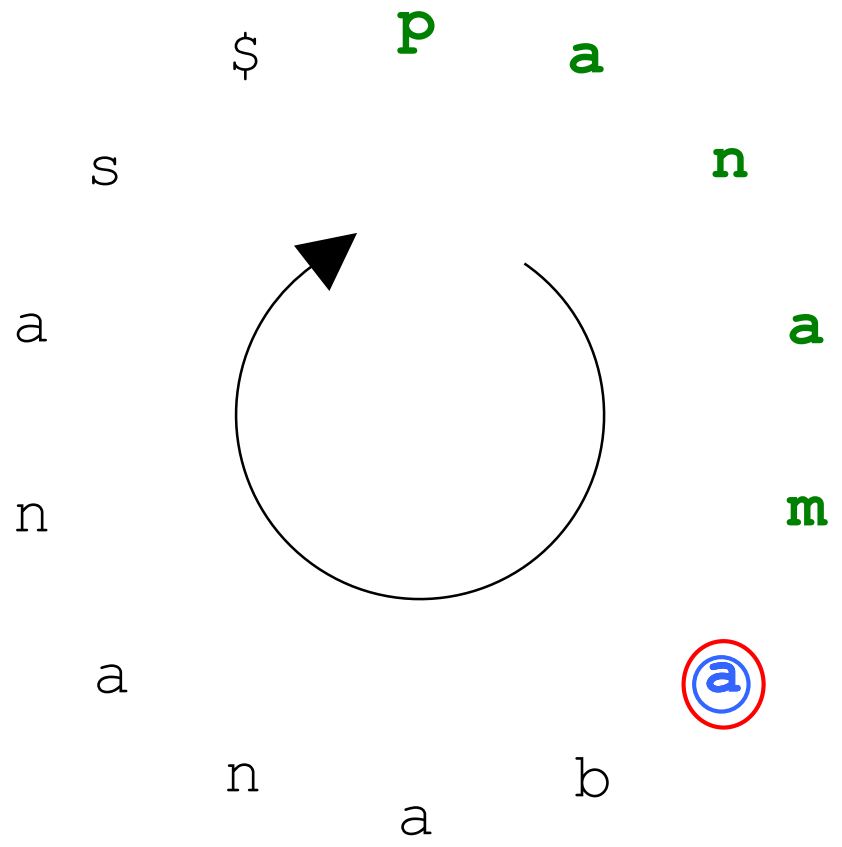
```
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana
```

Where is first "a" hiding inside the circle?



They Are Hiding at the Same Position!

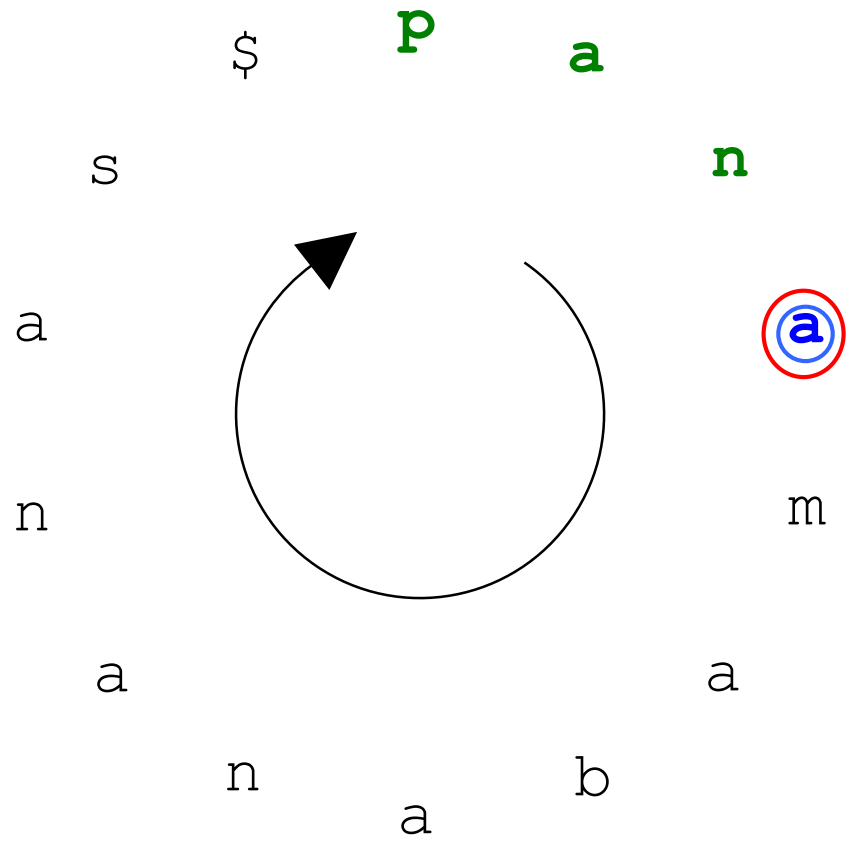
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana



1st **a** in *FirstColumn* and 1st **a** in *LastColumn*
are hiding at the same position along the cycle!

Another Strange Observation

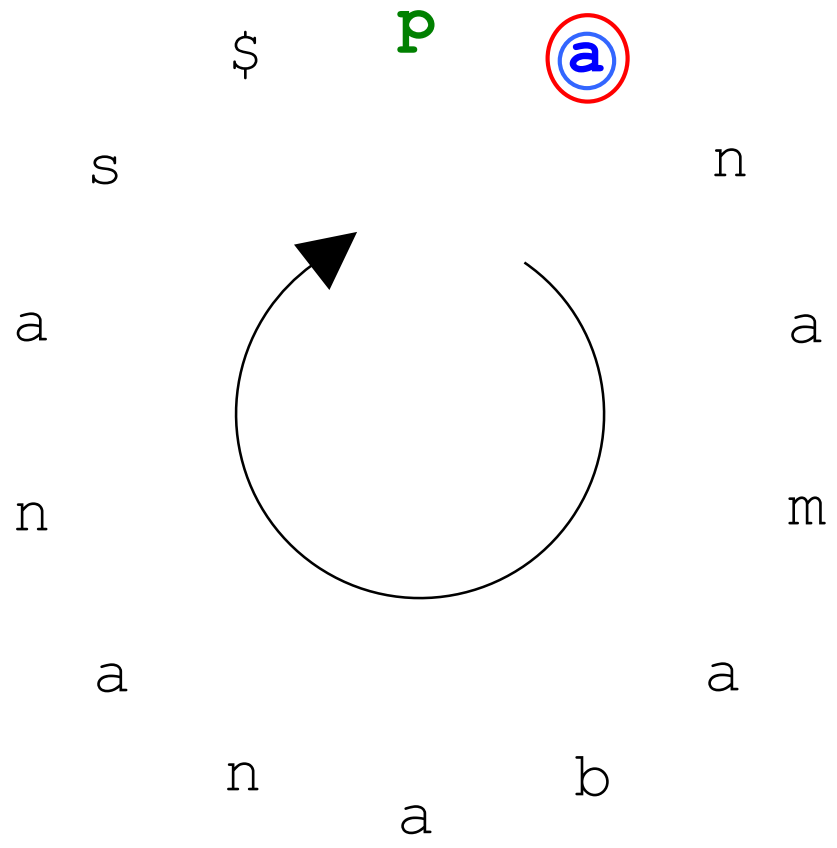
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana



2nd **a** in *FirstColumn* and 2nd **a** in *LastColumn*
are hiding at the same position along the cycle!

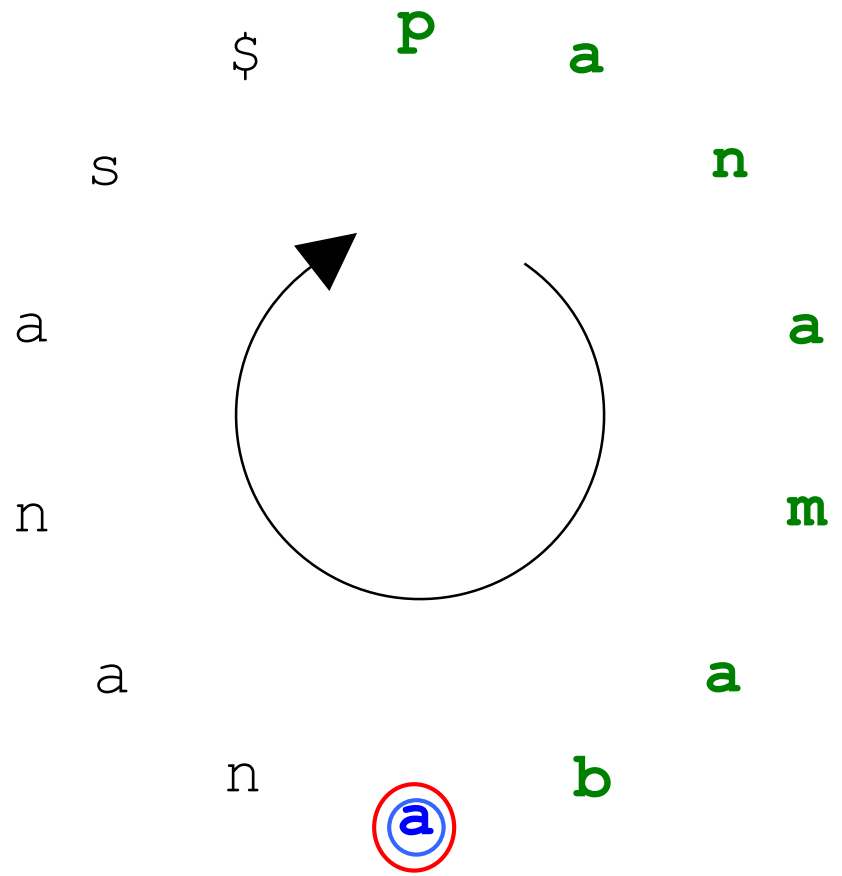
Another Strange Observation

```
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana
```



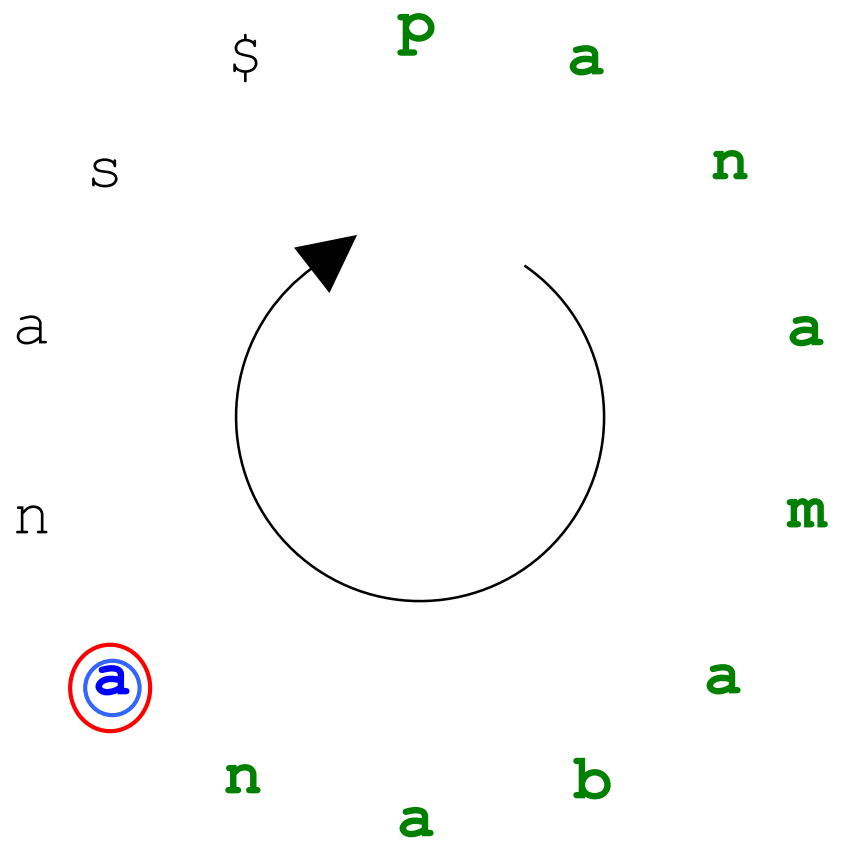
Another Strange Observation

```
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana
```



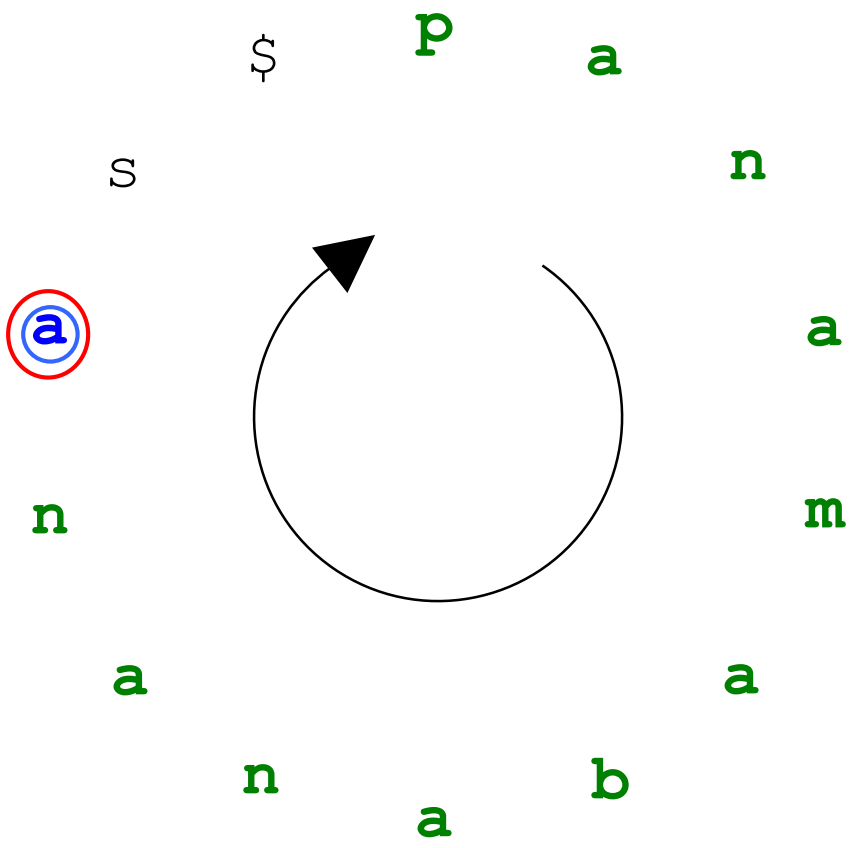
Another Strange Observation

```
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana
```



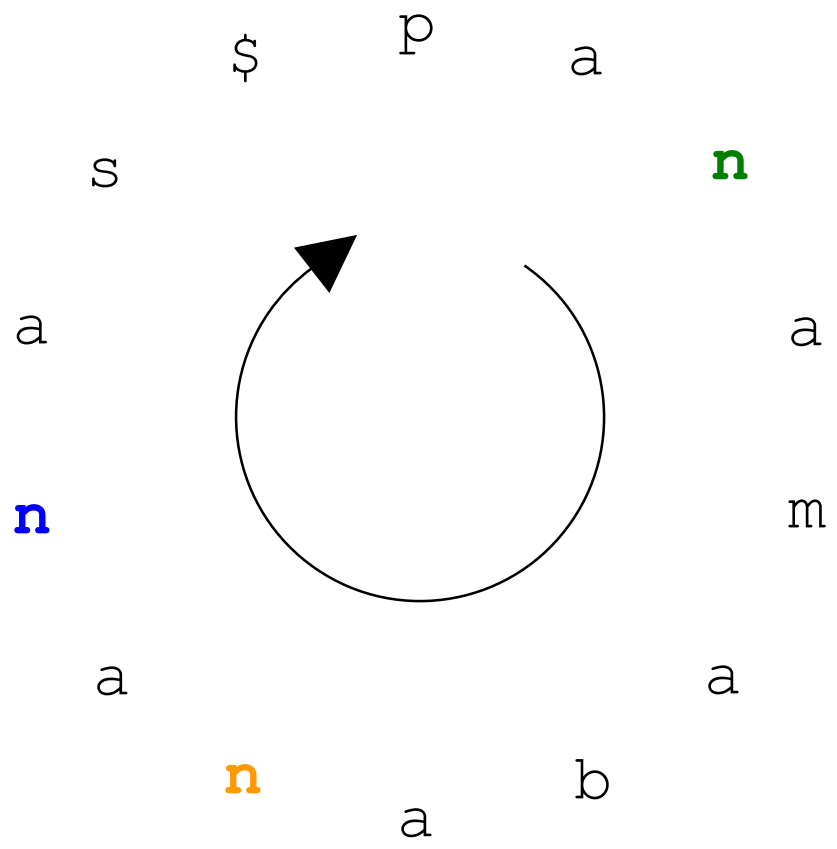
Another Strange Observation

```
$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana
```



Another Strange Observation

$panamabananas
abananas$panam
amabananas$pan
anamabananas$p
ananas$panamab
anas$panamaban
as$panamabanan
bananas$panama
mabananas$pana
namabananas$pa
nanas$panamaba
nas$panamabana
panamabananas$
s$panamabanana



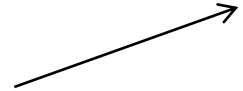
Is It True in General?

```
   $panamabananas
1  abananas$panam
2  amabananas$pan
3  anamabananas$p
4  ananas$panamab
5  anas$panamaban
6  as$panamabanan
   bananas$panama
   mabananas$pana
   namabananas$pa
   nanas$panamaba
   nas$panamabana
   panamabananas$
   s$panamabanana
```

These strings are sorted

Is It True in General?

	$panamabananas
1	abananas$panam
2	amabananas$pan
3	anamabananas$p
4	ananas$panamab
5	anas$panamaban
6	as$panamabanan
	bananas$panama
	mabananas$pana
	namabananas$pa
	nanas$panamaba
	nas$panamabana
	panamabananas$
	s$panamabanana



Chop off **a**

bananas$panam
mabananas$pan
namabananas$p
nanas$panamab
nas$panamaban
s$panamabanan

These strings are sorted

Is It True in General?

	$panamabananas
1	abananas$panam
2	amabananas$pan
3	anamabananas$p
4	ananas$panamab
5	anas$panamaban
6	as$panamabanan
	bananas$panama
	mabananas$pana
	namabananas$pa
	nanas$panamaba
	nas$panamabana
	panamabananas$
	s$panamabanana

These strings are sorted



Chop off **a**

```
bananas$panam
mabananas$pan
namabananas$p
nanas$panamab
nas$panamaban
s$panamabanan
```

Still sorted

Is It True in General?

	$panamabananas
1	abananas$panam
2	amabananas$pan
3	anamabananas$p
4	ananas$panamab
5	anas$panamaban
6	as$panamabanan
	bananas$panama
	mabananas$pana
	namabananas$pa
	nanas$panamaba
	nas$panamabana
	panamabananas$
	s$panamabanana

These strings are sorted

Chop off **a**

bananas$panam
mabananas$pan
namabananas$p
nanas$panamab
nas$panamaban
s$panamabanan

Still sorted

Add **a** to end

bananas$panam**a**
mabananas$pan**a**
namabananas$p**a**
nanas$panamab**a**
nas$panamaban**a**
s$panamabanan**a**

Is It True in General?

	$panamabananas
1	abananas$panam
2	amabananas$pan
3	anamabananas$p
4	ananas$panamab
5	anas$panamaban
6	as$panamabanan
	bananas$panama
	mabananas$pana
	namabananas$pa
	nanas$panamaba
	nas$panamabana
	panamabananas$
	s$panamabanana

These strings are sorted

Chop off **a**

bananas$panam
mabananas$pan
namabananas$p
nanas$panamab
nas$panamaban
s$panamabanan

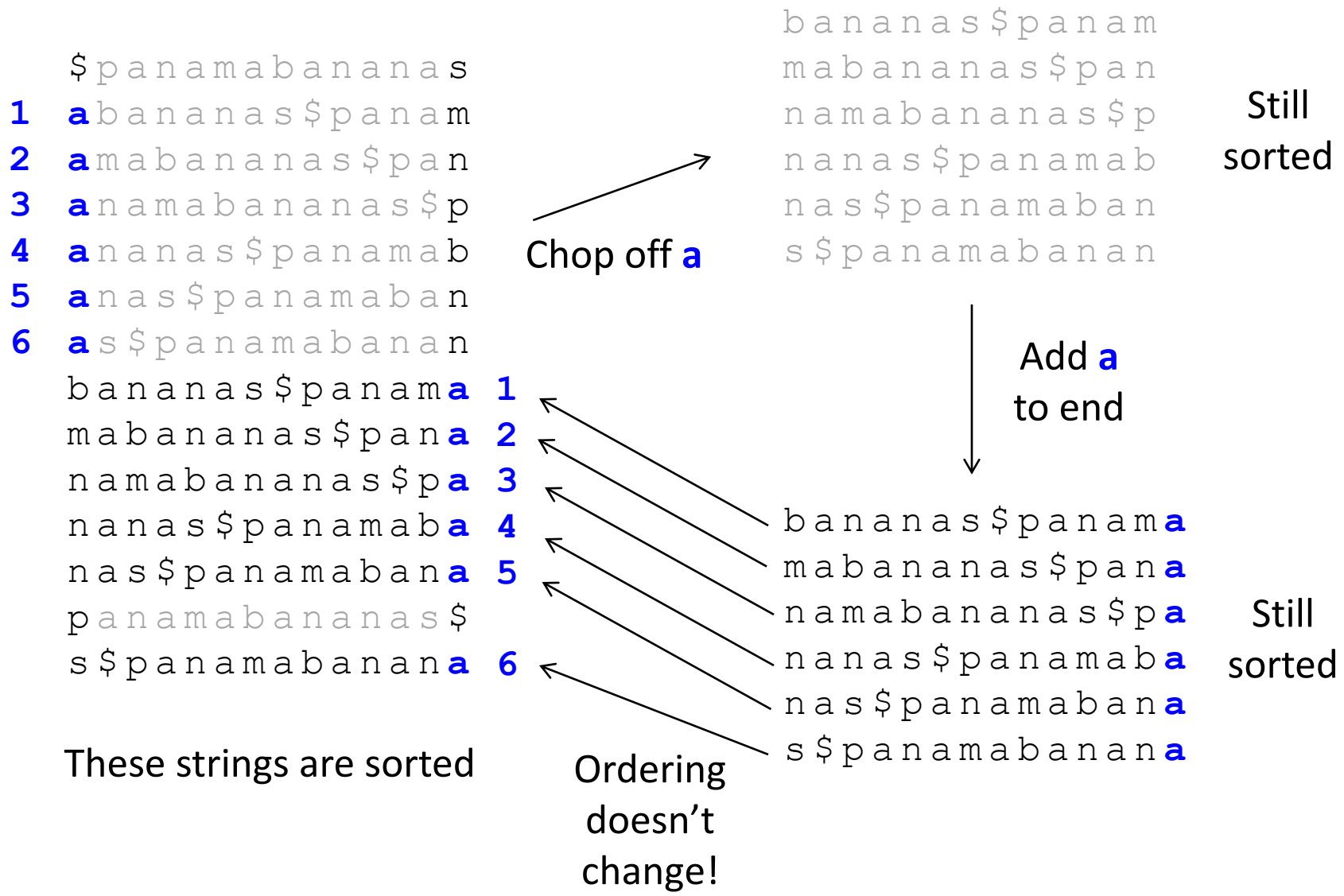
Still sorted

Add **a** to end

bananas$panam**a**
mabananas$pan**a**
namabananas$p**a**
nanas$panamab**a**
nas$panamaban**a**
s$panamabanan**a**

Still sorted

Is It True in General?



First-Last Property

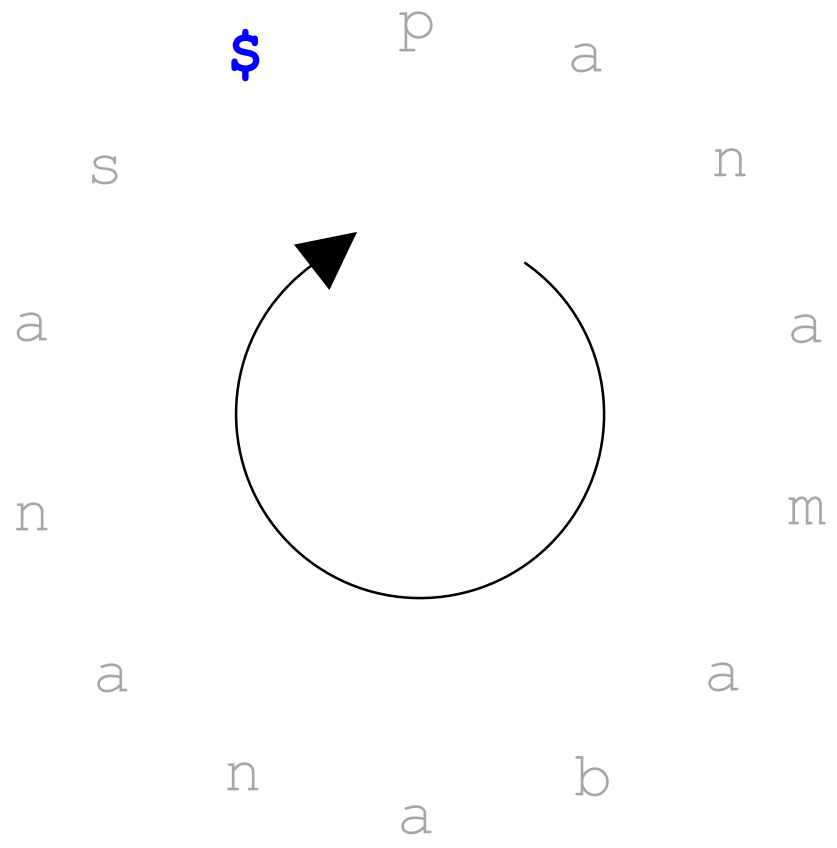
- the *k*-th occurrence of *symbol* in ***FirstColumn***
- and the *k*-th occurrence of *symbol* in ***LastColumn***
- correspond to appearance of *symbol* at the same position in *Text*.

$p_1 a_3 n_1 a_2 m_1 a_1 b_1 a_4 n_2 a_5 n_3 a_6 s_1 \$_1$

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$

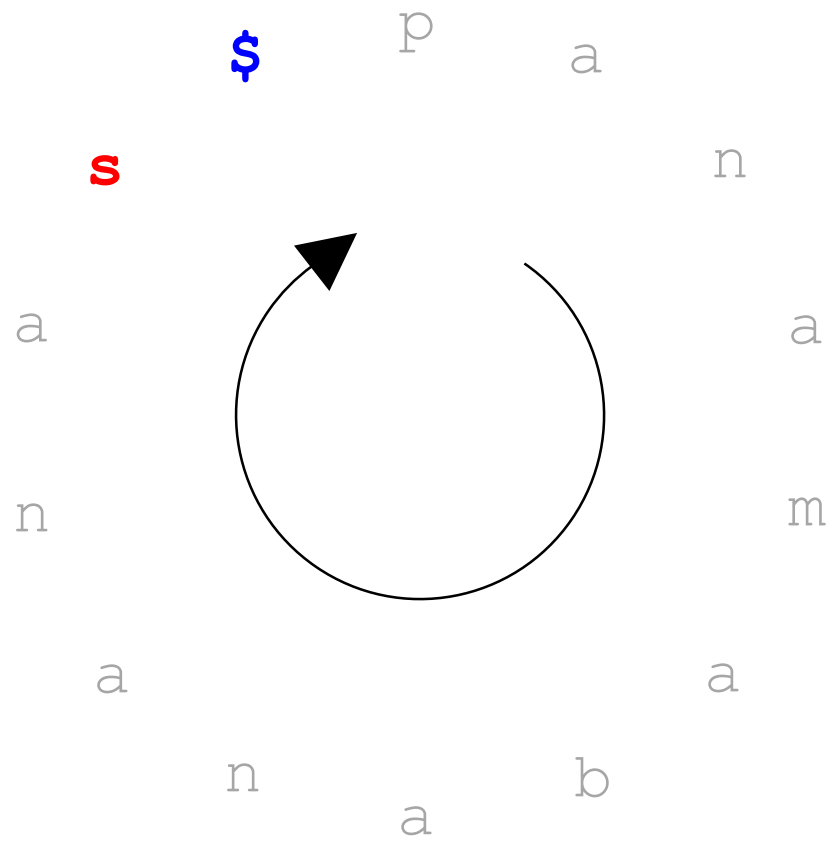
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$

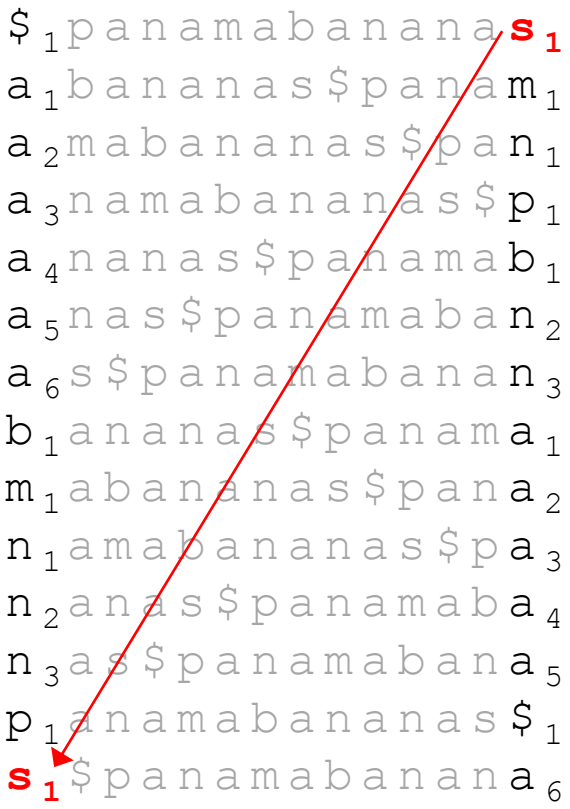


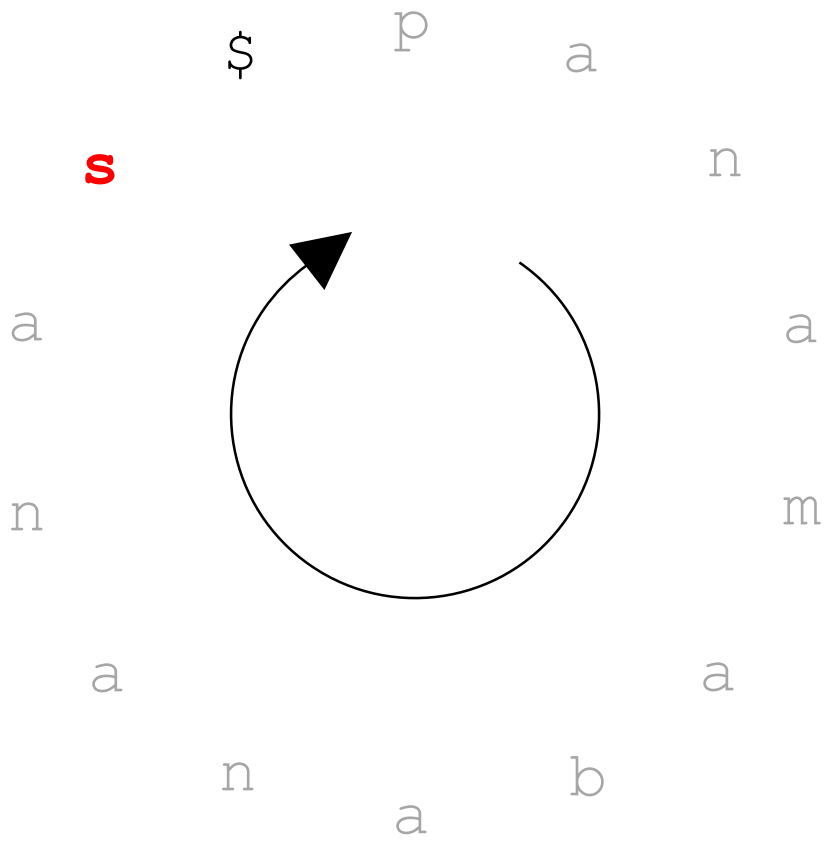
Inverting BWT Again

$_1$ p a n a m a b a n a n a s$_1$
a$_1$ b a n a n a s $ p a n a m$_1$
a$_2$ m a b a n a n a s $ p a n$_1$
a$_3$ n a m a b a n a n a s $ p$_1$
a$_4$ n a n a s $ p a n a m a b$_1$
a$_5$ n a s $ p a n a m a b a n$_2$
a$_6$ s $ p a n a m a b a n a n$_3$
b$_1$ a n a n a s $ p a n a m a$_1$
m$_1$ a b a n a n a s $ p a n a$_2$
n$_1$ a m a b a n a n a s $ p a$_3$
n$_2$ a n a s $ p a n a m a b a$_4$
n$_3$ a s $ p a n a m a b a n a$_5$
p$_1$ a n a m a b a n a n a s $ $_1$
s$_1$ $ p a n a m a b a n a n a$_6$



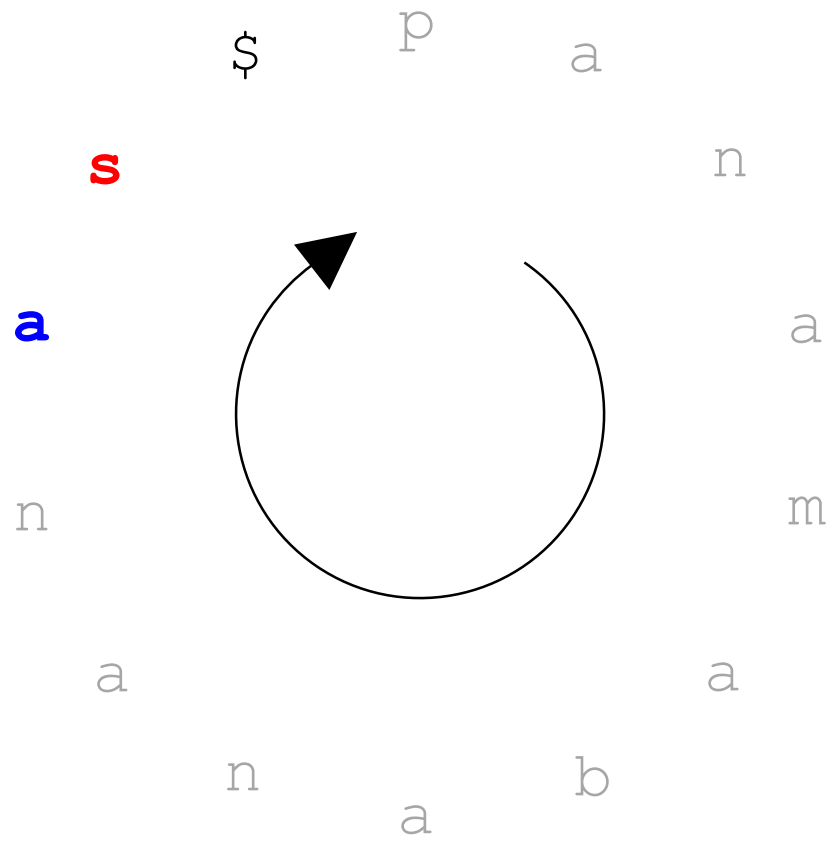
Inverting BWT Again



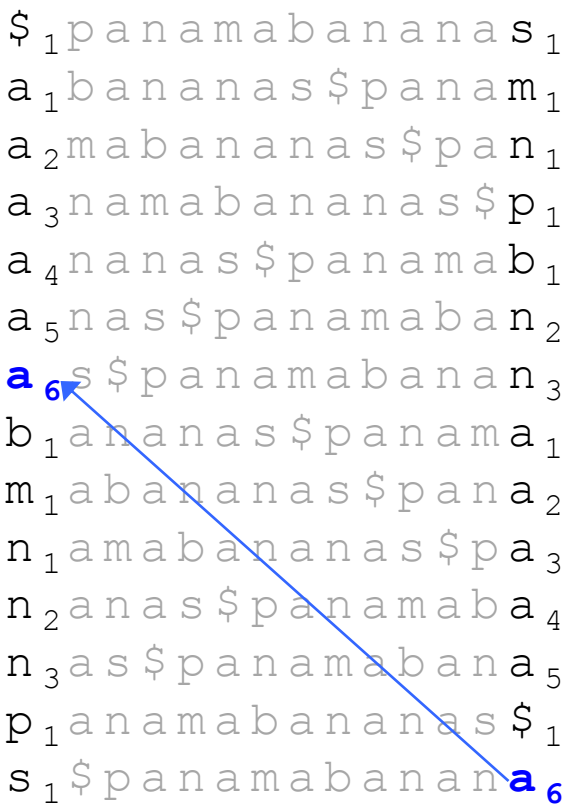


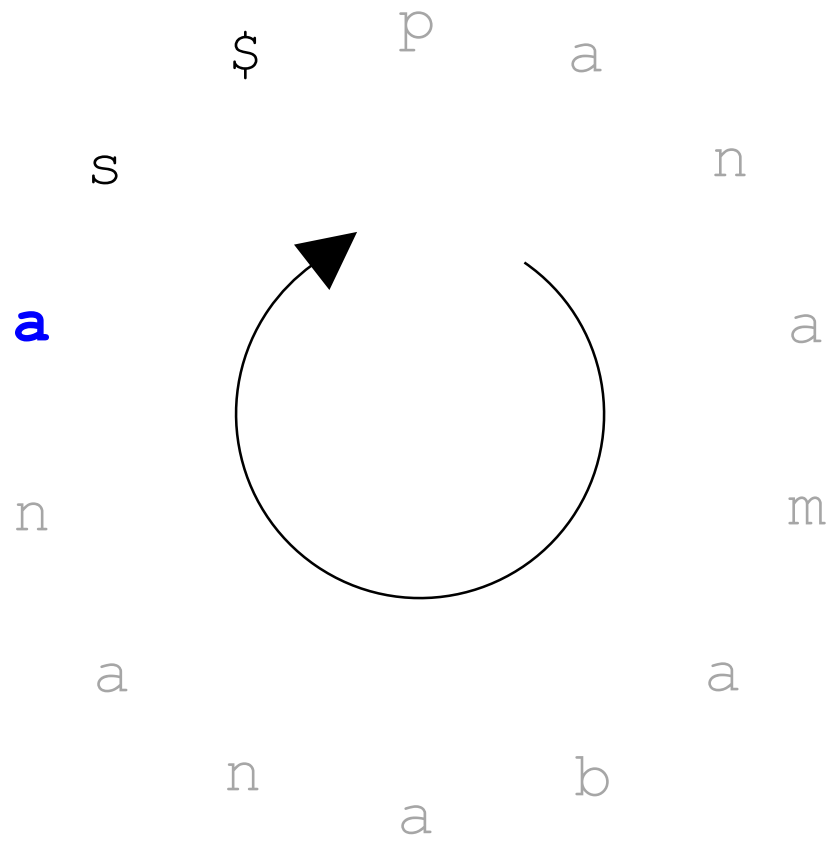
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n **a $_6$**



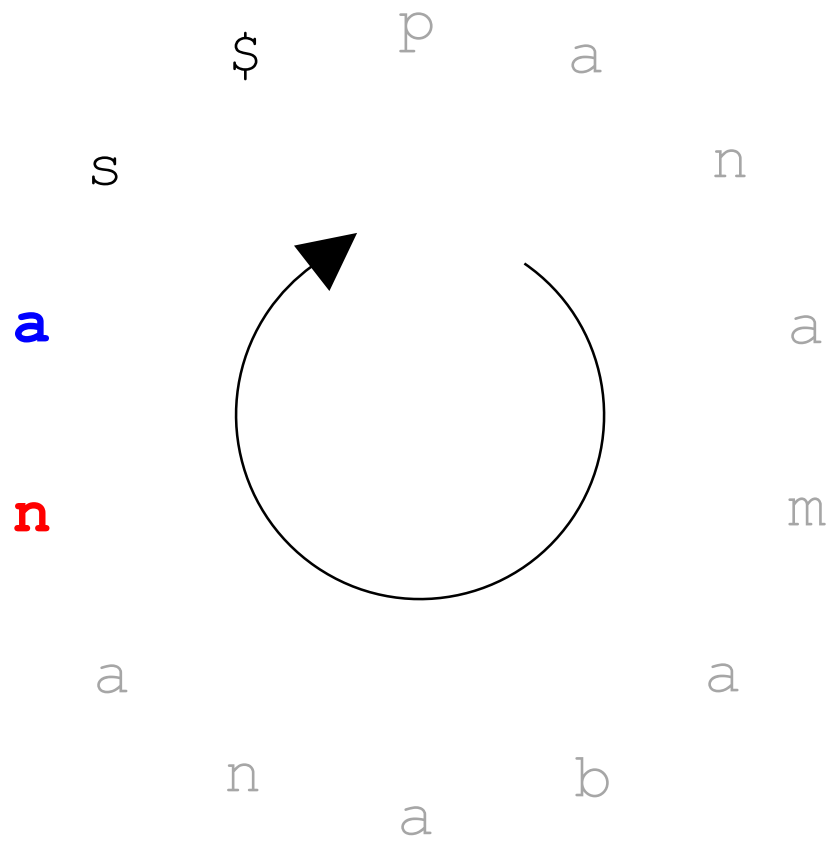
Inverting BWT Again



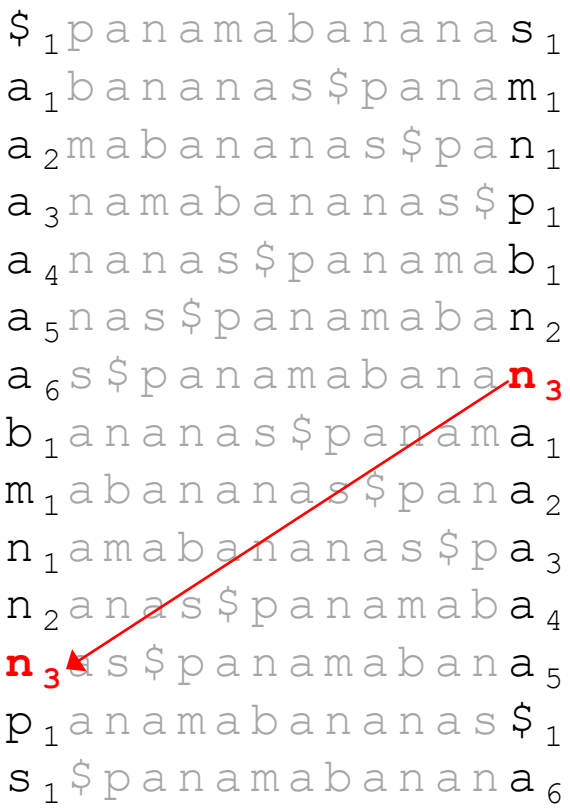


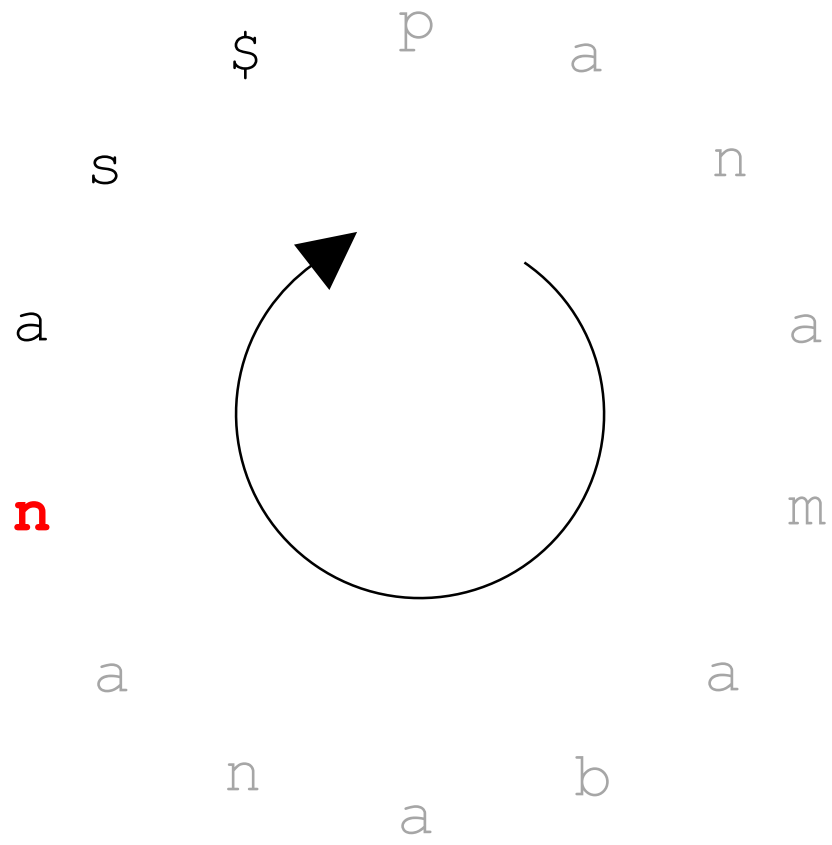
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a **n $_3$**
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



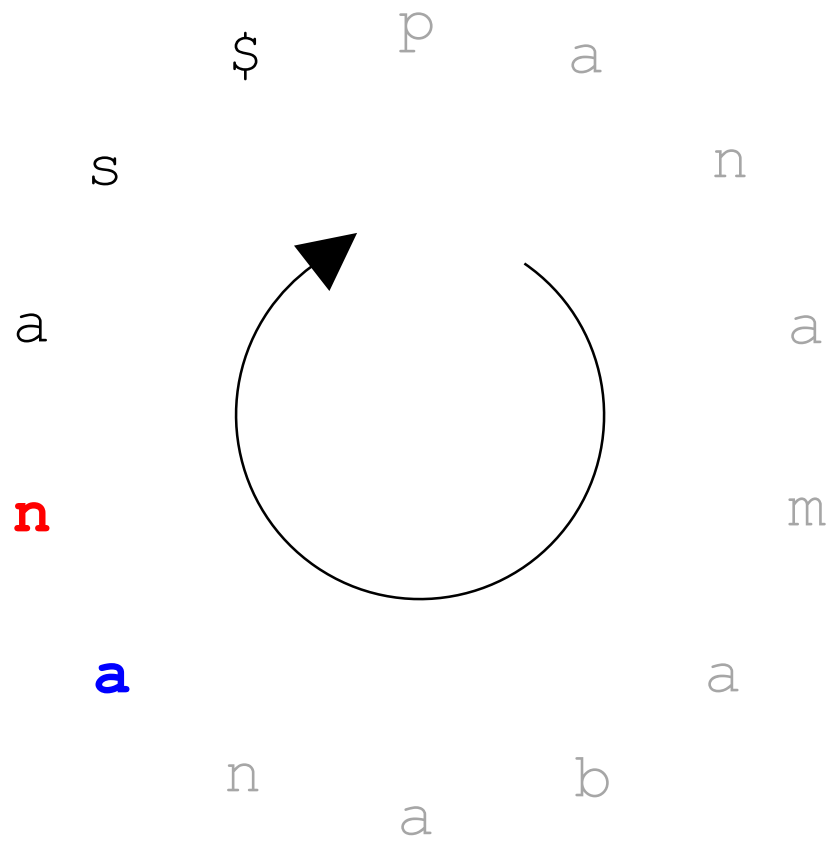
Inverting BWT Again



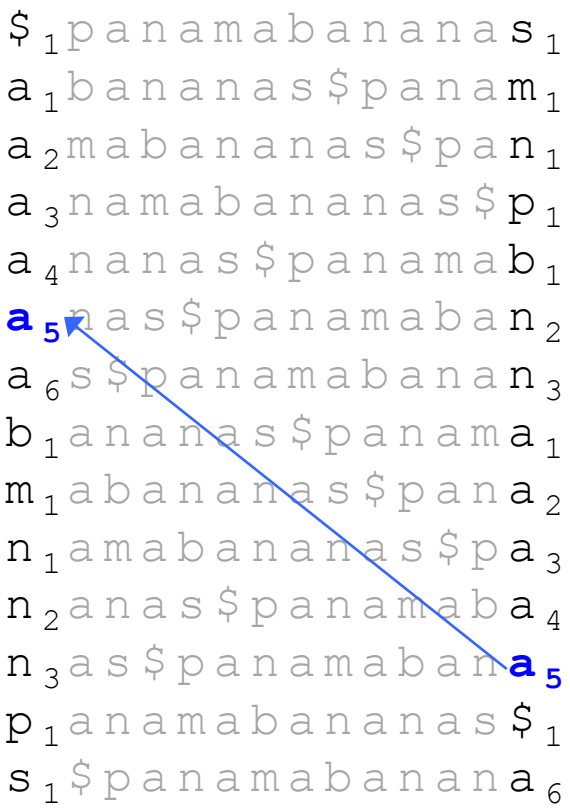


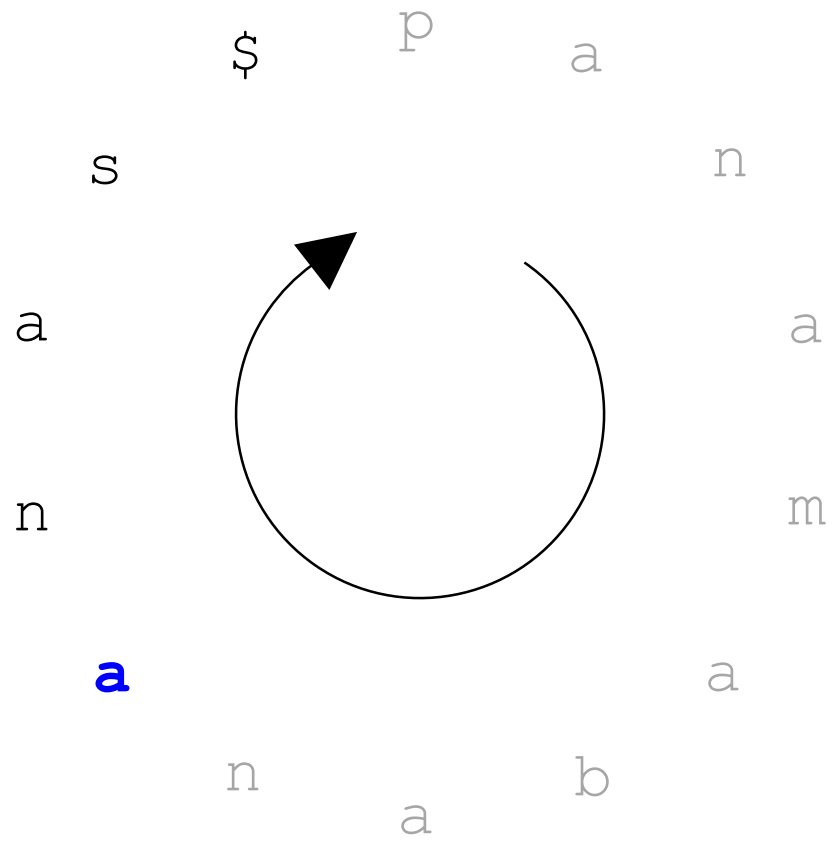
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n **a $_5$**
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



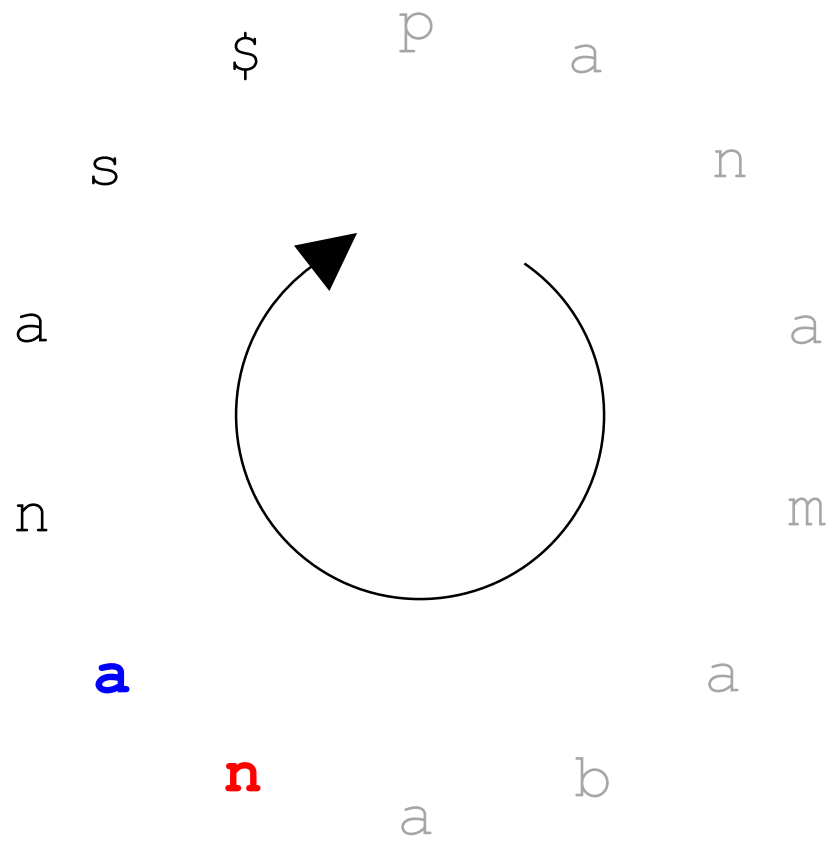
Inverting BWT Again



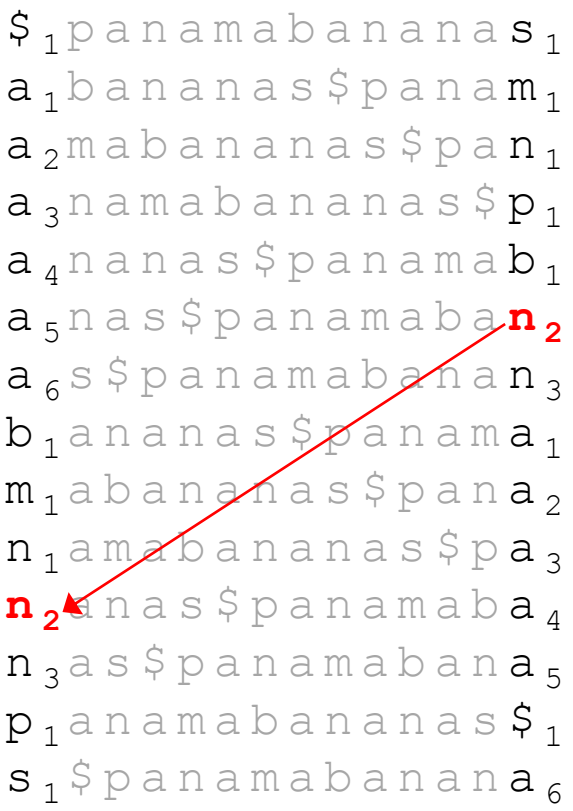


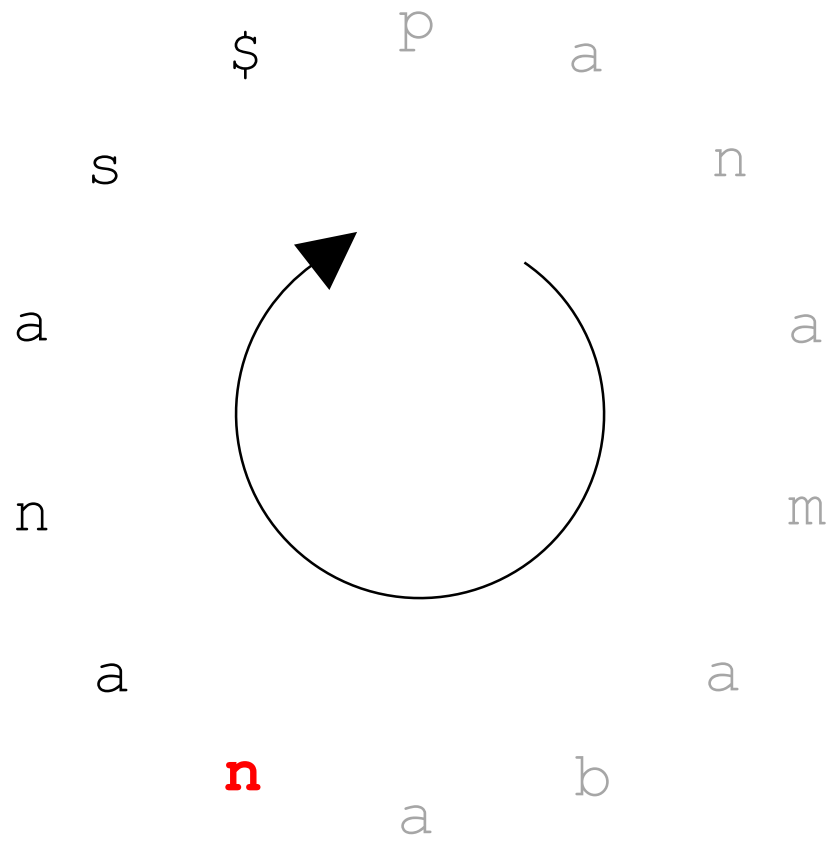
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a **n $_2$**
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



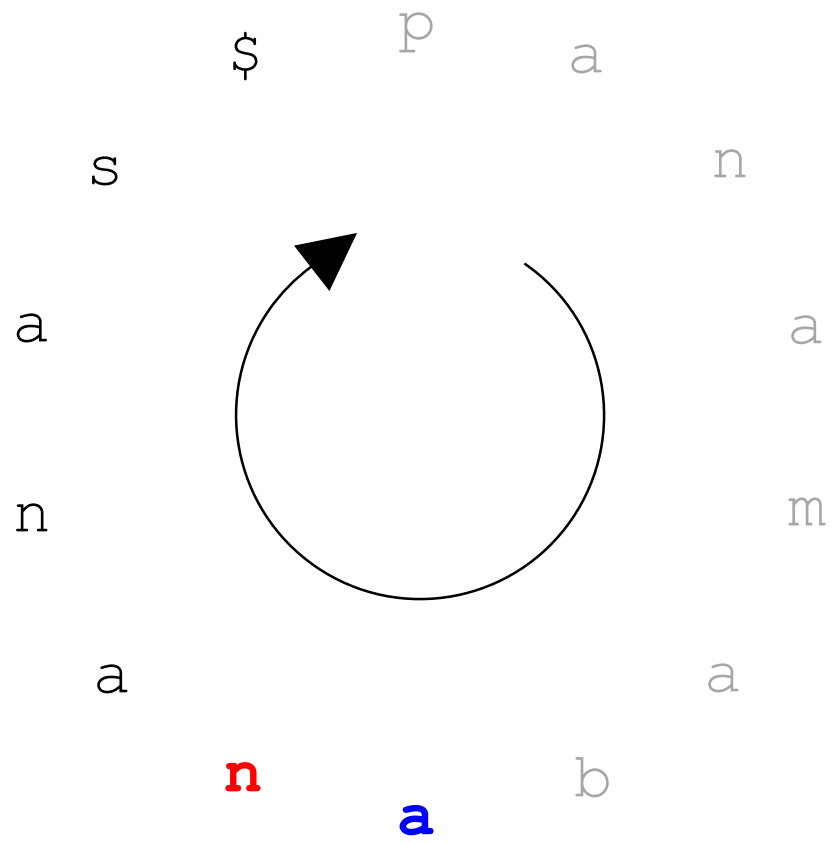
Inverting BWT Again

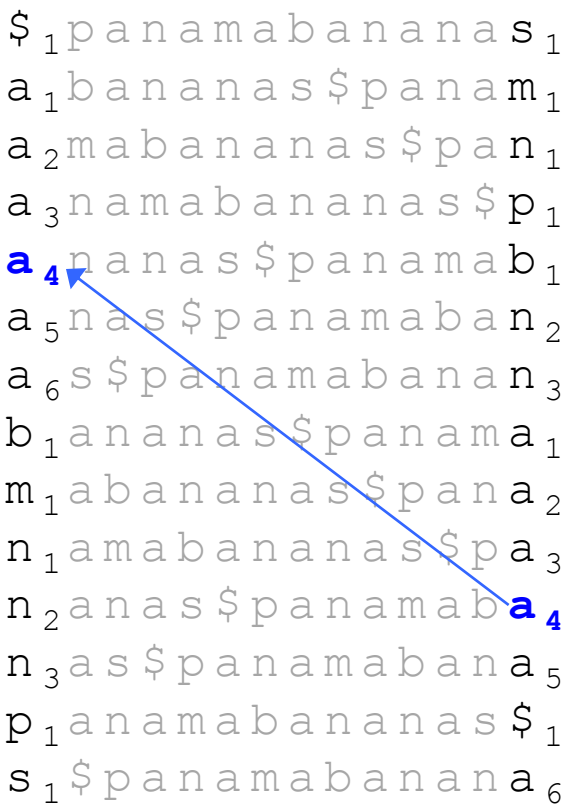


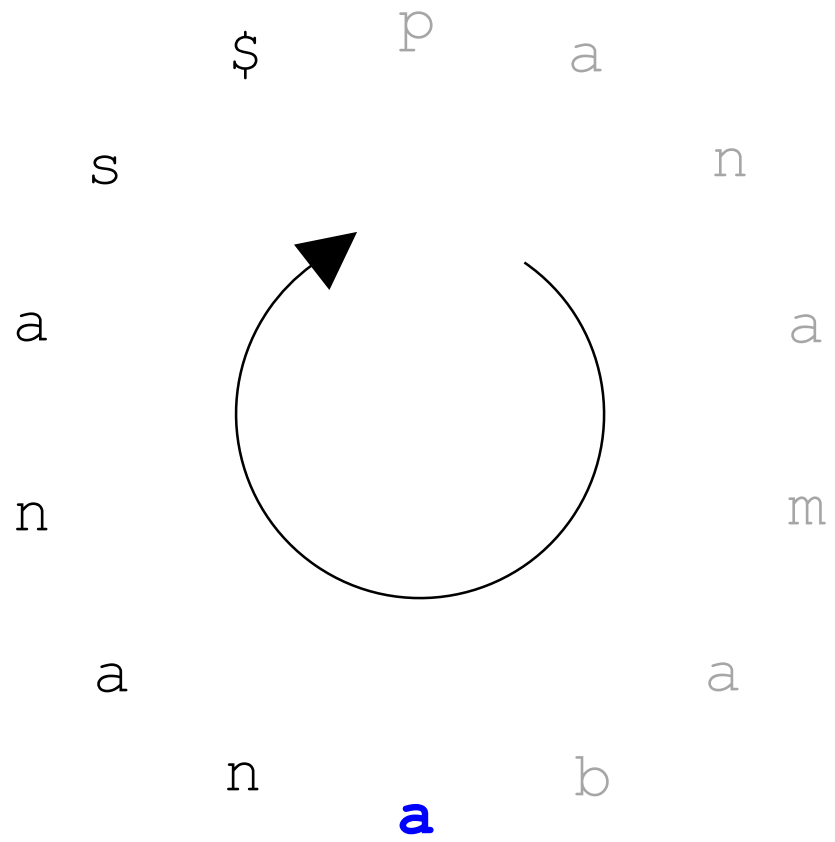


Inverting BWT Again

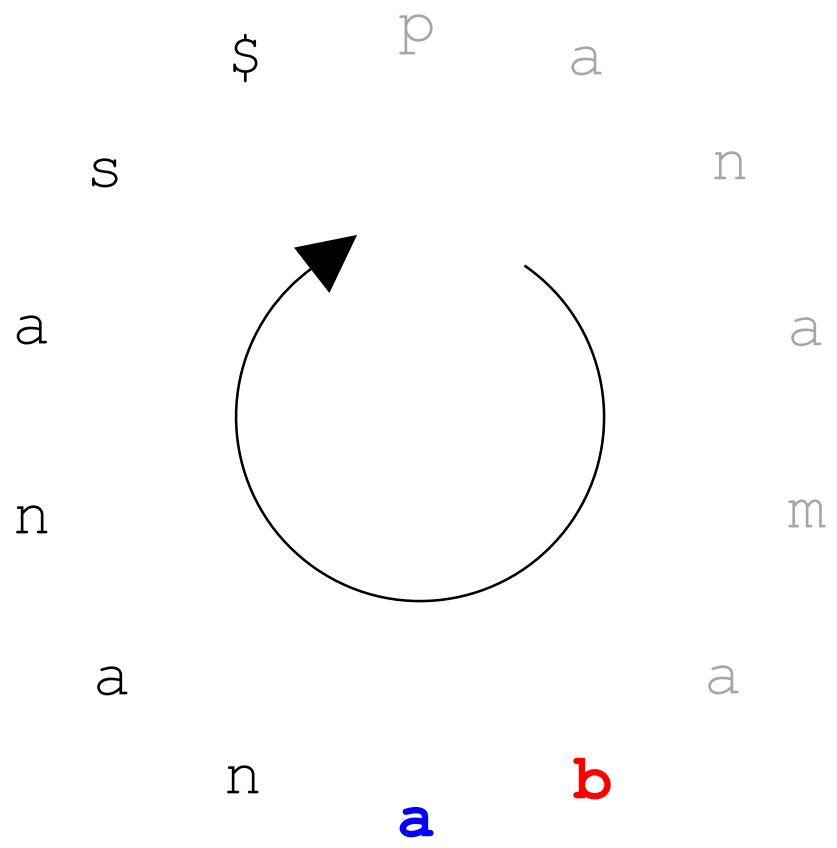
$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamab**a$_4$**
n$_3$as$panamaban**a**$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$




$1panamabananas1
a1bananas$panam1
a2mabananas$pan1
a3namabananas$p1
a4panas$panamab1
a5nas$panamaban2
a6s$panamabanan3
b1ananas$panama1
m1abananas$pana2
n1amabananas$pa3
n2anas$panamaba4
n3as$panamabana5
p1anamabananas$1
s1$panamabanana6

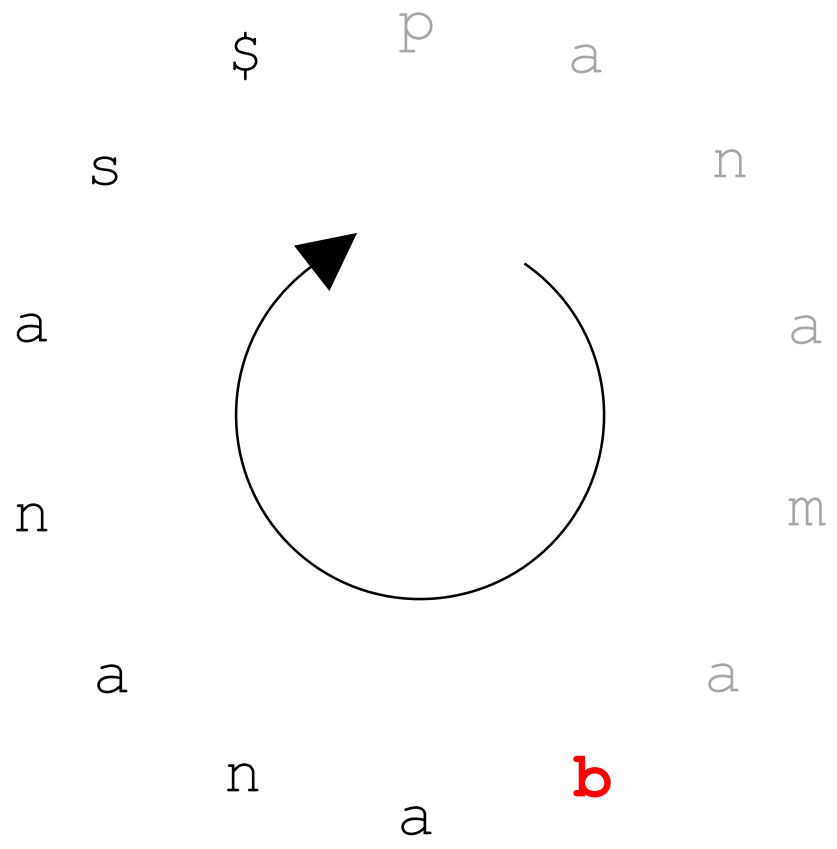

$
p
a
s
n
a
a
n
m
a
a
n
b
a

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panama**b$_1$**
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$



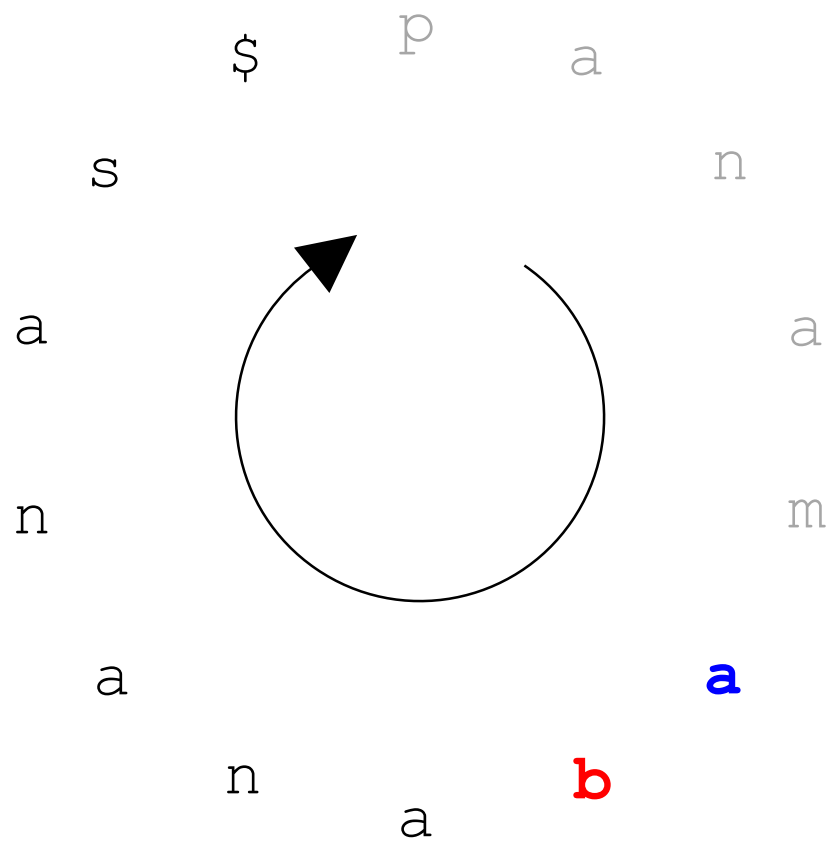
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a **b $_1$**
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$

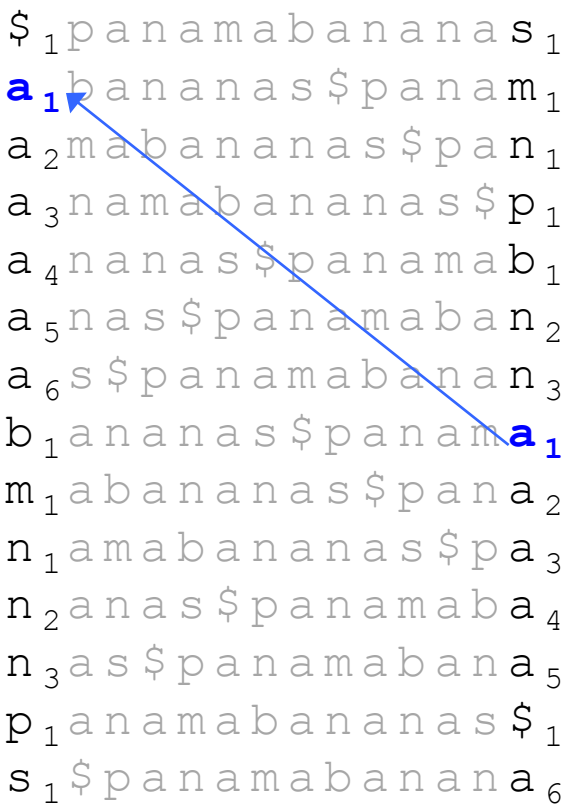


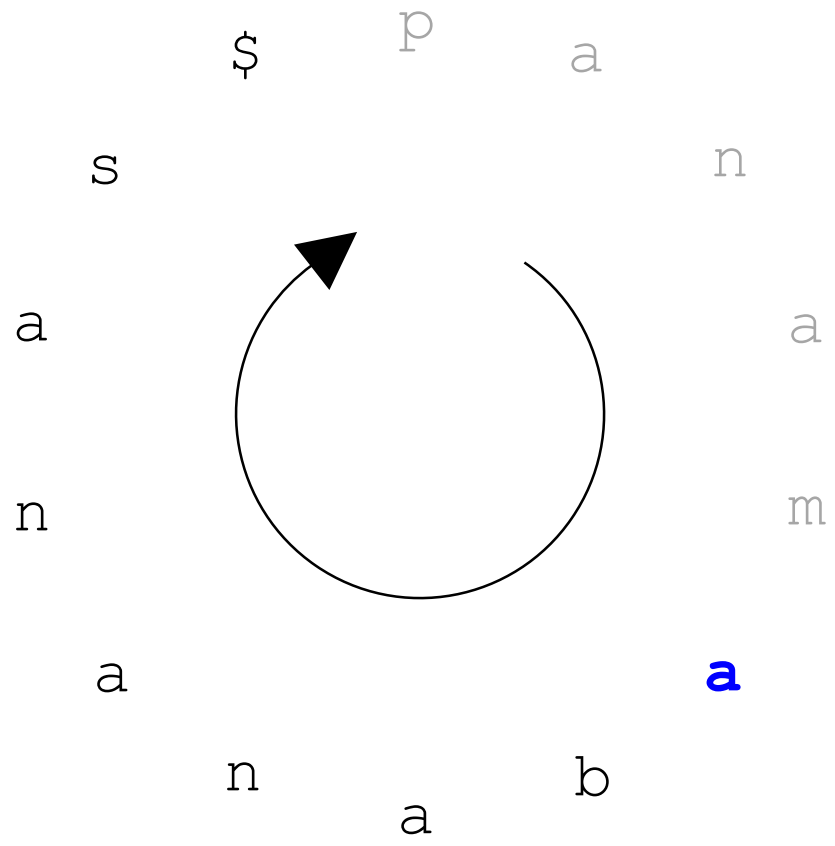
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m **a $_1$**
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



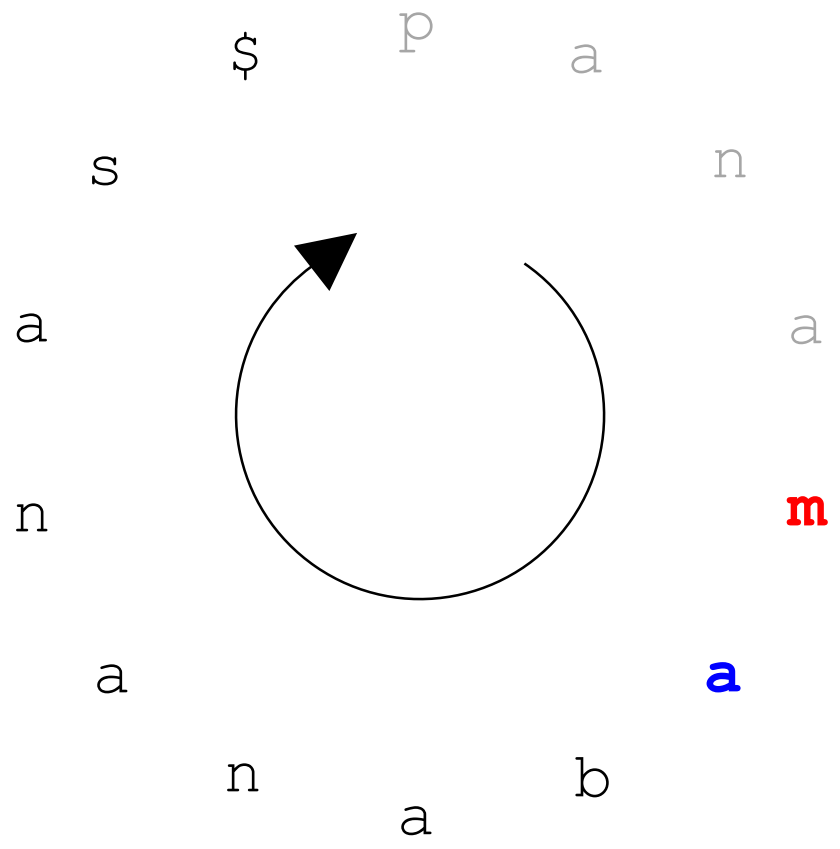
Inverting BWT Again





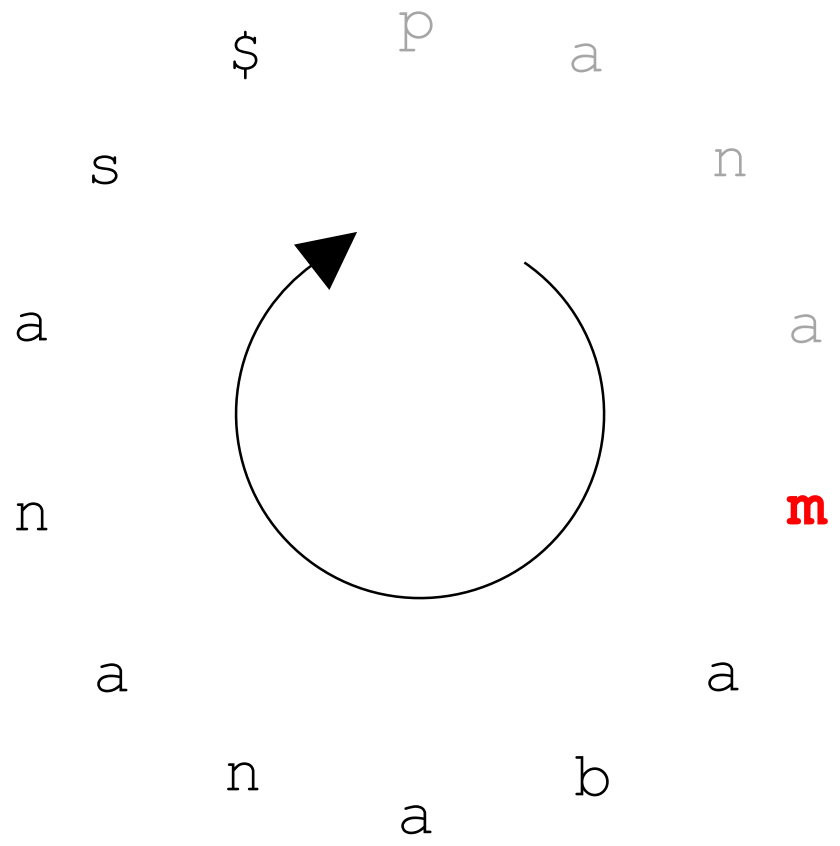
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a **m $_1$**
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



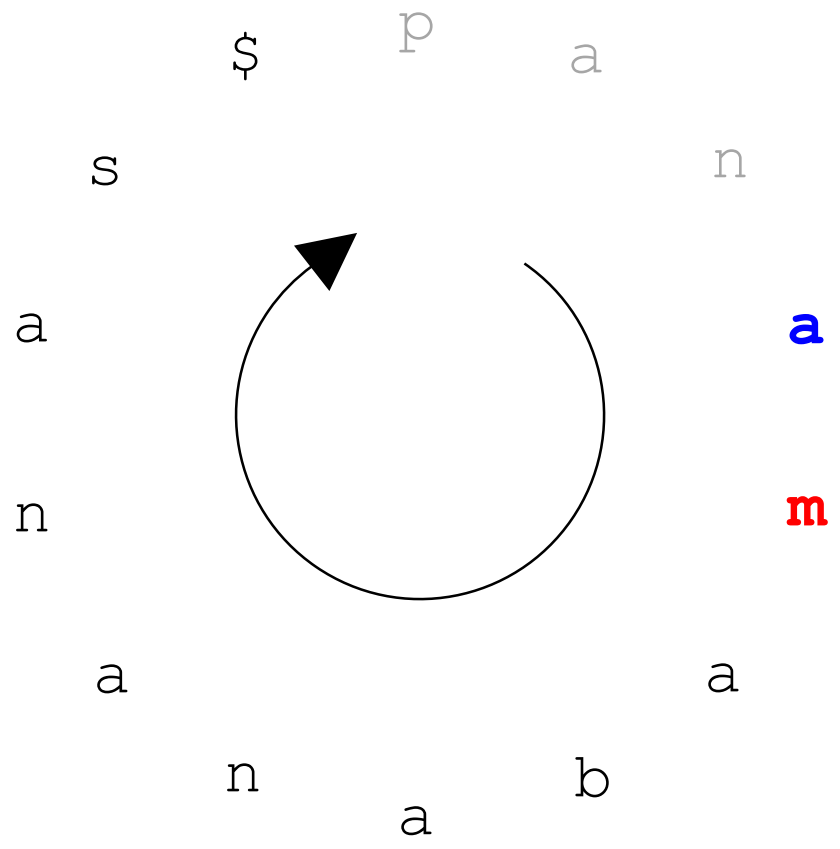
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a **m** $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$

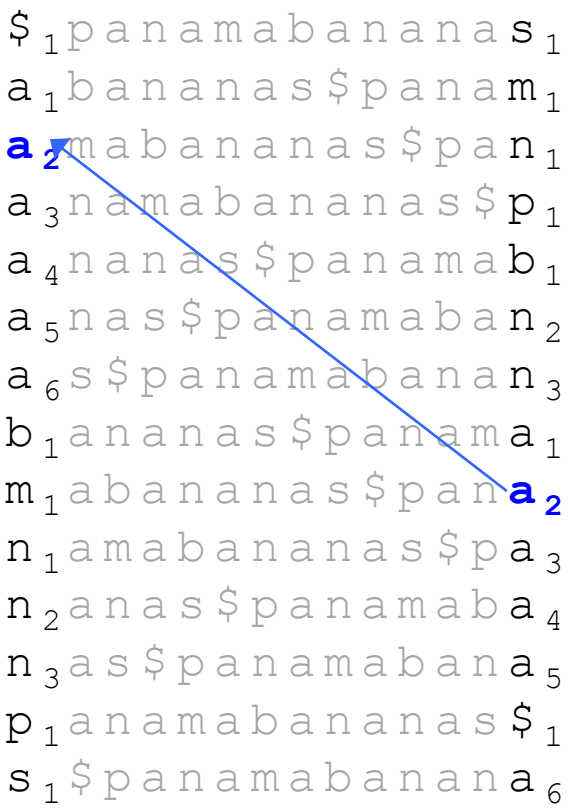


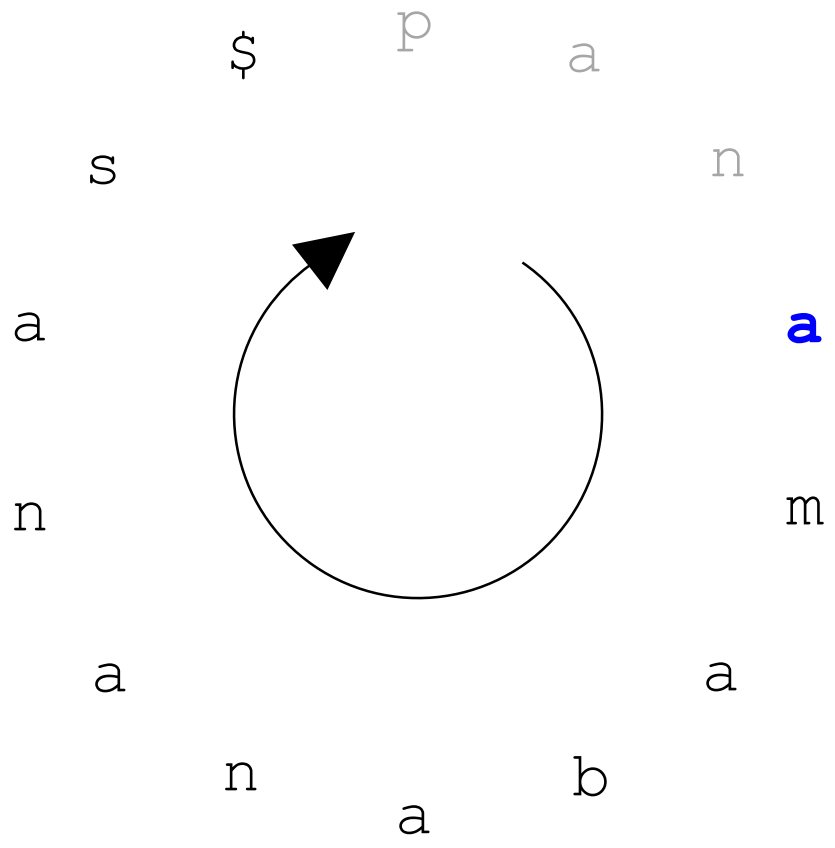
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n **a $_2$**
n $_1$ a m a b a n a n a s $ p a $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



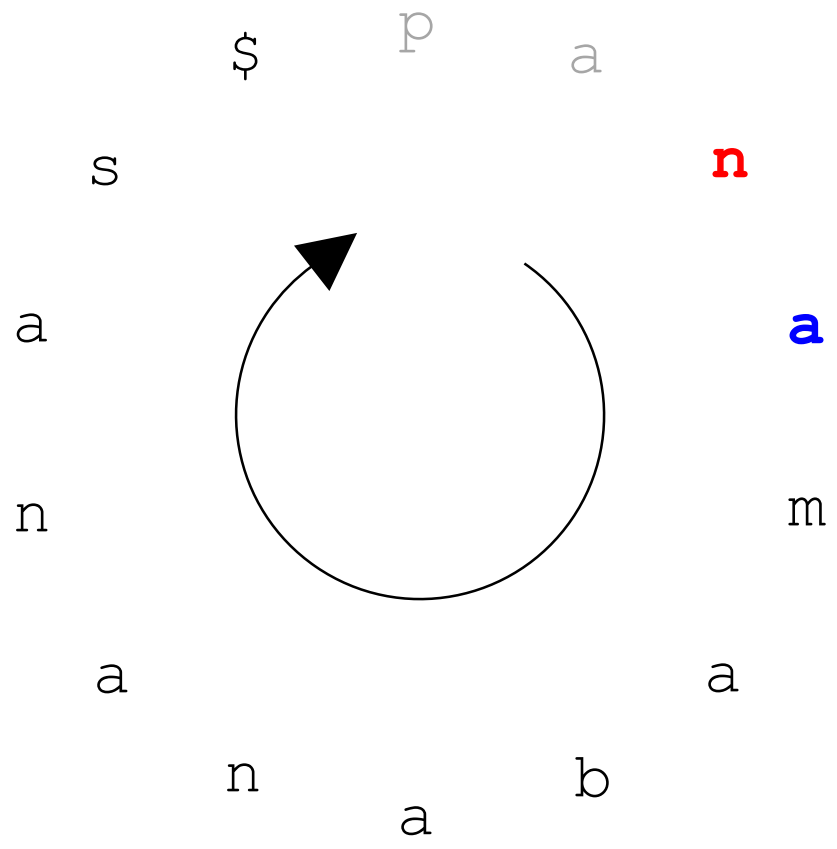
Inverting BWT Again



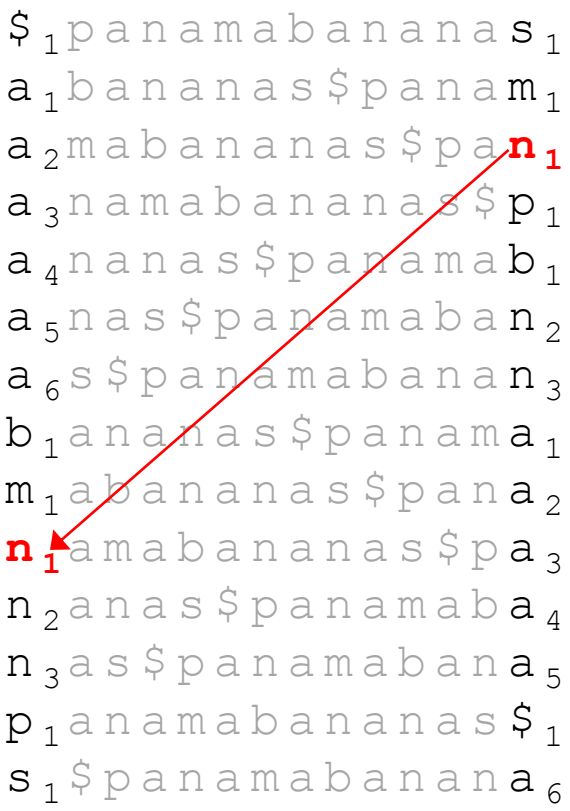


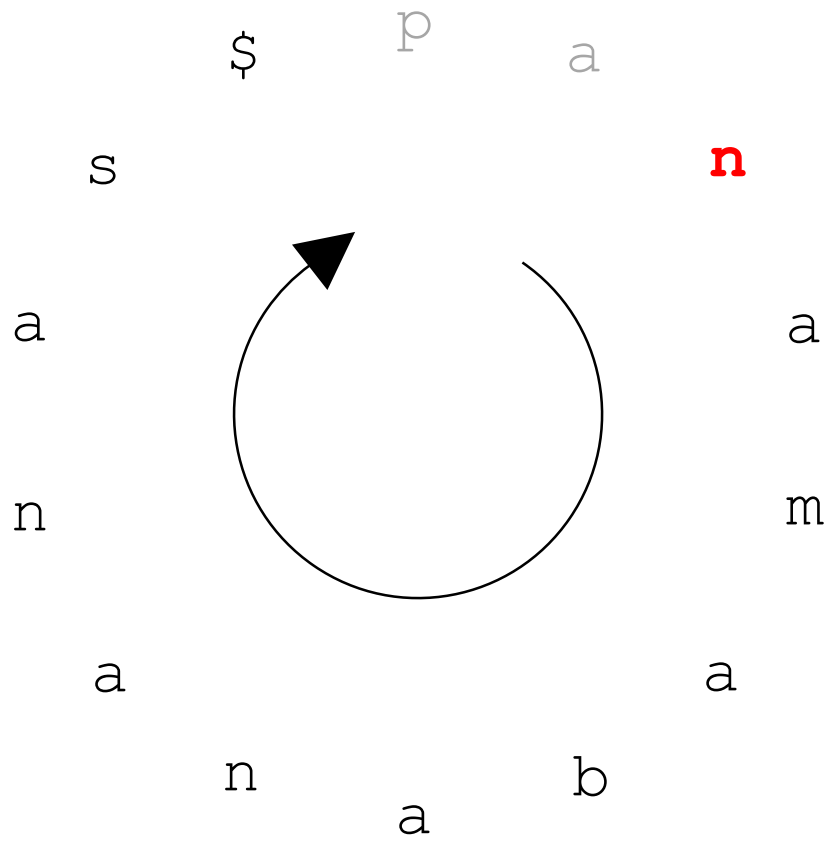
Inverting BWT Again

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananas$pa**n$_1$**
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$



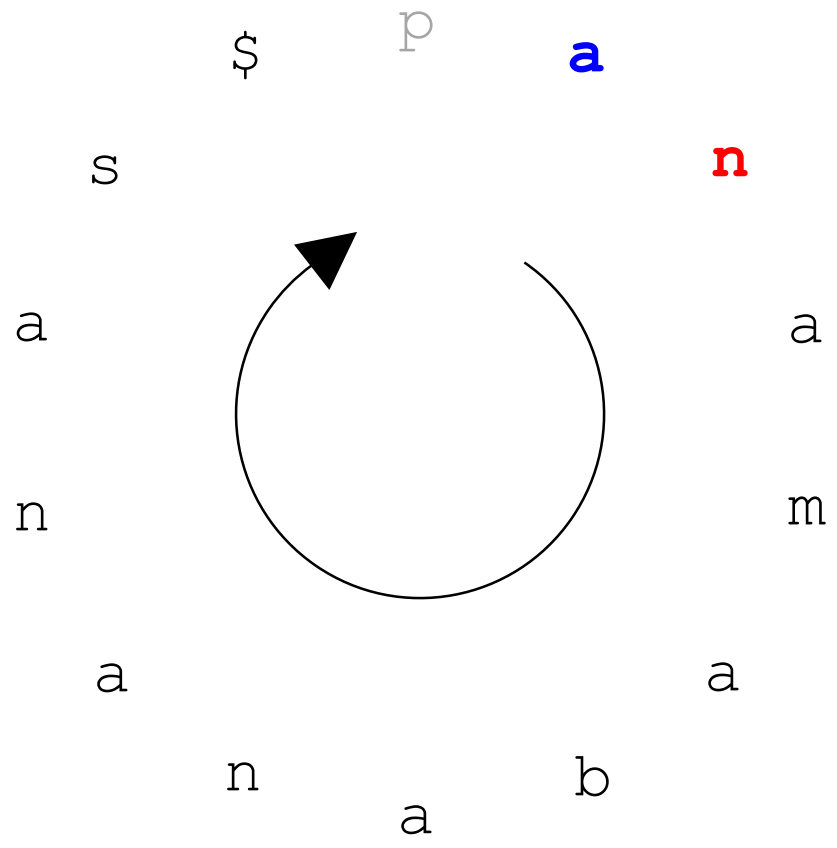
Inverting BWT Again



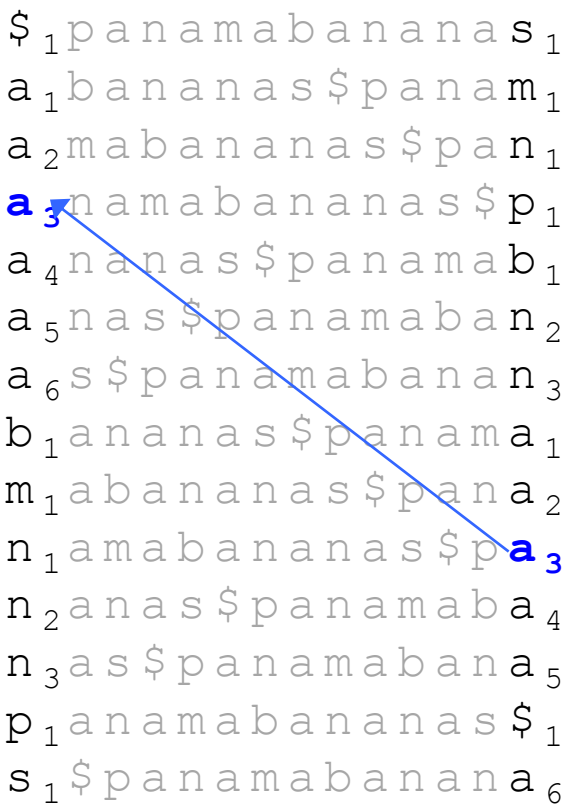


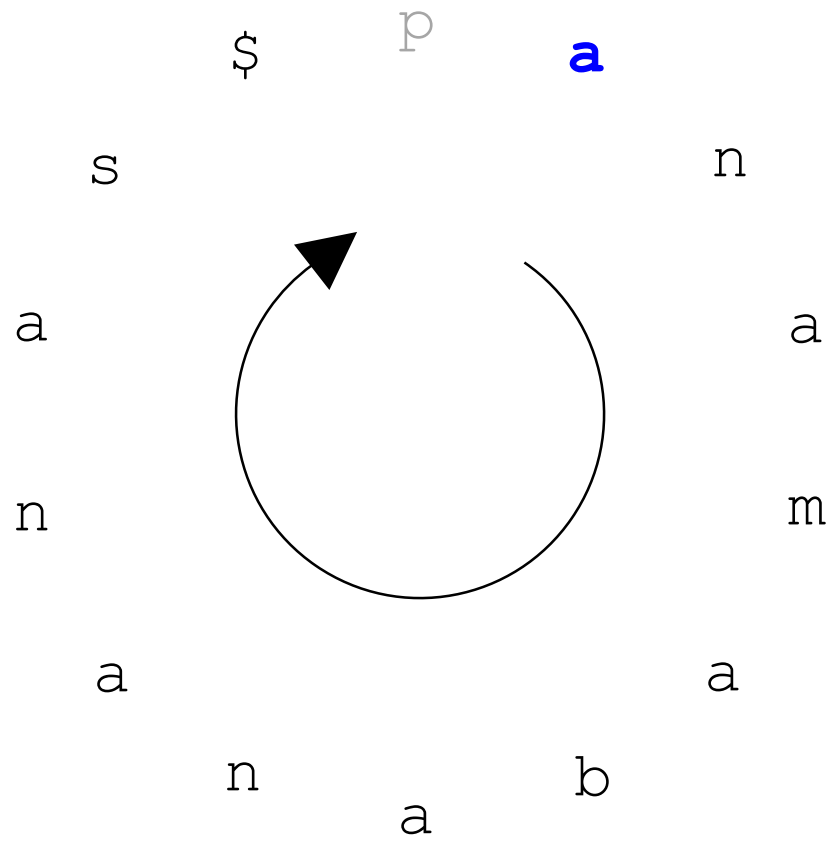
Inverting BWT Again

$_1$ p a n a m a b a n a n a s $_1$
a $_1$ b a n a n a s $ p a n a m $_1$
a $_2$ m a b a n a n a s $ p a n $_1$
a $_3$ n a m a b a n a n a s $ p $_1$
a $_4$ n a n a s $ p a n a m a b $_1$
a $_5$ n a s $ p a n a m a b a n $_2$
a $_6$ s $ p a n a m a b a n a n $_3$
b $_1$ a n a n a s $ p a n a m a $_1$
m $_1$ a b a n a n a s $ p a n a $_2$
n $_1$ a m a b a n a n a s $ p **a** $_3$
n $_2$ a n a s $ p a n a m a b a $_4$
n $_3$ a s $ p a n a m a b a n a $_5$
p $_1$ a n a m a b a n a n a s $ $_1$
s $_1$ $ p a n a m a b a n a n a $_6$



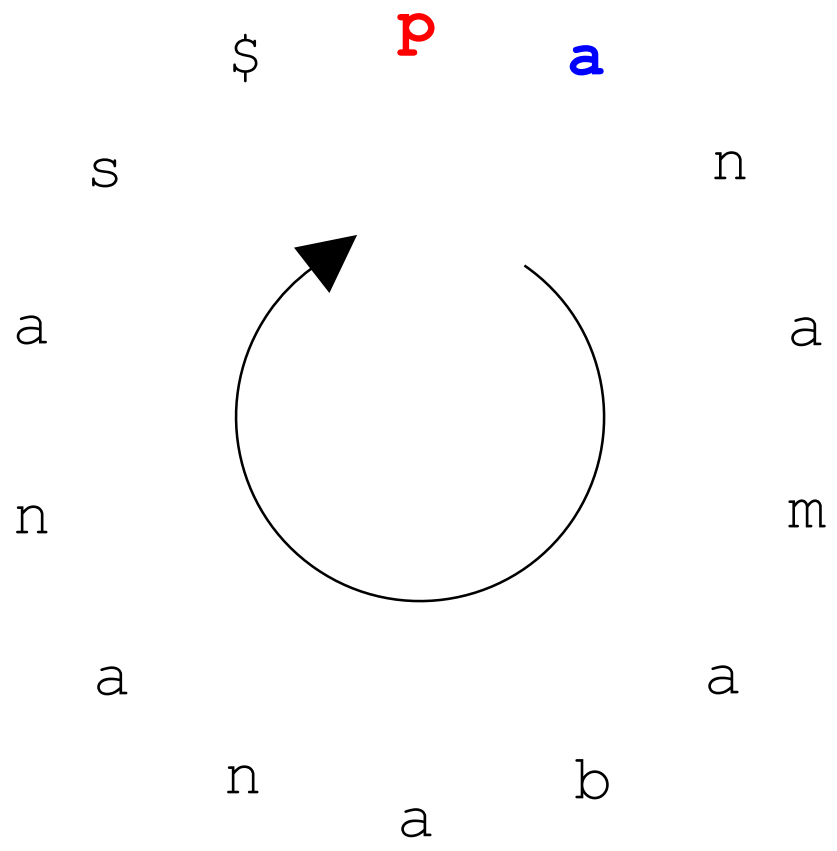
Inverting BWT Again



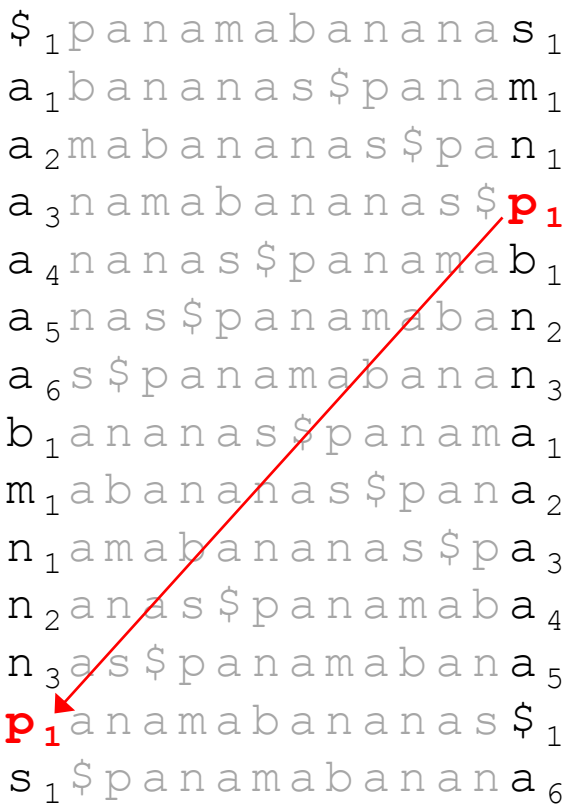


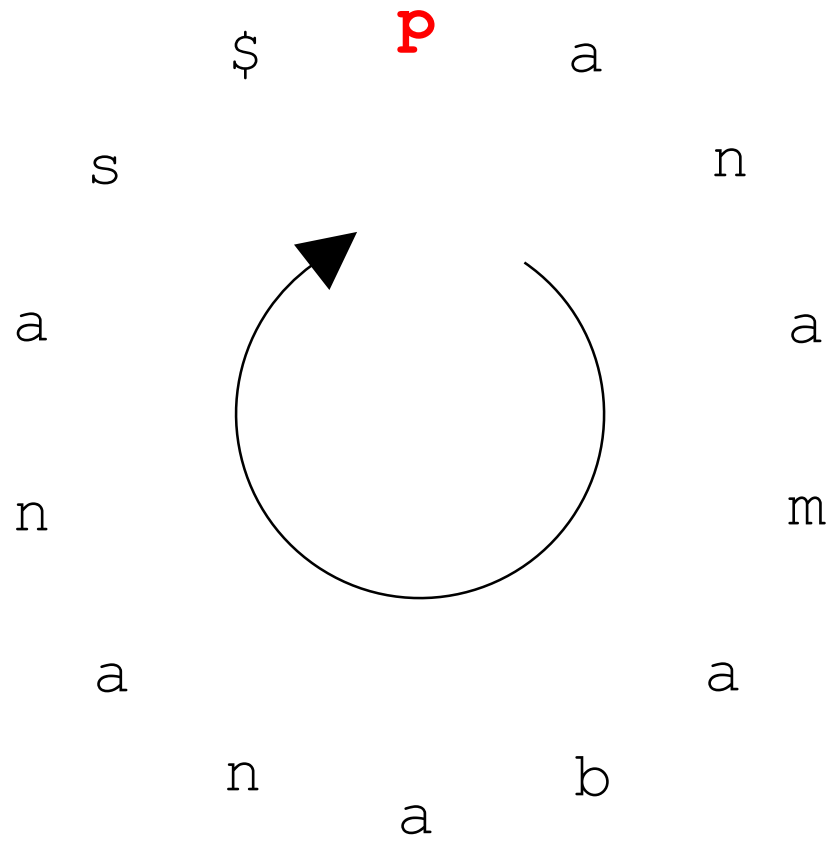
Inverting BWT Again

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$



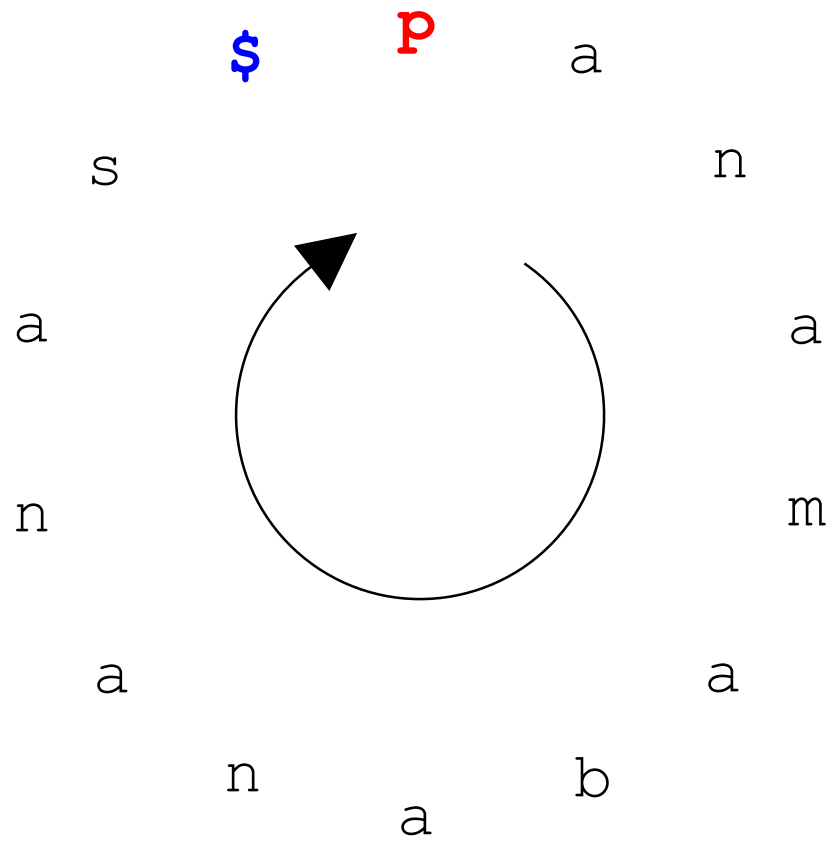
Inverting BWT Again



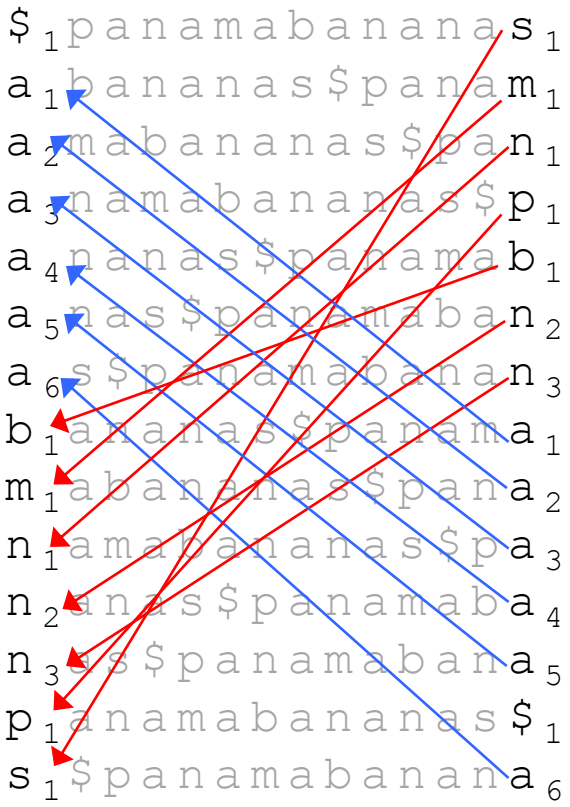


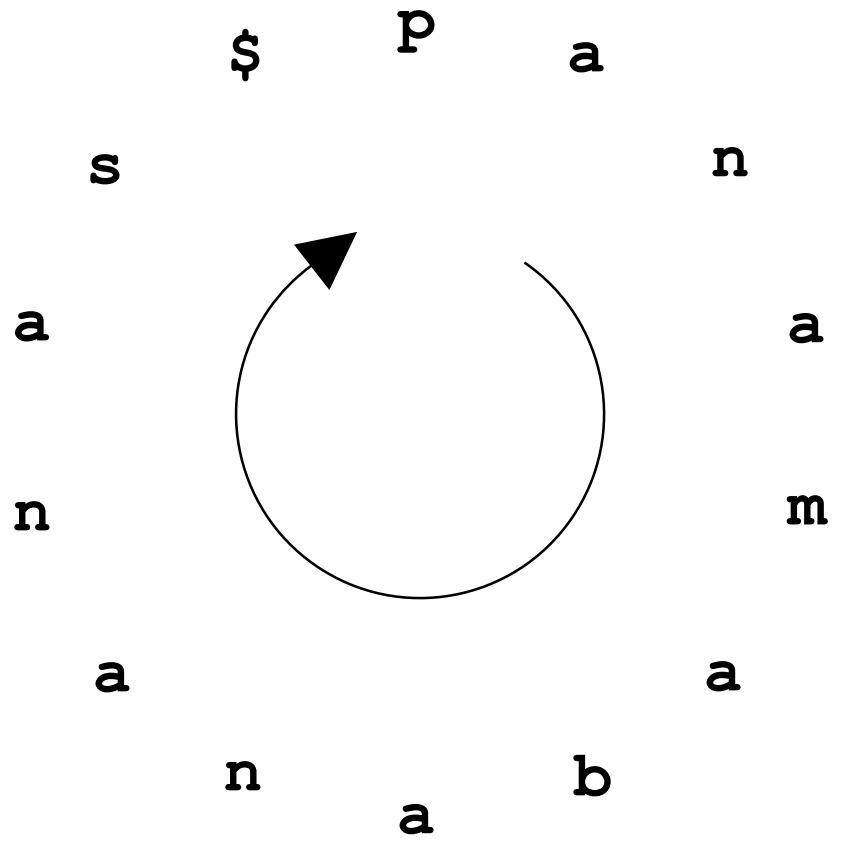
We Are Done!

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas**$$_1$**
s$_1$$panamabanana$_6$



This Was Fast!





- Memory: 2|*Text*|
- Time: O(|*Text*|)

Outline

- Burrows-Wheeler Transform
- Inverting Burrows-Wheeler Transform
- **Using BWT for Pattern Matching**
- Suffix Arrays
- Approximate Pattern Matching

Back to Pattern Matching

- Suffix Tree Pattern Matching:
  - Runtime: O(|*Text*| + |*Patterns*|)
  - Memory: 20•|*Text*|

For human genome:

- |*Text*|≈ $3*10^9$



- Can we use BWT(*Text*) to design a more memory efficient linear-time algorithm for Multiple Pattern Matching?

Finding Pattern Matches Using BWT

- Searching for **ana** in p**ana**mab**anana**s



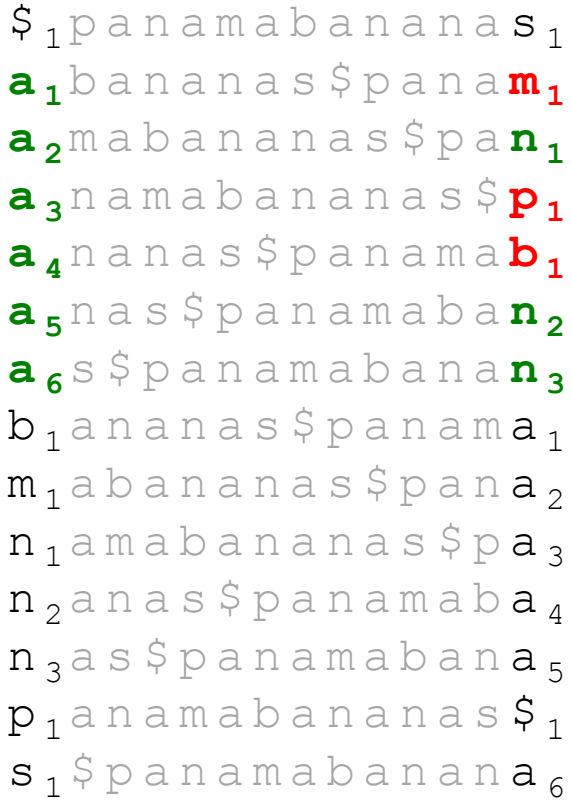
Lets Start by Matching the Last Symbol (**a**)

- Searching for an**a** in panamabananas

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$

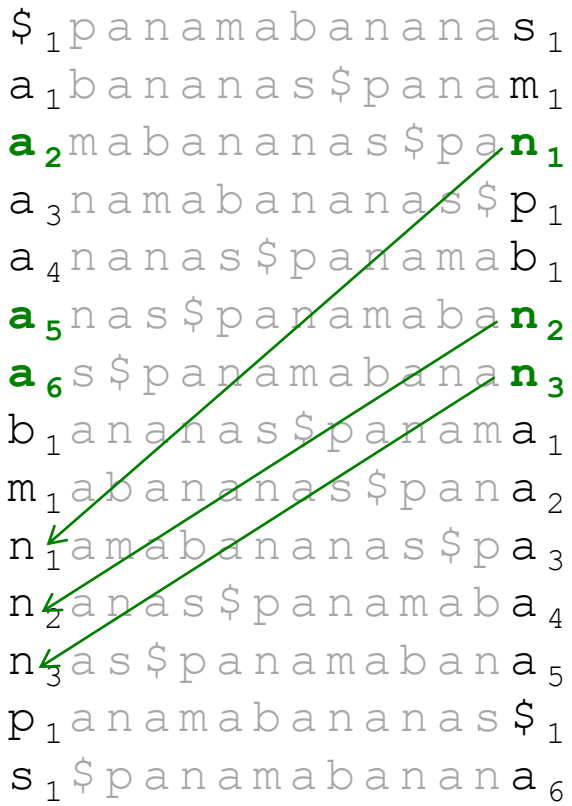
Matching the Last Two Symbols (na)

- Searching for a**na** in panamabananas



Three Matches of **na** Found!

- Searching for a**na** in panamabananas



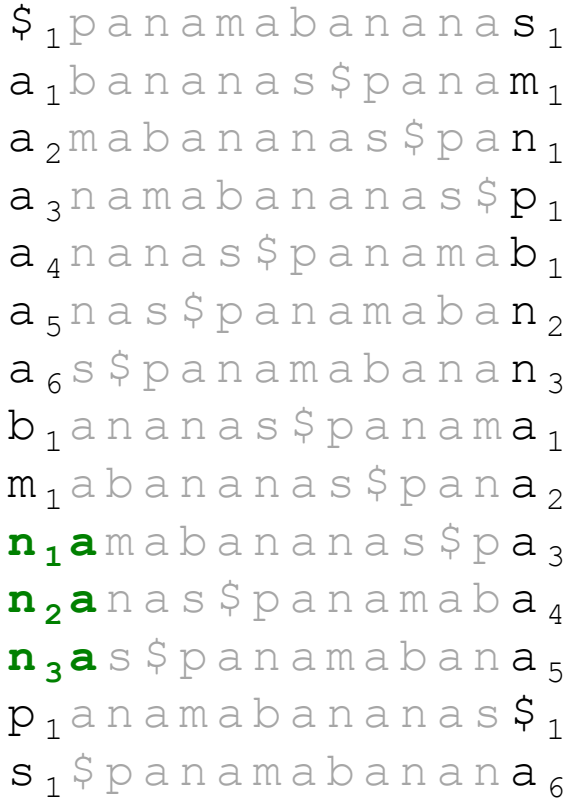
Three Matches of **na** Found!

- Searching for a**na** in panamabananas



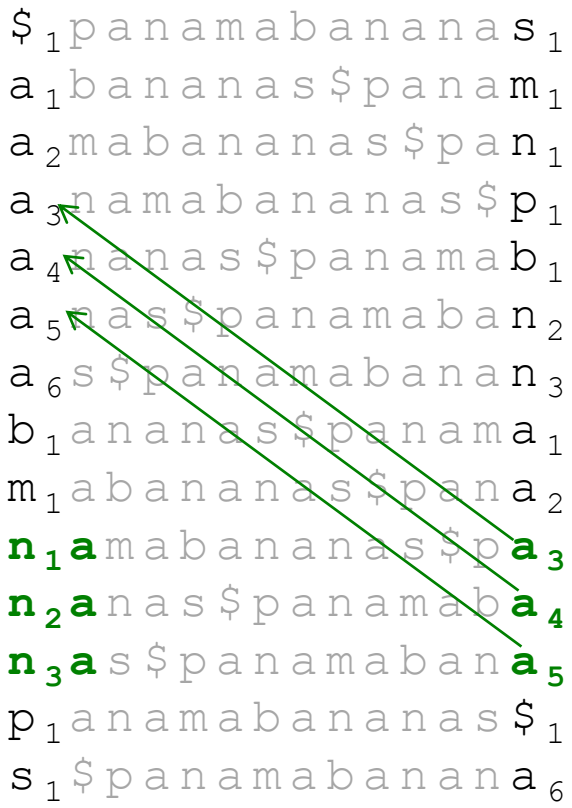
Three Matches of **na** Found!

- Searching for a**na** in panamabananas



Matching **ana**

- Searching for **ana** in panamabananas

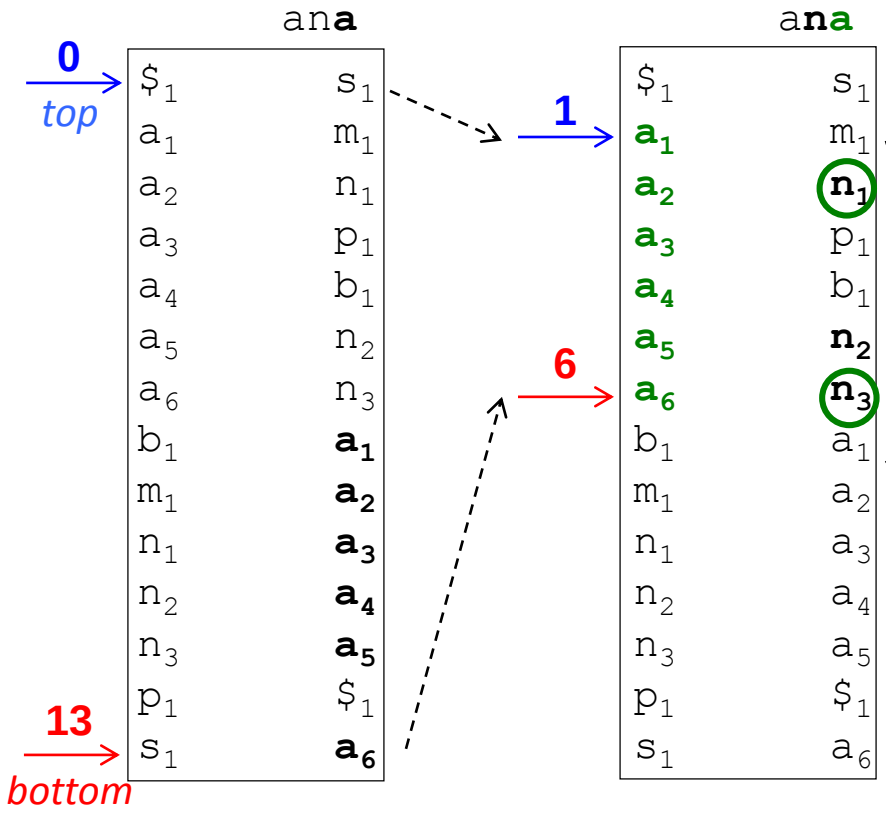


Three Matches of **ana** Found!

- Searching for **ana** in panamabananas



Searching for **ana** using *top* and *bottom* pointers



topIndex ← first position of *symbol* among positions from *top* to *bottom* in *LastColumn*

bottomIndex ← last position of *symbol* among positions from *top* to *bottom* in *LastColumn*

BWMatching

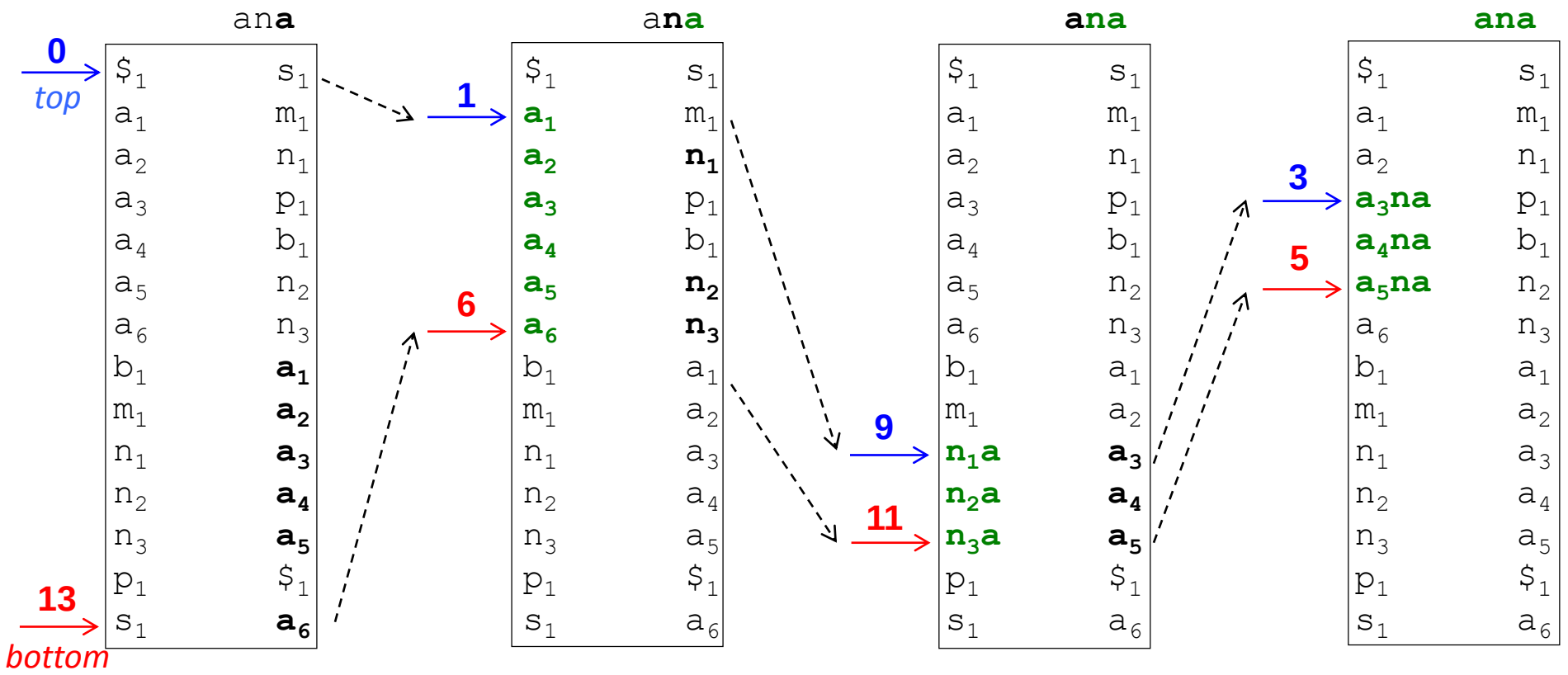
```
BWMATCHING(FirstColumn, LastColumn, Pattern, LASTTOFIRST)
    top ← 0
    bottom ← |LastColumn| − 1
    while top ≤ bottom
        if Pattern is nonempty
            symbol ← last letter in Pattern
            remove last letter from Pattern
            if positions from top to bottom in LastColumn contain symbol
                topIndex ← first position of symbol among positions from top to bottom
                           in LastColumn
                bottomIndex ← last position of symbol among positions from top to
                              bottom in LastColumn
                top ← LASTTOFIRST(topIndex)
                bottom ← LASTTOFIRST(bottomIndex)
            else
                return 0
        else
            return bottom − top + 1
```

Given a symbol at position *index* in *LastColumn,*

LastToFirst(*index*) defines the position of this symbol in *FirstColumn*

BWMatching is slow:

it analyzes every symbol from *top* to *bottom* in each step!



if positions from *top* to *bottom* in *LastColumn* contain *symbol*

topIndex ← first position of *symbol* among positions from *top* to *bottom* in *LastColumn*

bottomIndex ← last position of *symbol* among positions from *top* to *bottom* in *LastColumn*

Introducing *Count* Array

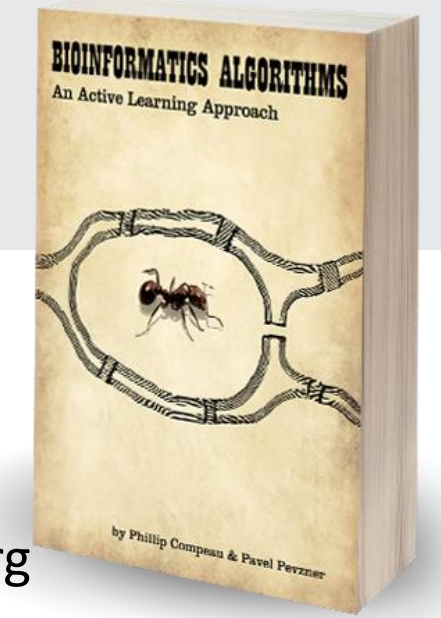
i	*FirstColumn*	*LastColumn*	LastToFirst(i)	Count						
				$	a	b	m	n	p	s
0	$\$_1$	s_1	13	0	0	0	0	0	0	0
1	a_1	m_1	8	0	0	0	0	0	0	1
2	a_2	n_1	9	0	0	0	1	0	0	1
3	a_3	p_1	12	0	0	0	1	1	0	1
4	a_4	b_1	7	0	0	0	1	1	1	1
5	a_5	n_2	10	0	0	1	1	1	1	1
6	a_6	n_3	11	0	0	1	1	2	1	1
7	b_1	a_1	1	0	0	1	1	3	1	1
8	m_1	a_2	2	0	1	1	1	3	1	1
9	n_1	a_3	3	0	2	1	1	3	1	1
10	n_2	a_4	4	0	3	1	1	3	1	1
11	n_3	a_5	5	0	4	1	1	3	1	1
12	p_1	$\$_1$	0	0	5	1	1	3	1	1
13	s_1	a_6	6	1	5	1	1	3	1	1
				1	6	1	1	3	1	1

$Count_{symbol}(i, LastColumn)$:

#occurrences of *symbol* in the first *i* positions of *LastColumn*

BetterBWMatching

```
BETTERBWMATCHING(FIRSTOCCURRENCE, LastColumn, Pattern, COUNT)
    top ← 0
    bottom ← |LastColumn| − 1
    while top ≤ bottom
        if Pattern is nonempty
            symbol ← last letter in Pattern
            remove last letter from Pattern
            top ← FIRSTOCCURRENCE(symbol) + COUNT_symbol(top, LastColumn)
            bottom ← FIRSTOCCURRENCE(symbol) + COUNT_symbol(bottom + 1,
                        LastColumn) − 1
        else
            return bottom − top + 1
    return
```



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Where Are the Matches?

- We know that **ana** occurs 3 times, but where does **ana** appear in *Text*???

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$

Outline

- Burrows-Wheeler Transform
- Inverting Burrows-Wheeler Transform
- Using BWT for Pattern Matching
- **Suffix Arrays**
- Approximate Pattern Matching

Where Are the Matches?

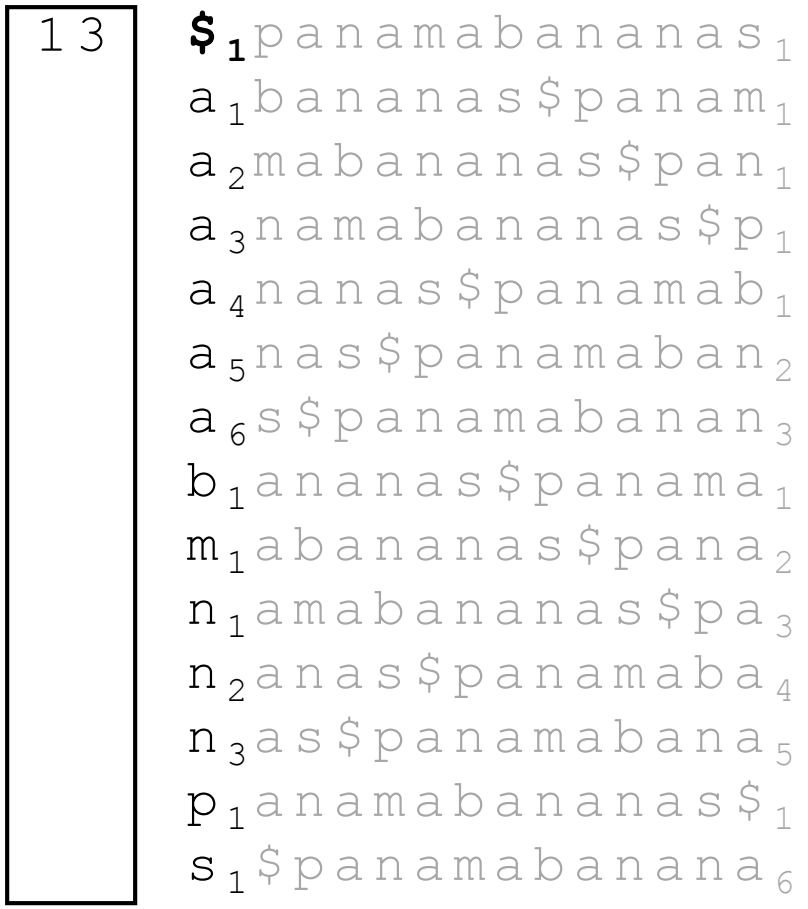
- **Suffix array** holds starting position of each suffix



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

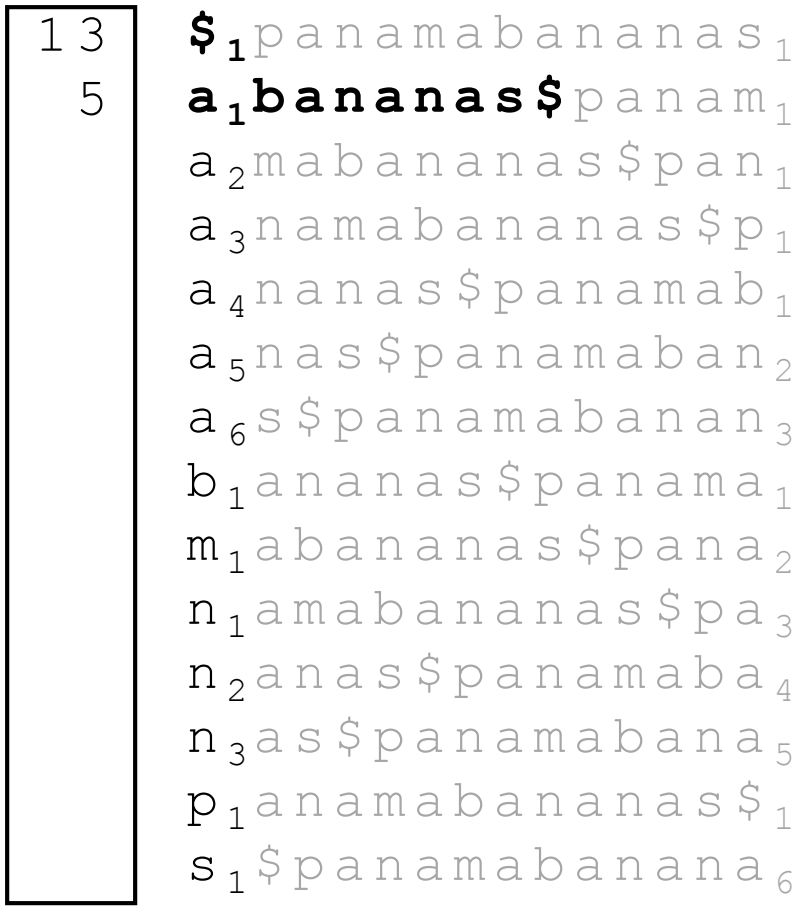
panamabananas**$**



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

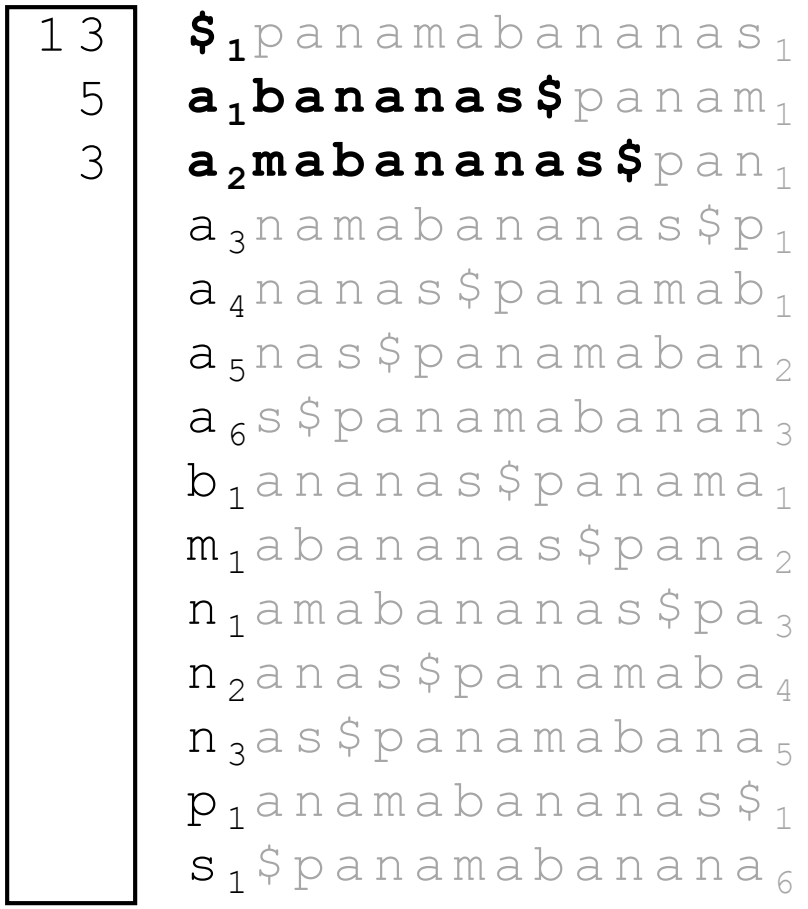
panam**abananas$**



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

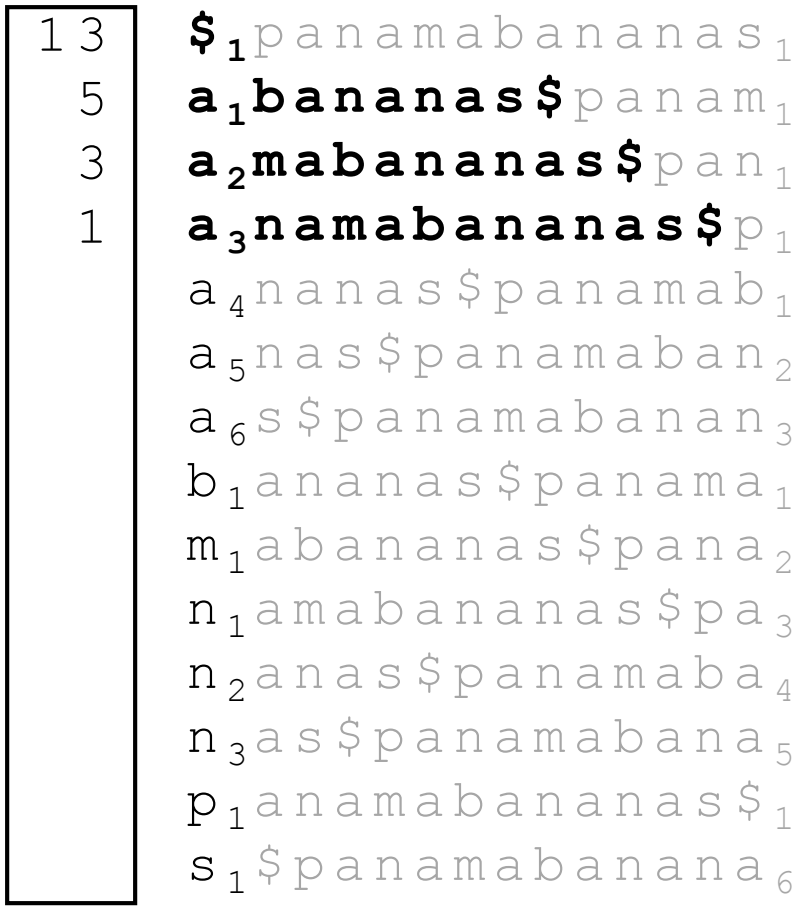
panamabananas$



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

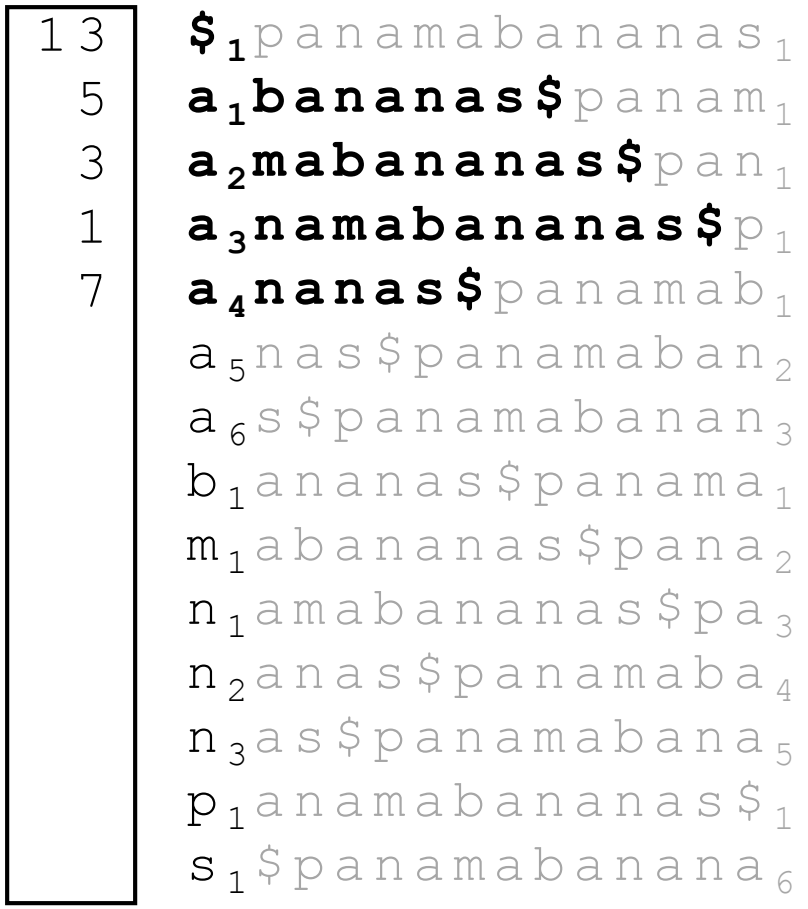
panamabananas$



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

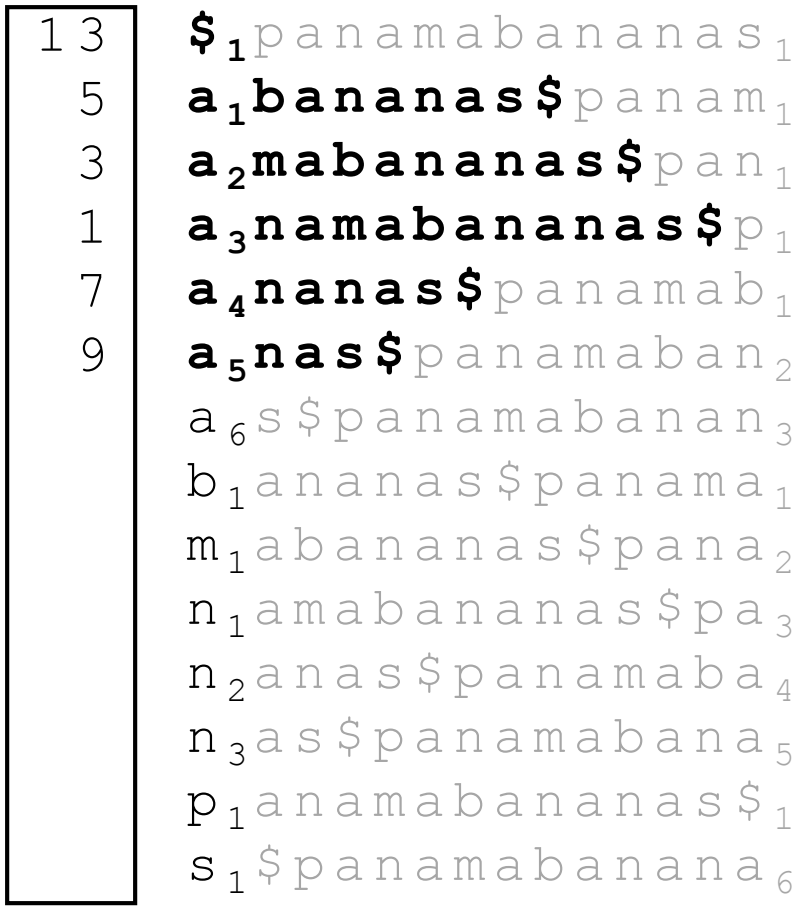
panamab**ananas$**



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamaban**anas$**



Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamabanan**as$**

13	$$_1$panamabananas$_1$
5	a$_1$bananas$panam$_1$
3	a$_2$mabananaspan_1$
1	a$_3$namabananas$p$_1$
7	a$_4$nanas$panamab$_1$
9	a$_5$nas$panamaban$_2$
11	a$_6$s$panamabanan$_3$
	b$_1$ananas$panama$_1$
	m$_1$abananas$pana$_2$
	n$_1$amabananaspa_3$
	n$_2$anas$panamaba$_4$
	n$_3$as$panamabana$_5$
	p$_1$anamabananas$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panama**bananas$**

13	$$_1$panamabananas$_1$
5	a$_1$**bananas$**panam$_1$
3	a$_2$**mabananas$**pan$_1$
1	a$_3$**namabananas$**p$_1$
7	a$_4$**nanas$**panamab$_1$
9	a$_5$**nas$**panamaban$_2$
11	a$_6$**s$**panamabanan$_3$
6	**b$_1$ananas$**panama$_1$
	m$_1$abananas$pana$_2$
	n$_1$amabananaspa_3$
	n$_2$anas$panamaba$_4$
	n$_3$as$panamabana$_5$
	p$_1$anamabananas$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panama**mabananas$**

13	**$**$_1$panamabananas$_1$
5	**a**$_1$**bananas$**panam$_1$
3	**a**$_2$**mabananas$**pan$_1$
1	**a**$_3$**namabananas$**p$_1$
7	**a**$_4$**nanas$**panamab$_1$
9	**a**$_5$**nas$**panamaban$_2$
11	**a**$_6$**s$**panamabanan$_3$
6	**b**$_1$**ananas$**panama$_1$
4	**m**$_1$**abananas$**pana$_2$
	n$_1$amabananaspa_3$
	n$_2$anas$panamaba$_4$
	n$_3$as$panamabana$_5$
	p$_1$anamabananas$$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamabananas$

13	$$_1$panamabananas$_1$
5	a$_1$bananas$panam$_1$
3	a$_2$mabananaspan_1$
1	a$_3$namabananas$p$_1$
7	a$_4$nanas$panamab$_1$
9	a$_5$nas$panamaban$_2$
11	a$_6$s$panamabanan$_3$
6	b$_1$ananas$panama$_1$
4	m$_1$abananas$pana$_2$
2	n$_1$amabananaspa_3$
	n$_2$anas$panamaba$_4$
	n$_3$as$panamabana$_5$
	p$_1$anamabananas$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamaba**nanas$**

13	**$**$_1$panamabananas$_1$
5	**a**$_1$**bananas$**panam$_1$
3	**a**$_2$**mabananas$**pan$_1$
1	**a**$_3$**namabananas$**p$_1$
7	**a**$_4$**nanas$**panamab$_1$
9	**a**$_5$**nas$**panamaban$_2$
11	**a**$_6$**s$**panamabanan$_3$
6	**b**$_1$**ananas$**panama$_1$
4	**m**$_1$**abananas$**pana$_2$
2	**n**$_1$**amabananas$**pa$_3$
8	**n**$_2$**anas$**panamaba$_4$
	n$_3$as$panamabana$_5$
	p$_1$anamabananas$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamabanana**nas$**

13	**$**$_1$panamabananas$_1$
5	**a**$_1$**bananas$**panam$_1$
3	**a**$_2$**mabananas$**pan$_1$
1	**a**$_3$**namabananas$**p$_1$
7	**a**$_4$**nanas$**panamab$_1$
9	**a**$_5$**nas$**panamaban$_2$
11	**a**$_6$**s$**panamabanan$_3$
6	**b**$_1$**ananas$**panama$_1$
4	**m**$_1$**abananas$**pana$_2$
2	**n**$_1$**amabananas$**pa$_3$
8	**n**$_2$**anas$**panamaba$_4$
10	**n**$_3$**as$**panamabana$_5$
	p$_1$anamabananas$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamabananas$

13	$$_1$panamabananas$_1$
5	a$_1$bananas$panam$_1$
3	a$_2$mabananaspan_1$
1	a$_3$namabananas$p$_1$
7	a$_4$nanas$panamab$_1$
9	a$_5$nas$panamaban$_2$
11	a$_6$s$panamabanan$_3$
6	b$_1$ananas$panama$_1$
4	m$_1$abananas$pana$_2$
2	n$_1$amabananaspa_3$
8	n$_2$anas$panamaba$_4$
10	n$_3$as$panamabana$_5$
0	p$_1$anamabananas$_1$
	s$_1$$panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

panamabanana**s$**

13	**$**$_1$panamabananas$_1$
5	**a**$_1$**bananas$**panam$_1$
3	**a**$_2$**mabananas$**pan$_1$
1	**a**$_3$**namabananas$**p$_1$
7	**a**$_4$**nanas$**panamab$_1$
9	**a**$_5$**nas$**panamaban$_2$
11	**a**$_6$**s$**panamabanan$_3$
6	**b**$_1$**ananas$**panama$_1$
4	**m**$_1$**abananas$**pana$_2$
2	**n**$_1$**amabananas$**pa$_3$
8	**n**$_2$**anas$**panamaba$_4$
10	**n**$_3$**as$**panamabana$_5$
0	**p**$_1$**anamabananas$**$_1$
12	**s**$_1$**$**panamabanana$_6$

Suffix Array

- **Suffix array:** holds starting position of each suffix beginning
  a row.

13	$_1$panamabananas$_1$
5	a$_1$bananas$panam$_1$
3	a$_2$mabananaspan_1$
1	a$_3$namabananas$p$_1$
7	a$_4$nanas$panamab$_1$
9	a$_5$nas$panamaban$_2$
11	a$_6$s$panamabanan$_3$
6	b$_1$ananas$panama$_1$
4	m$_1$abananas$pana$_2$
2	n$_1$amabananaspa_3$
8	n$_2$anas$panamaba$_4$
10	n$_3$as$panamaban$_5$
0	p$_1$anamabananas$$_1$
12	s$_1$$panamabanana$_6$

Using the Suffix Array to Find Matches

- Thus, **ana** occurs at positions **1**, **7**, **9**:
- panamab**anana**s$

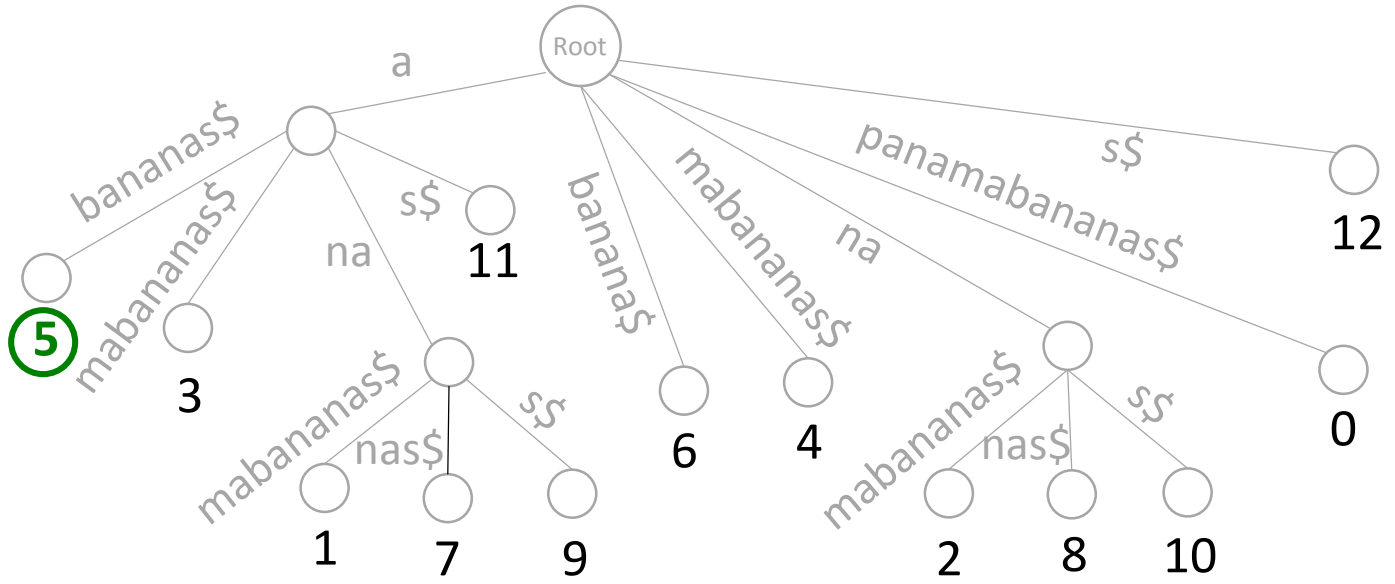


13	$$_1$panamabananas$_1$
5	a$_1$bananas$panam$_1$
3	a$_2$mabananaspan_1$
1	**a$_3$na**mabananas$p$_1$
7	**a$_4$na**nas$panamab$_1$
9	**a$_5$na**s$panamaban$_2$
11	a$_6$s$panamabanan$_3$
6	b$_1$ananas$panama$_1$
4	m$_1$abananas$pana$_2$
2	n$_1$amabananaspa_3$
8	n$_2$anas$panamaba$_4$
10	n$_3$as$panamabana$_5$
0	p$_1$anamabananas$$_1$
12	s$_1$$panamabanana$_6$

Naïve algorithm for constructing suffix array (sorting all suffixes of *Text*)

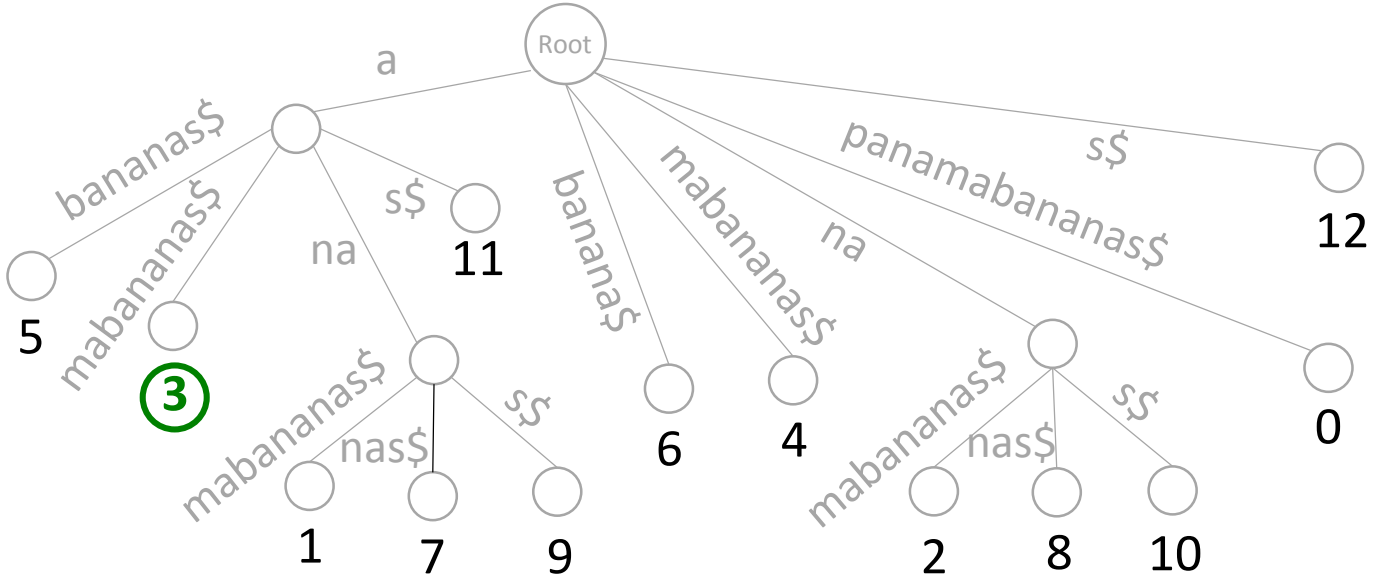
O(|*Text*| • log|*Text*|) comparisons

From Suffix Tree to Suffix Array: Depth-First Traversal



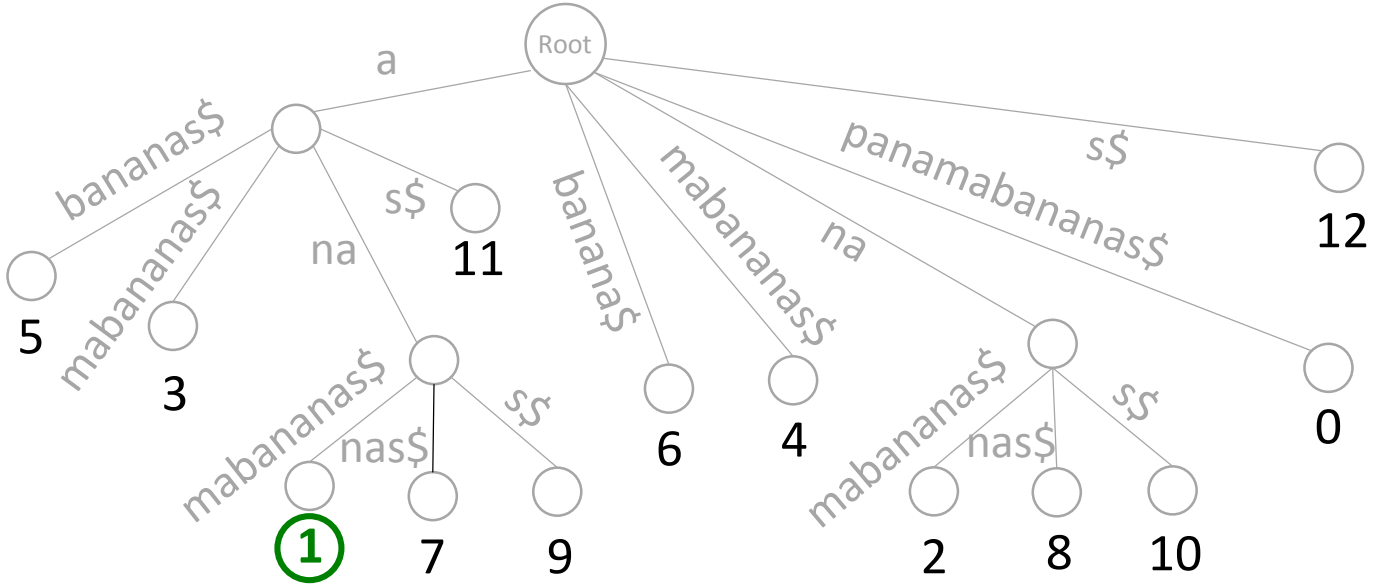
[13 **5** 3 1 7 9 11 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array: Depth-First Traversal



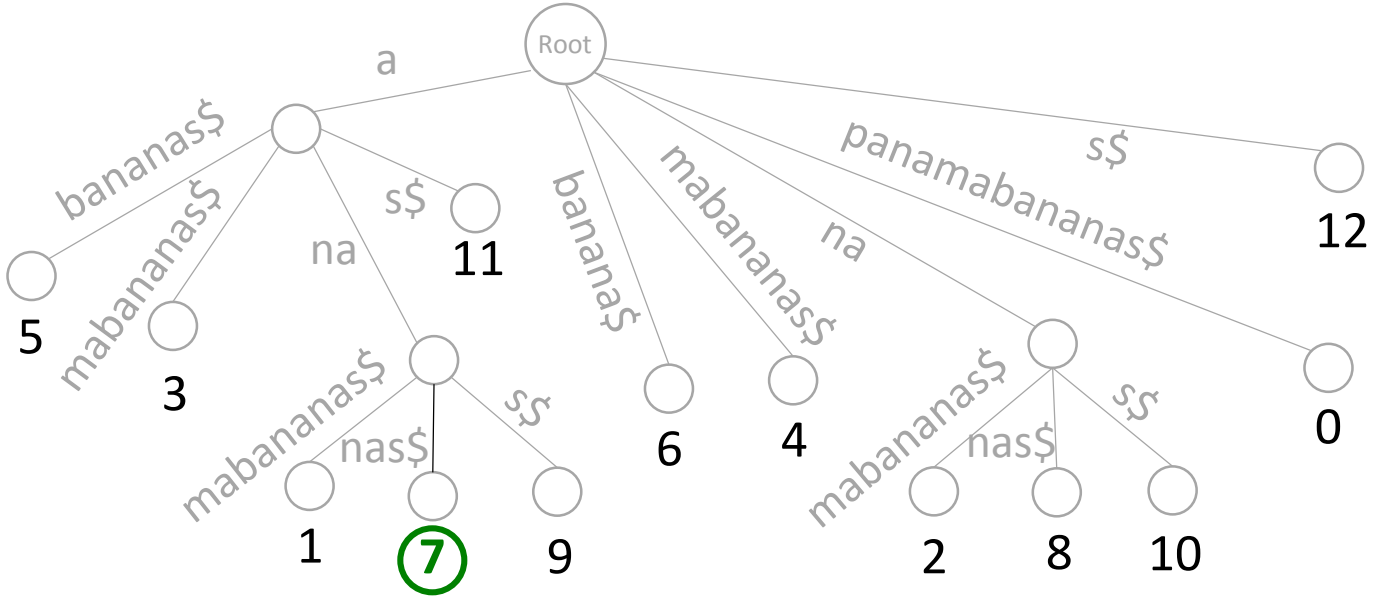
[13 5 **3** 1 7 9 11 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array: Depth-First Traversal



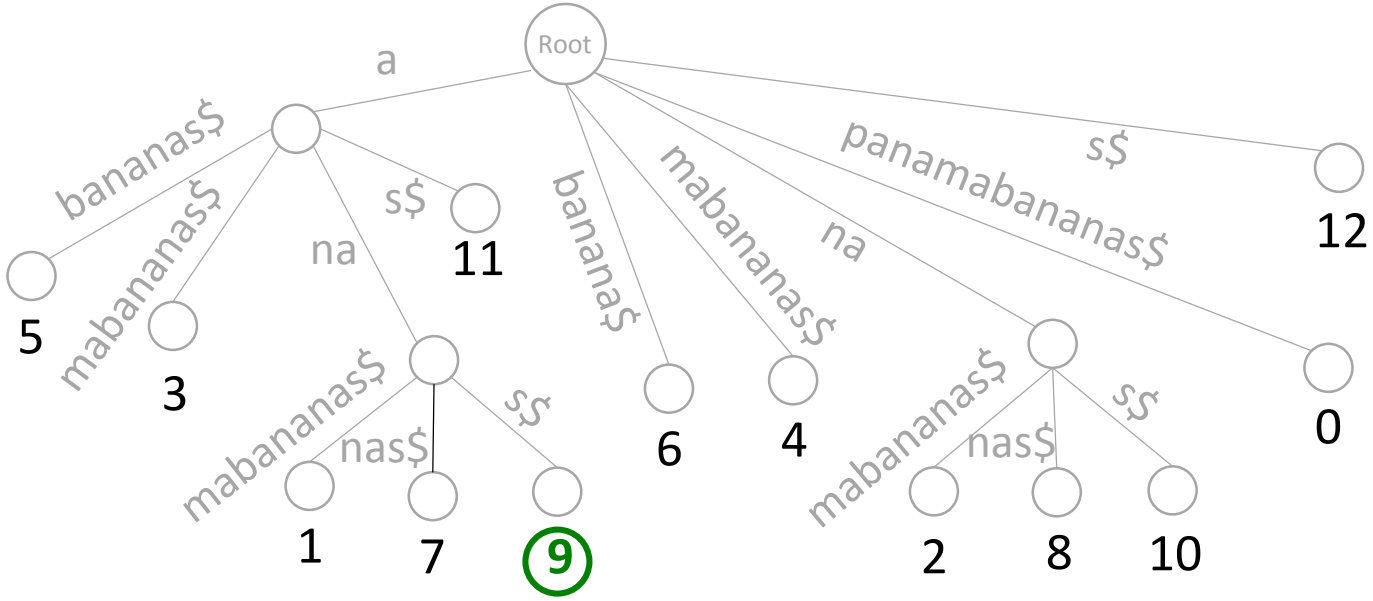
[13 5 3 1 7 9 11 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array



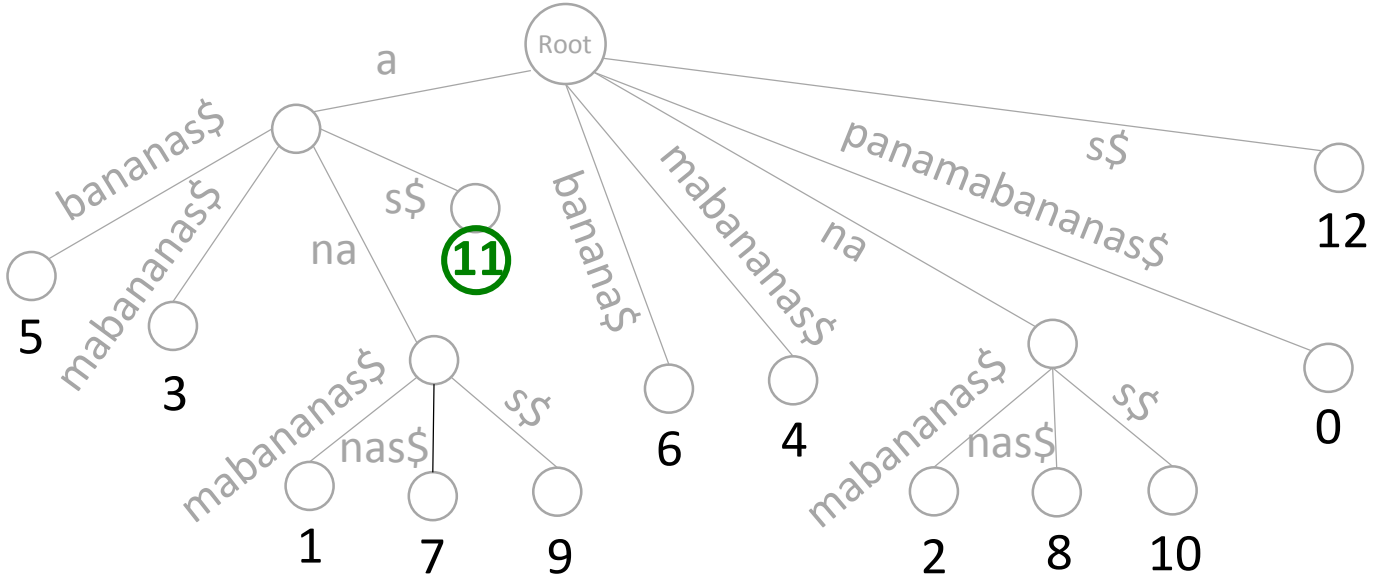
[13 5 3 1 7 9 11 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array



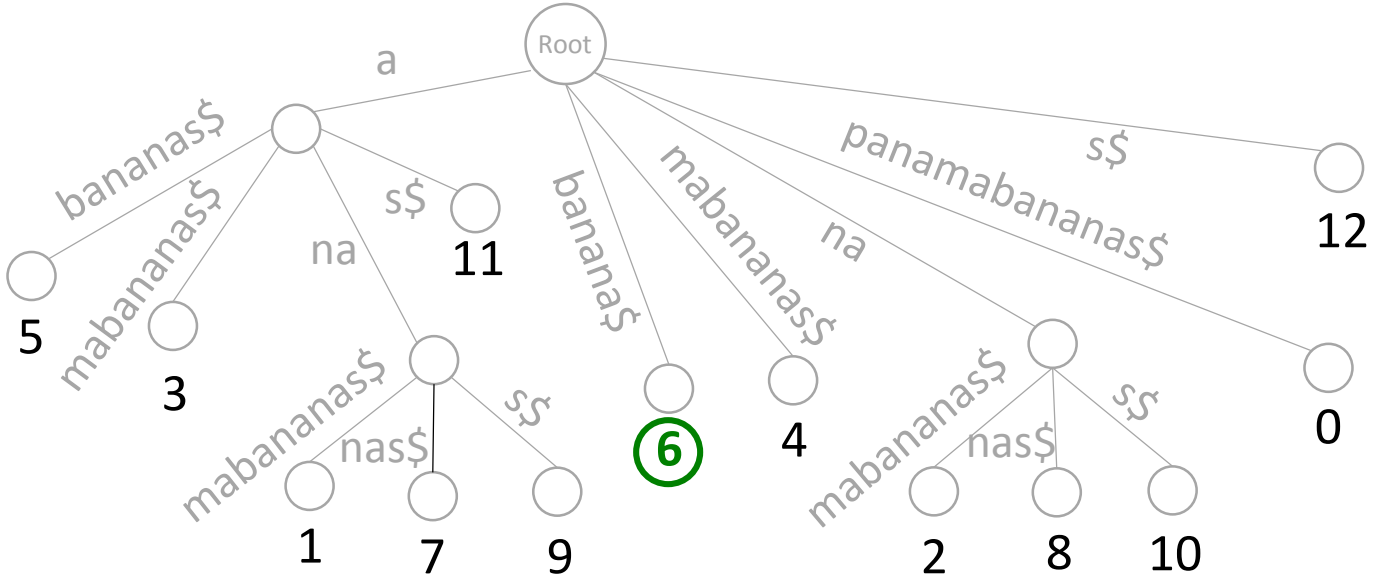
[13 5 3 1 7 **9** 11 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array



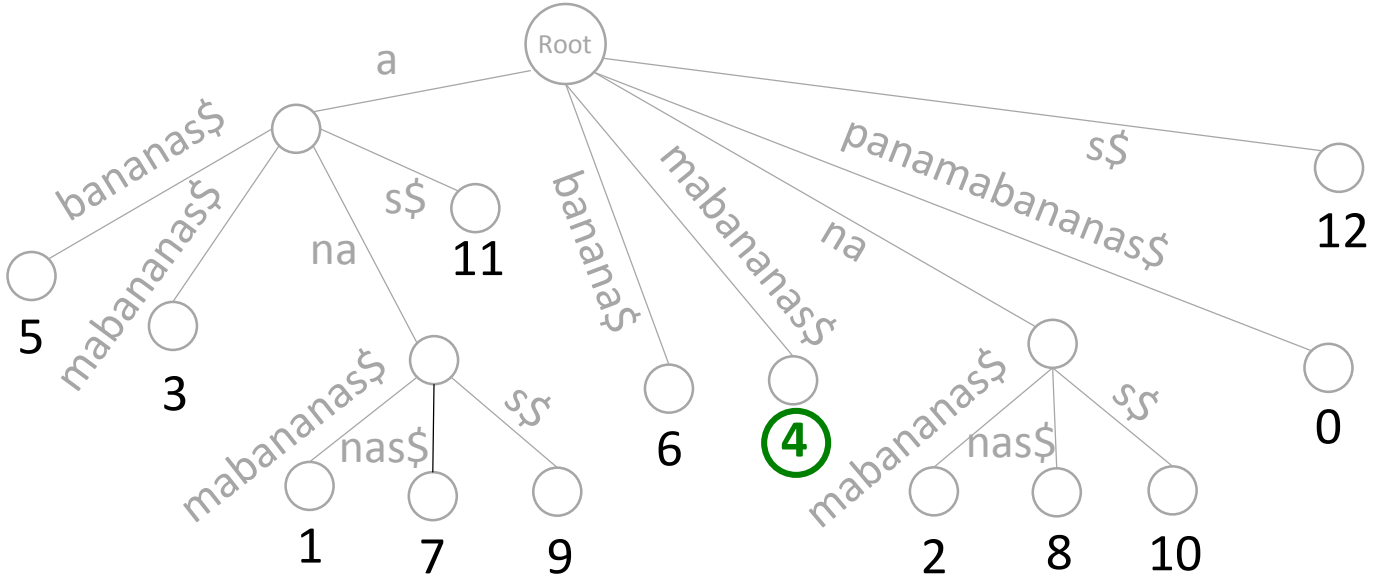
[13 5 3 1 7 9 **11** 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array



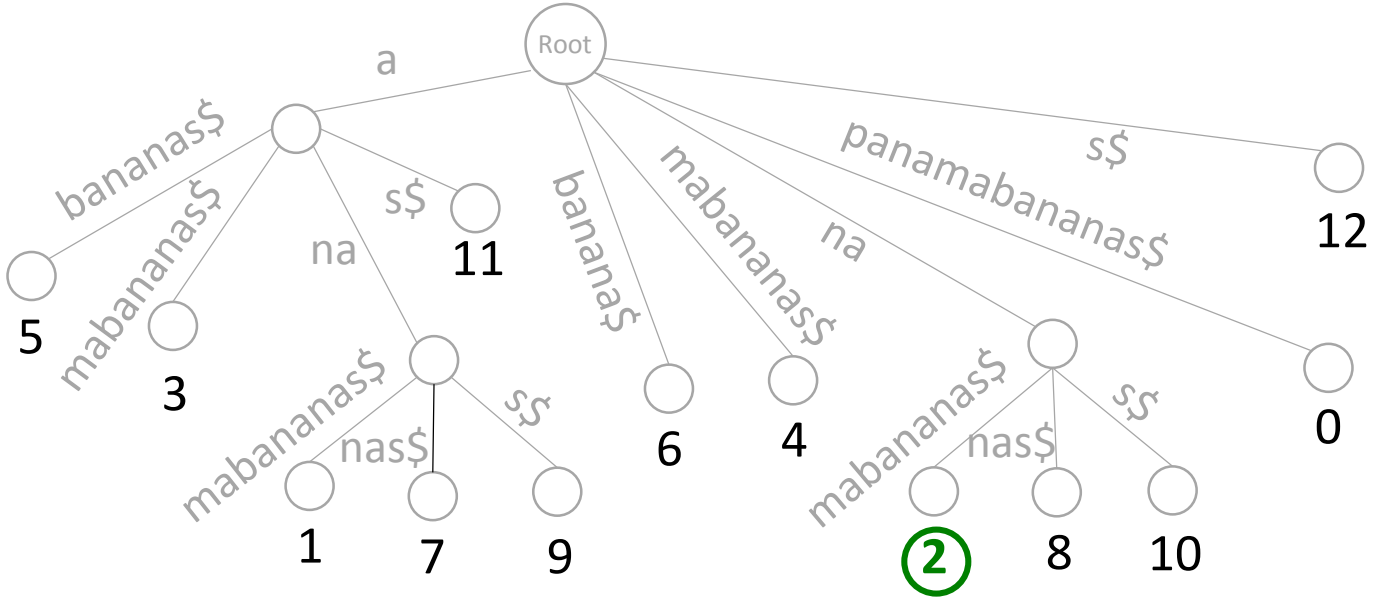
[13 5 3 1 7 9 11 **6** 4 2 8 10 0 12]

From Suffix Tree to Suffix Array



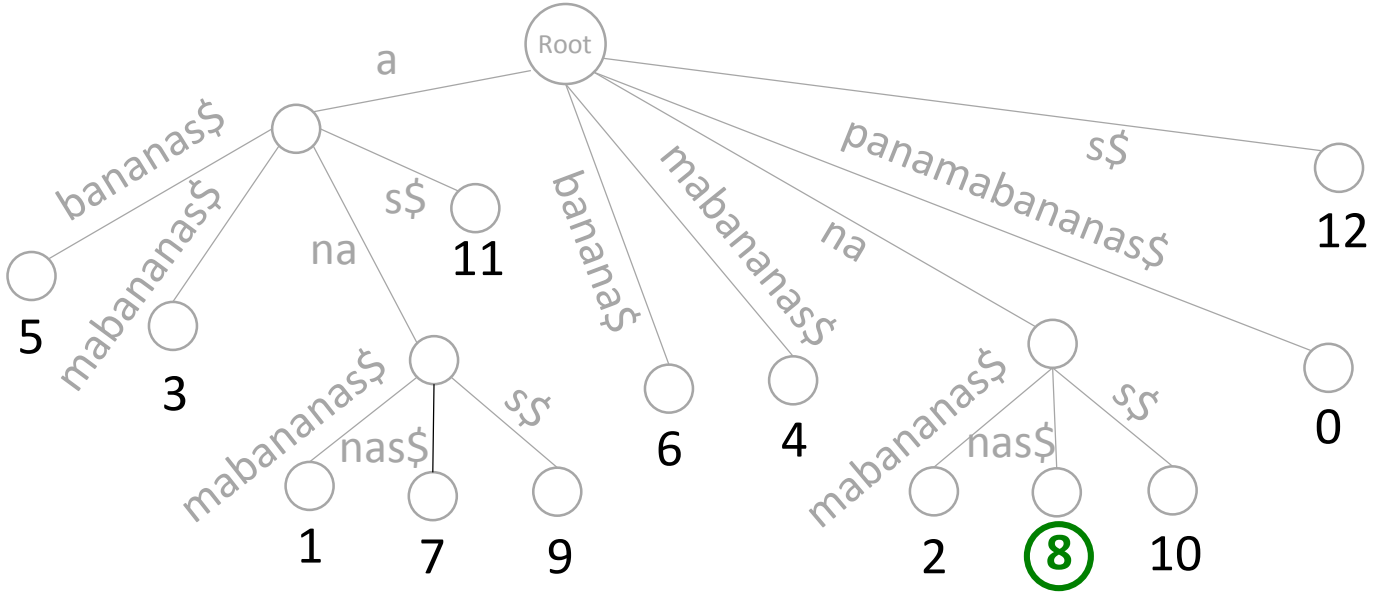
[13 5 3 1 7 9 11 6 **4** 2 8 10 0 12]

From Suffix Tree to Suffix Array



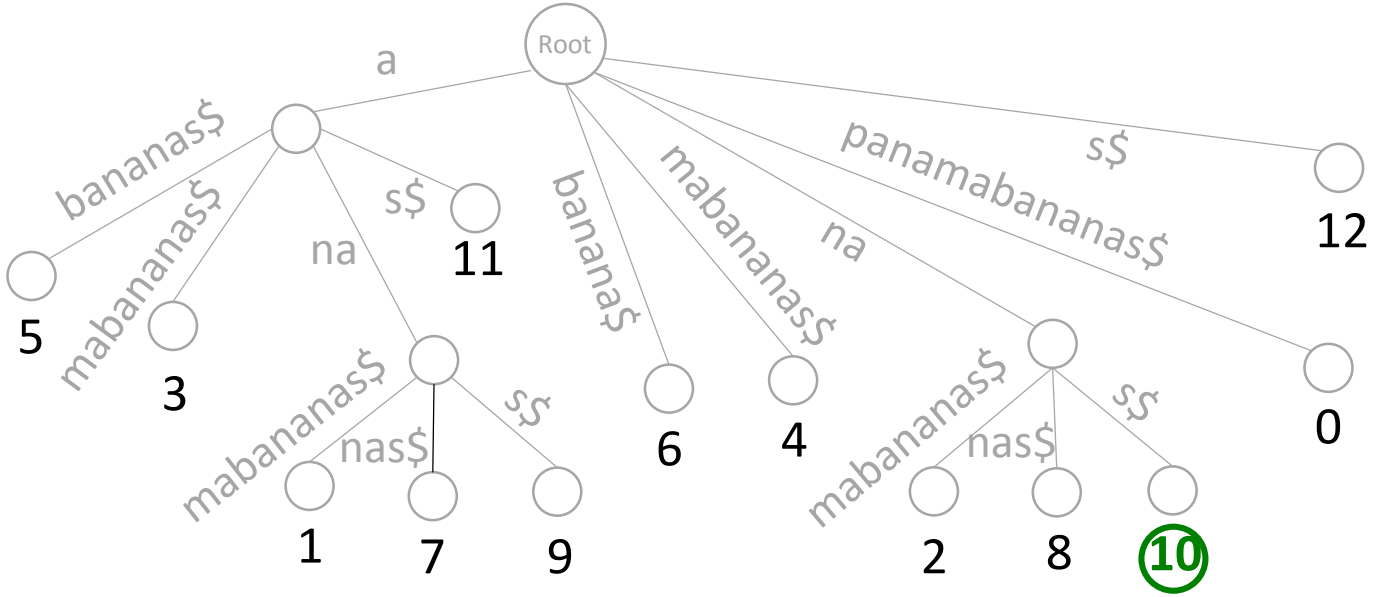
[13 5 3 1 7 9 11 6 4 **2** 8 10 0 12]

From Suffix Tree to Suffix Array



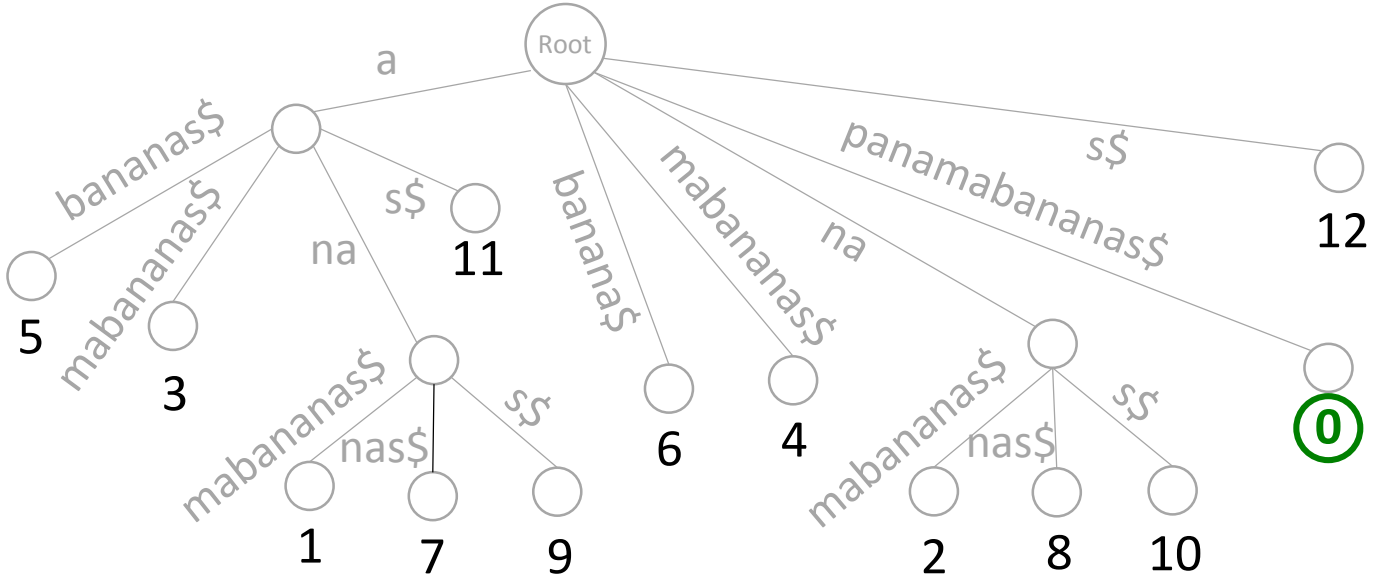
[13 5 3 1 7 9 11 6 4 2 8 10 0 12]

From Suffix Tree to Suffix Array



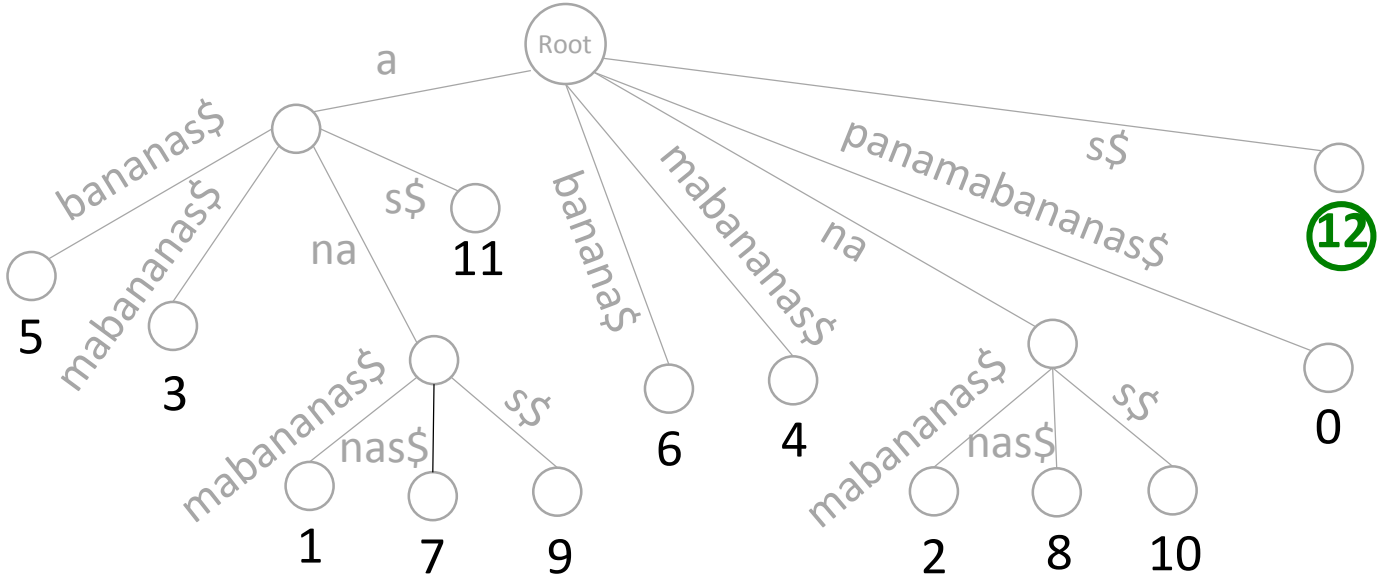
[13 5 3 1 7 9 11 6 4 2 8 **10** 0 12]

From Suffix Tree to Suffix Array



[13 5 3 1 7 9 11 6 4 2 8 10 **0** 12]

From Suffix Tree to Suffix Array



[13 5 3 1 7 9 11 6 4 2 8 10 0 12]

Constructing Suffix Array

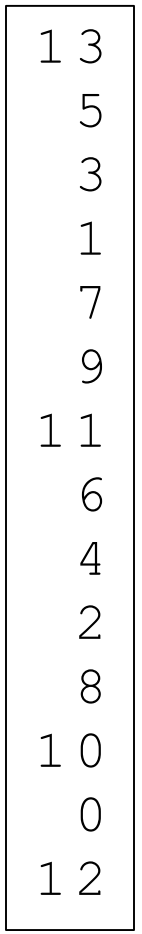
- Depth-first traversal of suffix tree
  - O(|*Text*|) time and ~20•|*Text*| space
- Manber-Myers algorithm (1990):
  - O(|*Text*|) time and ~4•|*Text*| space
- But memory footprint is still large for human genome!

We will learn how to quickly construct suffix array without relying on suffix tree later in this course



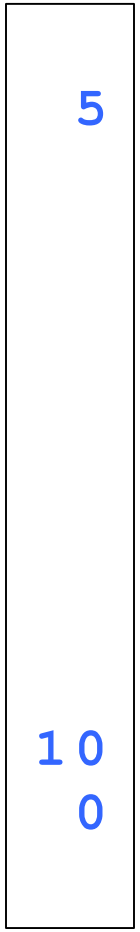
Reducing Memory Footprint for Suffix Array

- Can we store only a fraction of the suffix array but still do fast pattern matching?



Reducing Memory Footprint for Suffix Array

- Can we store only a fraction of the suffix array but still do fast pattern matching?
- Partial suffix array SuffixArray$_K$(*Text*) only contains values that are multiples of some integer *K*



Using the Suffix Array to Find Matches

	suffix array
$_1$panamabananas$_1$	13
a$_1$bananas$panam$_1$	5
a$_2$mabananaspan_1$	3
a$_3$namabananas$p$_1$	**1**
a$_4$nanas$panamab$_1$	**7**
a$_5$nas$panamaban$_2$	**9**
a$_6$s$panamabanan$_3$	11
b$_1$ananas$panama$_1$	6
m$_1$abananas$pana$_2$	4
n$_1$amabananaspa_3$	2
n$_2$anas$panamaba$_4$	8
n$_3$as$panamabana$_5$	10
p$_1$anamabananas$$_1$	0
s$_1$$panamabanana$_6$	12

Using the Partial Suffix Array to Find Matches

	partial suffix array
$_1$panamabananas$_1$	
a$_1$bananas$panam$_1$	**5**
a$_2$mabananaspan_1$	
a$_3$namabananas$p$_1$	
a$_4$nanas$panamab$_1$	
a$_5$nas$panamaban$_2$	
a$_6$s$panamabanan$_3$	
b$_1$ananas$panama$_1$	
m$_1$abananas$pana$_2$	
n$_1$amabananaspa_3$	
n$_2$anas$panamaba$_4$	
n$_3$as$panamabana$_5$	**1 0**
p$_1$anamabananas$$_1$	**0**
s$_1$$panamabanana$_6$	

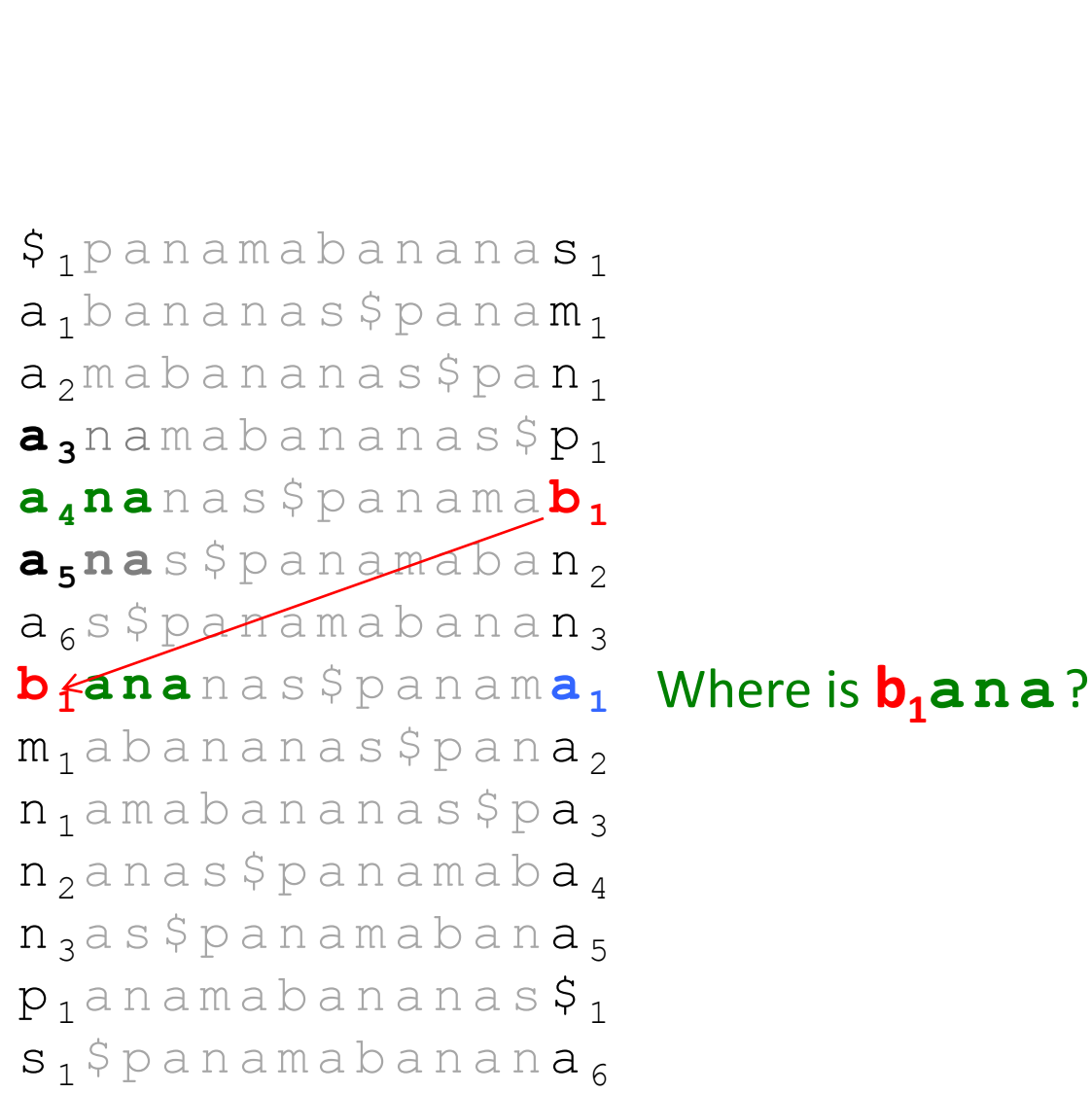
Using the Partial Suffix Array to Find Matches

		partial suffix array
$_1$panamabananas$_1$		
a$_1$bananas$panam$_1$		5
a$_2$mabananaspan_1$		
a$_3$namabananas$p$_1$		
a$_4$nanas$panamab$_1$	Where are these **ana** prefixes located in *Text*???	
a$_5$nas$panamaban$_2$		
a$_6$s$panamabanan$_3$		
b$_1$ananas$panama$_1$		
m$_1$abananas$pana$_2$		
n$_1$amabananaspa_3$		
n$_2$anas$panamaba$_4$		
n$_3$as$panamabana$_5$		1 0
p$_1$anamabananas$$_1$		0
s$_1$$panamabanana$_6$		

Focus on a_4na

		partial suffix array
$\$_1$panamabananas$_1$		
a$_1$bananas$panam$_1$		5
a$_2$mabananaspan_1$		
a$_3$namabananas$p$$_1$		
a$_4$nanas$panama**b$_1$**	Where is **a$_4$na**?	
a$_5$nas$panamaban$$_2$		
a$_6$s$panamabanan$$_3$		
b$_1$ananas$panama$$_1$		
m$_1$abananas$pana$$_2$		
n$_1$amabananas$pa$$_3$		
n$_2$anas$panamaba$$_4$		
n$_3$as$panamaban$$_5$		1 0
p$_1$anamabananas$$_1$		0
s$_1$$panamabanana$$_6$		

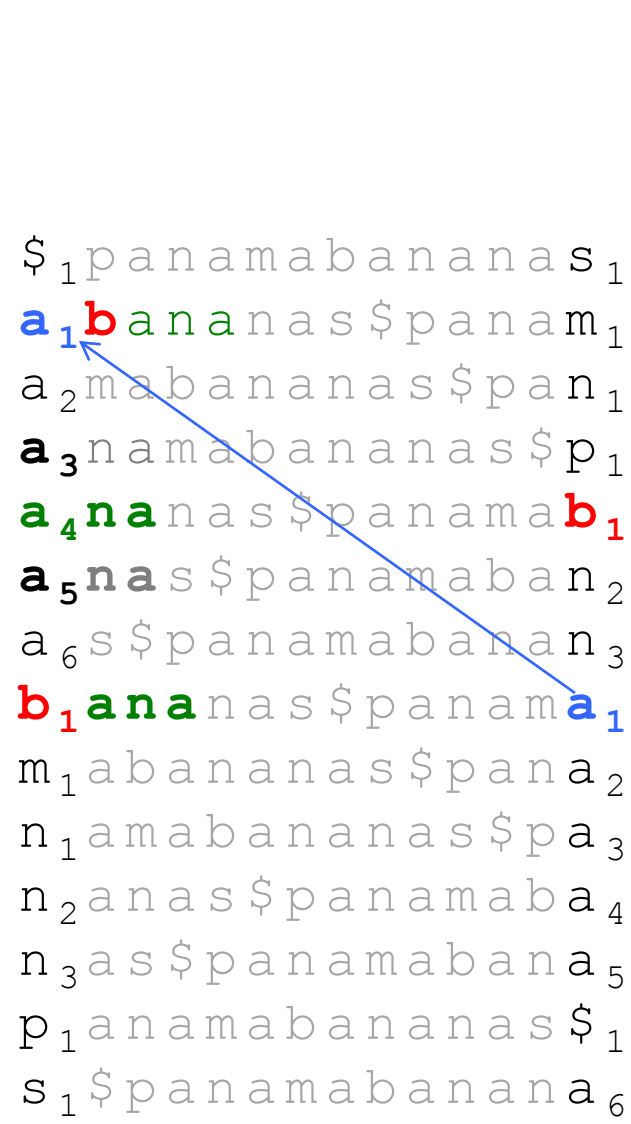
Focus on b_1ana



Where is b_1ana?

partial suffix array
5
1 0
0

Focus on a_1bana



Where is a_1bana?

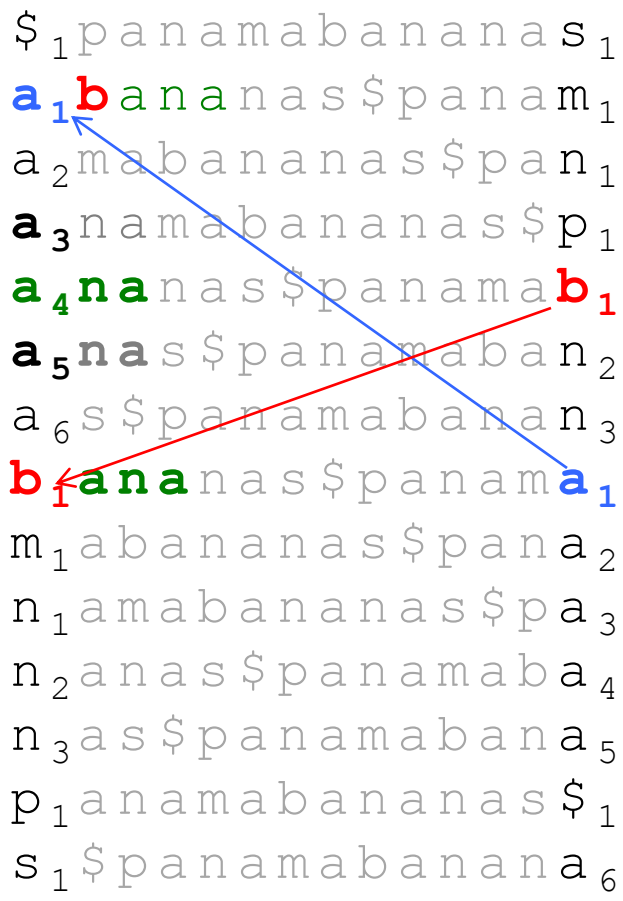
partial suffix array

5

1 0

0

Partial suffix array reveals position of a_1bana



a_1bana is at position 5

a_4na is at position 7

b_1ana is at position 6

partial suffix array

5	
7	
6	
1	0
	0

Outline

- Burrows-Wheeler Transform
- Inverting Burrows-Wheeler Transform
- Using BWT for Pattern Matching
- Suffix Arrays
- **Approximate Pattern Matching**

Returning to Search for Mutations

- **Approximate Pattern Matching Problem**:
  - **Input**: A string *Pattern*, a string *Text*, and an integer *d*.
  - **Output:** All positions in *Text* where the string *Pattern* appears as a substring with at most *d* mismatches.

Revealing Mutations by Analyzing Billions of Reads

- **Multiple Approximate Pattern Matching Problem**
  - **Input**: A **set** of strings *Patterns*, a string *Text*, and an integer *d*.
  - **Output:** All positions in *Text* where a string from *Patterns* appears as a substring with at most *d* mismatches.

BWT Saves the Day Again

- searching for `ana` in `panamabananas`

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$

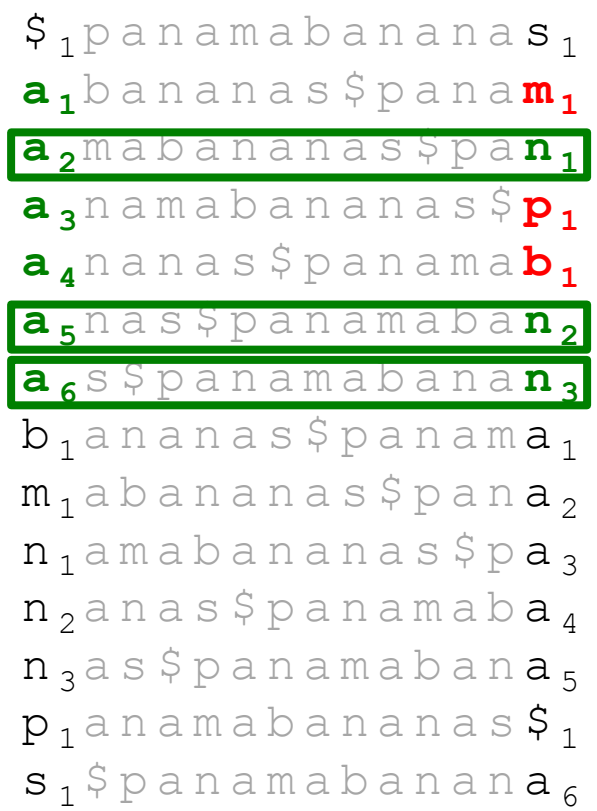
BWT Saves the Day Again

- searching for ana in panamabananas

$_1$panamabananas$_1$
a$_1$bananas$panam$_1$
a$_2$mabananaspan_1$
a$_3$namabananas$p$_1$
a$_4$nanas$panamab$_1$
a$_5$nas$panamaban$_2$
a$_6$s$panamabanan$_3$
b$_1$ananas$panama$_1$
m$_1$abananas$pana$_2$
n$_1$amabananaspa_3$
n$_2$anas$panamaba$_4$
n$_3$as$panamabana$_5$
p$_1$anamabananas$$_1$
s$_1$$panamabanana$_6$

BWT Saves the Day Again

- searching for **ana** in panamabananas

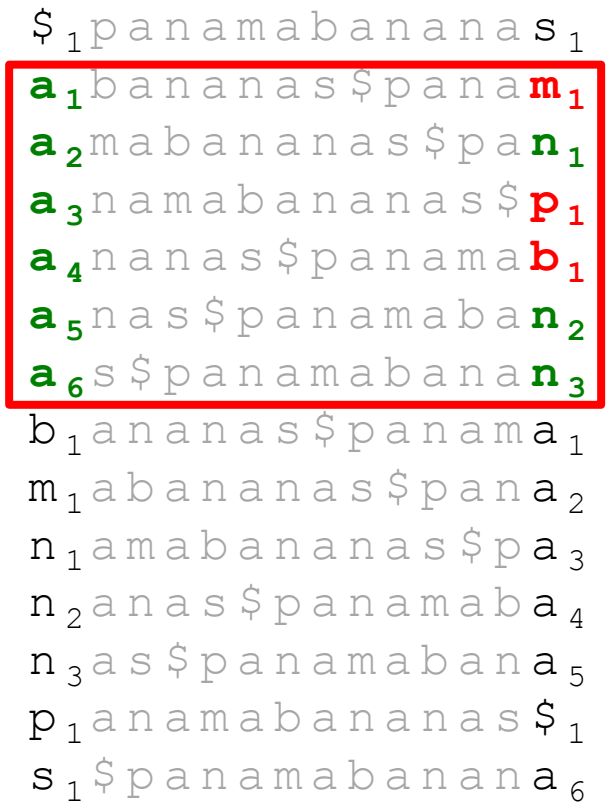


Exact matching

BWT Pattern Matching with 1 Mismatch

- searching for **ana** in panamabananas

To allow for 1 mismatch, we need to analyze the rows ending in red letters as well.



Approximate matching with at most 1 mismatch

BWT Pattern Matching with 1 Mismatch

- searching for **ana** in panamabananas

To allow for 1 mismatch, we need to analyze the rows ending in red letters as well.

	# Mismatches
$$_1$panamabananas$_1$	
a$_1$bananas$panam**a$_1$**	1
a$_2$mabananas$pan**a$_1$**	0
a$_3$namabananas$**p$_1$**	1
a$_4$nanas$panamab**a$_1$**	1
a$_5$nas$panamab**an$_2$**	0
a$_6$s$panamaban**a$_3$**	0
b$_1$ananas$panama$_1$	
m$_1$abananas$pana$_2$	
n$_1$amabananaspa_3$	
n$_2$anas$panamaba$_4$	
n$_3$as$panamabana$_5$	
p$_1$anamabananas$$_1$	
s$_1$$panamabanana$_6$	

BWT Pattern Matching with 1 Mismatch

- searching for **ana** in panamabananas

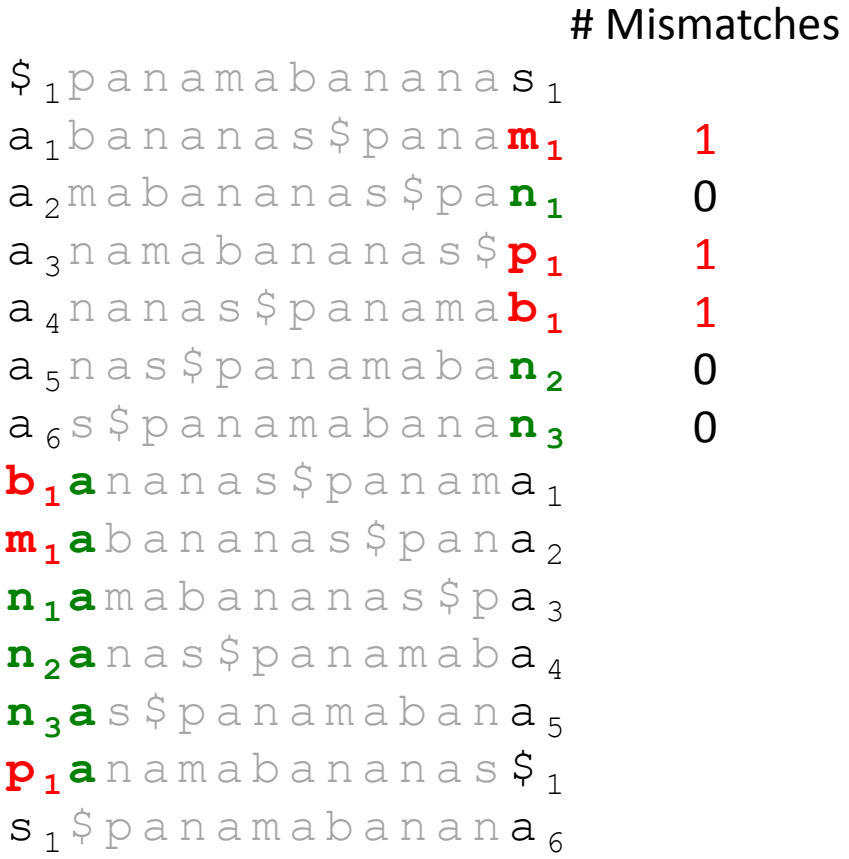
Now we analyze all rows with at most 1 mismatch using the First-Last property.

	# Mismatches
$$_1$panamabananas$_1$	
a$_1$bananas$panam**a$_1$**	1
a$_2$mabananas$pan**a$_1$**	0
a$_3$namabananas$p**a$_1$**	1
a$_4$nanas$panamab**a$_1$**	1
a$_5$nas$panamaban**a$_2$**	0
a$_6$s$panamabanan**a$_3$**	0
b$_1$ananas$panama$_1$	
m$_1$abananas$pana$_2$	
n$_1$amabananaspa_3$	
n$_2$anas$panamaba$_4$	
n$_3$as$panamabana$_5$	
p$_1$anamabananas$$_1$	
s$_1$$panamabanana$_6$	

BWT Pattern Matching with 1 Mismatch

- searching for **ana** in panamabananas

Now we analyze all rows with at most 1 mismatch using the First-Last property.



Mismatches

1
0
1
1
0
0

BWT Pattern Matching with 1 Mismatch

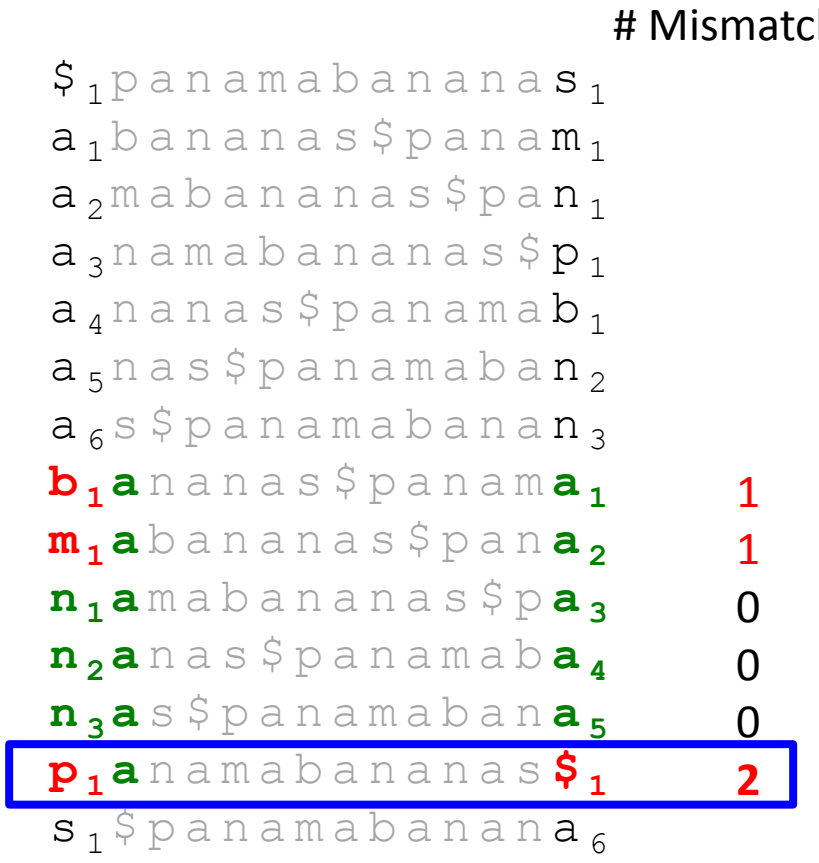
- searching for **ana** in panamabananas

Now we analyze all rows with at most 1 mismatch using the First-Last property.

	# Mismatches
$$_1$panamabananas$_1$	
a$_1$bananas$$_1$panam$_1$	
a$_2$mabananas$$_1pan_1$	
a$_3$namabananas$$_1p_1$	
a$_4$nanas$$_1$panamab$_1$	
a$_5$nas$$_1$panamaban$_2$	
a$_6$s$$_1$panamabanan$_3$	
b$_1$ananas$$_1$panama$_1$	1
m$_1$abananas$$_1$pana$_2$	1
n$_1$amabananas$$_1pa_3$	0
n$_2$anas$$_1$panamaba$_4$	0
n$_3$as$$_1$panamabana$_5$	0
p$_1$anamabananas$$_1$$_1$	1
s$_1$$$_1$panamabanana$_6$	

BWT Pattern Matching with 1 Mismatch

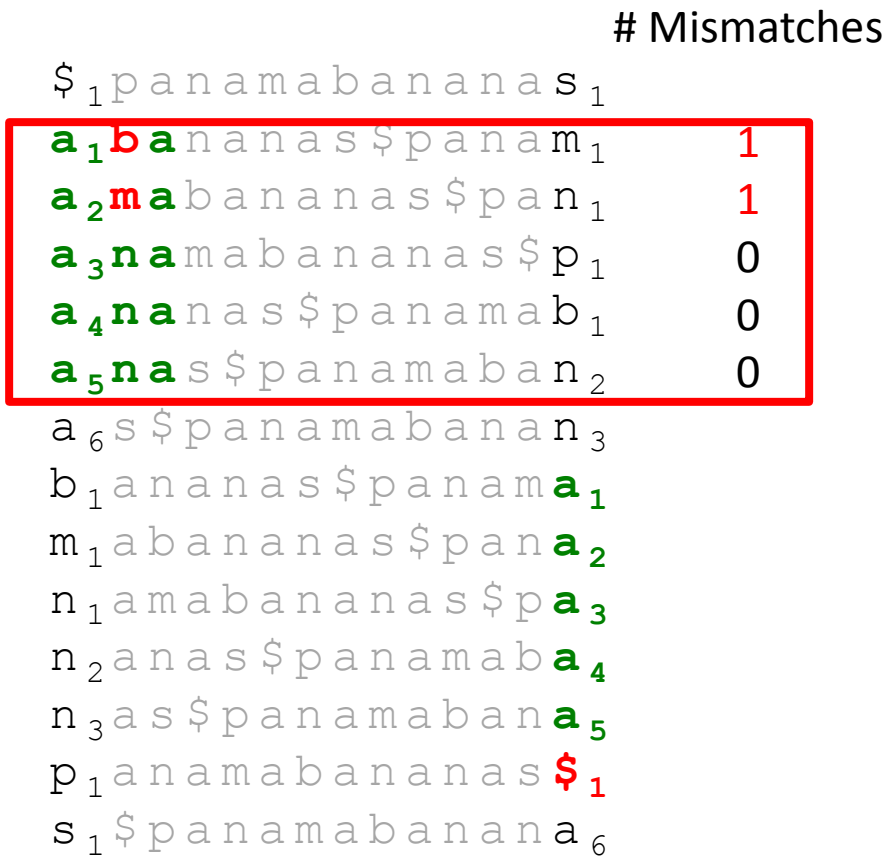
- searching for **ana** in panamabananas



This row results in a 2nd mismatch (the $), so we discard it.

Five Approximate Matches Found!

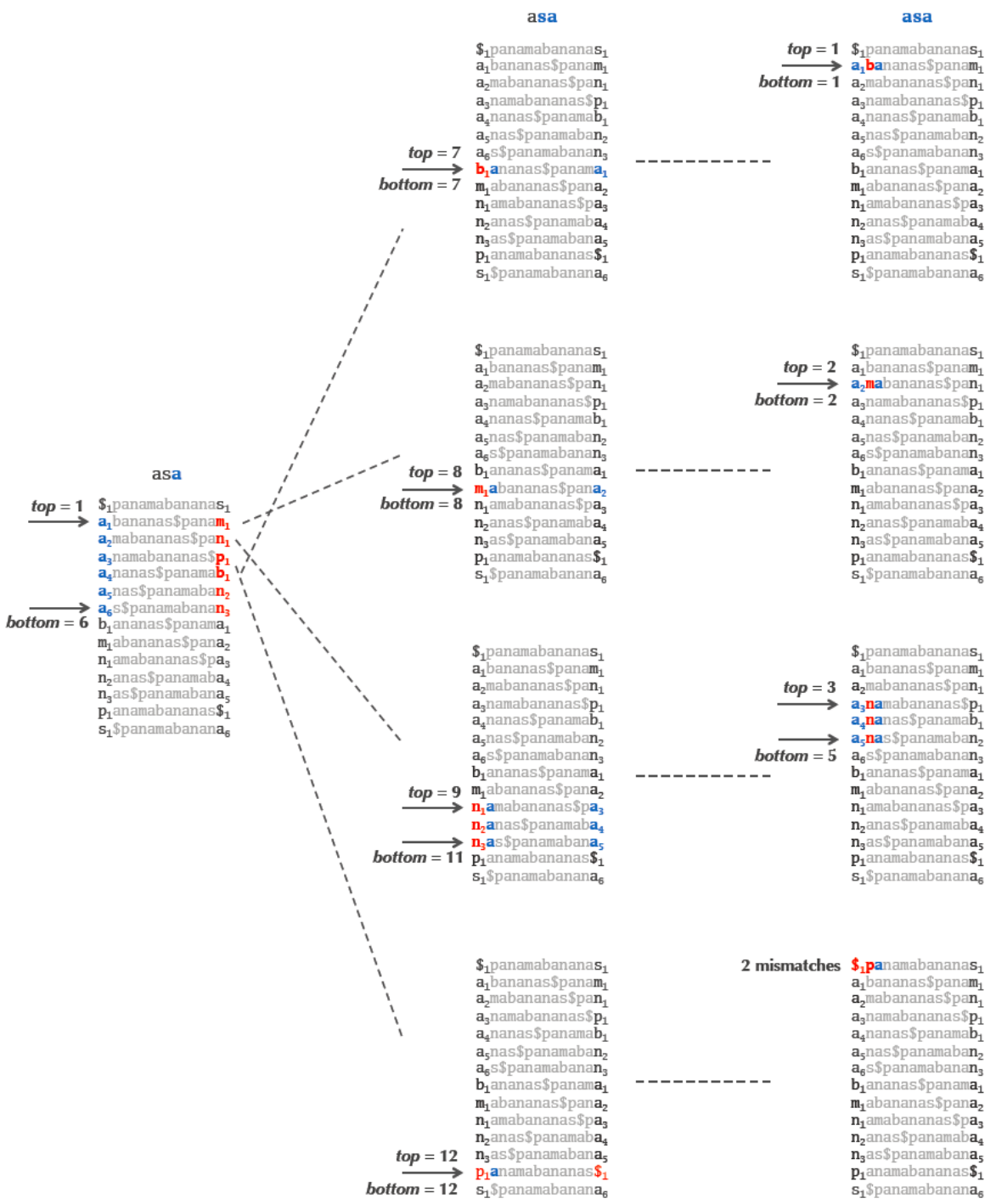
- searching for **ana** in panamabananas



Where Are The Matches?

- searching for **ana** in panamabananas

	Suffix Array
$$_1$panamabananas$_1$	
a$_1$bananas$panam$_1$	5
a$_2$mabananaspan_1$	3
a$_3$namabananas$p$_1$	1
a$_4$nanas$panamab$_1$	7
a$_5$nas$panamaban$_2$	9
a$_6$s$panamabanan$_3$	
b$_1$ananas$panama$_1$	
m$_1$abananas$pana$_2$	
n$_1$amabananaspa_3$	
n$_2$anas$panamaba$_4$	
n$_3$as$panamabana$_5$	
p$_1$anamabananas$$_1$	
s$_1$$panamabanana$_6$	



In reality, approximate pattern matching with BWT is more complex (we omitted various details)

